

Conditional Response Types and the Identification of Multivalued Treatment Effects

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Abstract

I study treatment effects under multiple options that lack a clear ranking. When the identifying variation stems from multiple instruments, agents can switch into different options and from many initial states. I discuss how to define and employ conditional response types (combinations of potential choices given by one instrument that depend on the variation of other instruments) to identify the shares of agents switching at well-defined margins of choice and their treatment effects. I present an empirical strategy consistent with this framework and apply it to: subsidies for malaria treatment in Kenya, childcare choice and children's development in Colombia, and merit- and need-based scholarships for higher education in Colombia. While standard methods identify the local average treatment effect of one option versus the next-best (a combination of fallback alternatives), I show how combining multiple sources of variation and defining conditional response types identifies effects of pairwise combinations of available options.

1 INTRODUCTION

Economic decisions often require individuals to choose among multiple options, such as selecting healthcare or childcare providers, or pursuing different paths in higher education. When agents can endogenously choose from multiple unordered alternatives, standard instrumental variables methods identify a combination of local treatment effects with possibly negative weights, or, using the variation from one instrument alone, the effect of one option *versus* the next-best (*i.e.*, a mixture of other options that agents would have chosen in the absence of their preferred option).¹ Although the effect of

¹On the particular issue of standard IV methods (namely, two-stage least squares, 2sls) assigning negative weights or leading to estimates lacking clear interpretations, see [Bhuller and Sigstad \(2024\)](#). In terms of the effect of one option *versus* the next-best and developments on identification and estimation, see, for example, [Heckman et al. \(2006, 2008\)](#); [Mountjoy \(2022\)](#).

an option versus the next-best can answer policy questions in some contexts, it often masks important heterogeneity among agents with distinct fallback alternatives.² If the fallback option is observed (*e.g.*, using rankings), pairwise treatment effects can be identified under additional testable behavioral assumptions (Kirkebøen et al., 2016; Heinesen et al., 2024). Otherwise, identification might require multiple continuous instruments (Mountjoy, 2022) with large support (Heckman et al., 2008), or additional assumptions on potential outcomes (Kline and Walters, 2016; Hull, 2018; Lee and Salanié, 2024).

I study the treatment effects of multiple unordered options in scenarios where discrete or continuous instruments are available, but option rankings are not observed. I propose an identification strategy that combines the variation of multiple discrete or continuous instruments such that, under some conditions, the set of next-best states changes as the instruments change, without requiring direct observation of fallback options or rankings. To fix ideas, suppose an experiment offers subsidies encouraging households to use drug shops (option 1) instead of visiting public healthcare centers (option 2) or doing nothing (option 3) to care for their health, and the researcher also observes the distance to the healthcare center. When distance is high, households are unlikely to visit healthcare centers, in which case the experimental variation primarily shifts choices between drug shops and choosing no care. Within my framework, I allow for the response to the variation in one instrument—*e.g.*, random assignment to subsidies encouraging households to use drug shops—to differ depending on other instruments—*e.g.*, proximity to healthcare centers—that affect the same choice margins. I formalize this notion by defining *conditional response types* consisting of combinations of potential choices given by one instrument that differ depending on the variation of other instruments. I derive conditions under which *conditional response types* help identify pairwise treatment effects—local and marginal treatment effects of one option relative to another option.

Recent theoretical and empirical work has focused on the identification and estimation of treatment effects with instrumental variables in contexts where agents can choose from multiple unordered alternatives.³ Because the options often lack a clear ordering, the variation in the instruments can induce agents to switch into different options and from many initial states (Angrist et al., 1996; Heckman et al., 2006). Importantly, the set of potential responses to variation in (a binary) instrument goes beyond the always-takers, never-takers, compliers, and defiers in Imbens and Angrist (1994). For instance,

²For instance, in the context of healthcare choices, a researcher may seek to estimate the average treatment effects of different options—for example, do drug shops improve health outcomes relative to healthcare centers or to receiving no care?—rather than the effect of choosing drug shops relative to a scenario in which they are unavailable.

³For an overview of recent developments in the literature of unordered treatments and instrumental variables, see Mogstad and Torgovitsky (2024). Theoretical work can be found in Heckman and Pinto (2018); Lee and Salanié (2018, 2024); Xie (2020); Caetano and Escanciano (2020); Feng (2024); Kamat et al. (2023), and for identification results with empirical applications, see Kline and Walters (2016); Mountjoy (2022).

Kirkebøen et al. (2016) and Heinesen et al. (2024) examine identification by combining instrumental variables and rankings with additional assumptions on the set of potential responses, namely, no next-best defiers nor irrelevance defiers.⁴ In the absence of rankings, different contexts might require more or less restrictive assumptions on choice behavior, or instruments with enough variation to eliminate some next-best options (Heckman et al., 2006, 2008).⁵ While jointly modeling responses to multiple instruments may have identification implications, as in the binary choice-binary IV case in Mogstad et al. (2021, 2024), current methods for multiple options implicitly assume that the behavior of compliers (or responses) to one instrument is the same across the distribution of other IVs.⁶

I contribute to the literature on multiple unordered treatments in three ways. First, I follow the methodological approach of Heckman and Pinto (2018) in which revealed preference analysis defines vectors used to identify the prevalence of compliers (i.e., agents who change their behavior as the IV changes) and their counterfactuals. I extend this notion to the case of *conditional response types* which are defined by variation in an instrument but can change across the support of other IVs. The building block for defining *conditional response types* is the assumption that instruments affect choices through their effect on potential costs. I model agents' decisions with a latent utility model with additive separability between the unobserved preference heterogeneity and the cost function. This is standard in the unobserved heterogeneity literature (Vytlacil, 2002). Recently, Heckman and Pinto (2018) extended this notion for unordered choices; Lee and Salanié (2024) model changes in relative mean utility through changes in relative costs. Following long-standing literature of shape restrictions that stem from economic theory, I assume that the cost function determining individual's decisions is convex.⁷ The role of convexity in identification is twofold. First, convexity imposes restrictions such that once an option becomes too costly, its likelihood of being the next-best state goes to zero thereafter. Second, without any restrictions on the shape of the function, multiple instruments can affect the costs in different directions. Convexity imposes a smoothness condition such that, although the instruments enter the cost function nonlinearly, agents who are in-

⁴Kirkebøen et al. (2016) and Heinesen et al. (2024) examine the returns to fields of study in Norway and Denmark using admission cutoffs and students' reported rankings. They define next-best defiers (students who would not choose their fallback option when below the cutoff) and irrelevance defiers (students whose cutoff status affects other fields). The authors assess the validity of assuming away these two groups, propose testable implications, and assess the resulting bias. For an alternative approach to estimate local treatment effects when rankings are available, see Nibbering et al. (2022).

⁵For instance, Kline and Walters (2016); Hull (2018); Lee and Salanié (2024) employ assumptions on homogeneity of counterfactuals to disentangle treatment effects along well-defined margins of choice. Kamat (2024) focuses on partial identification of effects of program offers and exploits changes in the choice structure (comparing choice sets with and without the offer).

⁶The authors show that for the binary treatment case, the standard monotonicity assumption in Imbens and Angrist (1994) imposes homogeneity in the responses to multiple instruments.

⁷See, for example, a nonparametric model of multinomial choice Matzkin (1991) with restrictions of concavity and monotonicity to achieve identification. Recent work by Freyberger and Horowitz (2015) imposes shape restrictions such as convexity and monotonicity on the functional form of the outcome equation. For a review of shape restrictions in applied work, see Chetverikov et al. (2018).

duced to change their behavior do so in the same direction. As a result, the combinations of potential choices in *conditional response types* are consistent with the restrictions of convexity on the cost function.

Second, I show that *conditional response types* satisfy a weaker version of monotonicity than in [Imbens and Angrist \(1994\)](#). Specifically, they satisfy partial (unordered) monotonicity for multiple treatments as defined by [Mountjoy \(2022\)](#).⁸ This assumption requires that agents who react to the variation in one instrument do so in the same direction (towards or away from the same choices), conditional on the variation in other IVs. Although this assumption has been previously employed, I further exploit a well-known, but frequently overlooked, property of monotonicity. It requires that changes in an instrument from z to z' induce all agents in the same direction; meanwhile, changes from z' to z'' do not have to induce agents in the same direction as changes in z to z' . Nonetheless, it is common to assume the same patterns of behavior at z to z' and z' to z'' . I show how allowing for more flexibility in the responses across multiple values of an IV, while unwarranted for the binary choice case, can aid in identification in contexts of multiple unordered choices. That is, with *conditional response types*, I identify treatment effects for a richer set of compliers along multiple margins of choice that result from allowing agents' responses at z versus z' and z' versus z'' to differ.

Third, I discuss the role of *conditional response types* in identifying shares of compliers at different margins of choice and their treatment effects. To do so, I assume that the set of *conditional response types* changes as the conditioning instrument changes, as implied by the restrictions on convexity of costs. If sufficient next-best states disappear because, under convexity of costs, one or several options are becoming too costly as the instruments change, the researcher can identify pairwise treatment effects of one option *versus* the remaining next-best state. [Heckman et al. \(2006\)](#) and [Heckman et al. \(2008\)](#) showed that the multiple choice problem can be reduced to a standard binary choice case provided that, as one instrument increases or decreases, the probability of one option being the next-best state goes to one. They employ an identification argument requiring instruments with extreme values. In my framework, this argument arises from incorporating economic theory to the identification problem and provides testable implications to empirically assess the validity of *conditional response types*. Importantly, the more sensitive the cost function is to changes in the instruments, the less extreme values of the IVs might be required for identification. That is, identification can be achieved by conditioning on binary IVs, if under one of the two IV values the costs of choosing one option increase dramatically, or on continuous IVs with marginal changes in costs where identification depends on the slope of the cost function.

⁸Alternative monotonicity assumptions have been proposed in the literature. [Das \(2005\)](#) imposes an assumption conditional on covariates for nonlinear IVs; [Heckman et al. \(2006\)](#) impose conditional monotonicity on a second instrument, for the binary treatment case. More recently, [Mogstad et al. \(2021, 2024\)](#) and [Goff \(2024\)](#) present identification results for the binary treatment case.

Lastly, the strategy I present can be applied, but is not limited, to experimental designs with noncompliance when costs are also available. Noncompliance is frequent in applied work—*e.g.*, [Kline and Walters, 2016](#); [Dean and Jayachandran, 2020](#)—and poses the challenge of identifying counterfactual choices or fallback options that can largely vary across individuals. I show that the strategy can be implemented with a semiparametric approach as in [Mountjoy \(2022\)](#), without imposing restrictions on the relation between the instruments on the choice equation.^{9,10} This approach fits with nonlinear instrumental variables estimation for discrete endogenous variables presented in [Das \(2005\)](#) and more generally, in [Imbens and Newey \(2009\)](#); [Chen et al. \(2014\)](#); [D’Haultfœuille and Février \(2015\)](#) and [Chernozhukov et al. \(2007\)](#). In particular, I show the identification and estimation of conditional Local Average Treatment Effects (LATEs) and Marginal Treatment Effects (MTEs) for pairwise comparisons of choices.

I use the identification results to empirically assess the effect of: (i) choice of care for illness in Kenya; (ii) childcare choice and children’s development; and (iii) higher education enrollment choices on students’ labor market outcomes in Colombia. These settings are relevant for analyzing IVs in contexts of multiple treatments because options lack a clear ranking, and agents with distinct fallback options might derive different gains or losses from their choices. Across the three applications, I combine discrete and continuous IVs that jointly affect the same margins of choice. In particular, I exploit experimental (or quasi-experimental) variation jointly with the geographical distance to one of the options. Consistent with the general framework of this paper, I assume that the instruments affect choices through their effect on costs. I impose a shape restriction of convex and increasing costs on the distance to one option. My framework allows the response to the experimental variation to change with distance, which serves as the building block to define *conditional response types* in each context. Yet, the set of treatment effects that can be identified by merely applying the logic of *conditional response types* differs across these settings. In particular, achieving identification of margin-specific treatment effects depends on the sensitivity of the cost function to the conditioning instrument (namely, distance to one of the options in each application).

In the first application, I focus on the case of subsidies for malaria treatment in Kenya using data from [Cohen et al. \(2015a\)](#). Randomized assignment provided some households with subsidies to purchase antimalarial artemisinin-base combination therapies (ACTs) at drug shops. When facing an illness episode, being assigned to receive the subsidy induces households to buy a subsidized drug at the drug shop, and away from seeking

⁹Nonparametric estimation of IV strategies such as 2SLS suffers from the ill-posed inverse problem. Nonetheless, the strategy I present here deals with discrete endogenous variables and does not suffer from this problem.

¹⁰The method can also be implemented with a linear decision model by incorporating interactions between the instruments. An earlier version of this paper employs a linear estimation approach and explains its implementation in detail. See Chapter 2, Section 2.3 in this [link](#).

care at a healthcare center or doing nothing—or seeking other types of treatment. The random subsidy alone identifies the effect of drug shops *versus* the next-best. I combine the experimental variation with the geographic distance to the healthcare center and define conditional response types to identify treatment parameters at two specific margins: seeking care at drug shops *versus* healthcare centers, and seeking care at drug shops *versus* no care. Healthcare choices exhibit high sensitivity to distance from the healthcare center. The share of compliers seeking care at drug shops *versus* healthcare centers (no care) goes from 20% (0%) when distance to the nearest healthcare center is very small and drops (rises) towards 0% (5-10%) for distances around 7-9 kilometers. Next, I estimate local average treatment effects and marginal treatment effects across these two margins of choice. I find that households choosing drug shops but who would not have sought care for their health in the absence of the subsidy might be worse off in terms of illness episodes than households who would have sought care at health centers.

Second, I study the case of childcare choices in Colombia and their impact on the cognitive, socio-emotional, and nutritional development of children with a subsample from [Bernal et al. \(2019\)](#). In the context of an expansion of public care provision that took place in 2011, the researchers ran a lottery among existing small centers, providing information and encouraging winning parents to switch to large centers. Hence, parents could choose to keep their children in small centers, care for them at home, or send them to large centers. Response types defined by the experimental variation, conditional on the geographical distance between the child’s home and the nearest large center, allow identification of the effect of care at small *versus* large centers; however, additional assumptions—*e.g.*, homogeneity—are necessary to disentangle effects for a second margin of choice: small centers *versus* home care.¹¹ I show that the standard LATE is contaminated by a bias term introduced by the latter margin, as some careworkers from winning small centers transition to large centers, indirectly reducing the availability of small-center care. Estimates of LATEs and MTEs suggest that, on average, switching to large centers might not benefit children on any developmental dimension.

Lastly, I apply the logic of conditional response types to higher education enrollment choices in Colombia. In particular, I use a random subsample from [Londoño-Vélez et al. \(2025\)](#) to study the first cohort of the *Ser Pilo Paga* program (SPP) which provided scholarships for enrollment in high-quality institutions to high school graduates based on need (*i.e.*, being below a poverty score threshold) and merit (*i.e.*, being above a cutoff on the score of the high school exit exam). Following [Londoño-Vélez et al. \(2020\)](#), I divide the sample into two estimating subsamples based on whether students are eligible based on need or merit. To estimate treatment effects of higher education choices, I use

¹¹Convenience (*i.e.*, low travel time or distance) has been recognized as an important predictor of childcare choice ([Attanasio et al., 2013](#); [Bernal and Fernández, 2013](#)). Therefore, despite winning the lottery, some parents might be discouraged from choosing large centers if the distance from home is relatively large.

the remaining dimension of SPP eligibility in each subsample as a source of exogenous variation, combined with variation from the distance between the students' high school and the nearest low-quality higher education institution. I assume that, regardless of the subsample, SPP eligibility would induce students to enroll in high-quality higher education institutions instead of enrolling in low-quality higher education institutions or not enrolling.

Unlike the first two applications, students show little sensitivity to changes in distance to one of the options—namely, low-quality higher education institutions. However, among need-eligible students, nearly all of the variation comes from those who, in the absence of *Ser Pilo Paga*, would have chosen not to enroll, rather than from those who would have attended low-quality institutions. Thus, conditional response types help identify the effect of non-enrollment *versus* enrollment in a high-quality university among the need-eligible. I also present results for the enrollment margin between low- and high-quality universities among merit-eligible students, which is only separately identified under the homogeneity assumption. Overall, I find that students on the margin between choosing high-quality universities or the next-best benefit from the former in terms of a higher likelihood of working formally in 2021. Nonetheless, this effect may be weaker among individuals facing greater family or financial constraints that prevent them from choosing high-quality institutions.

2 GENERAL FRAMEWORK: MULTIPLE UNORDERED TREATMENTS

Consider a decision maker who can choose from a finite number of alternatives in choice set $\mathcal{K}$. Let k represent an alternative or option, $k \in \mathcal{K}$, and let K denote the total number of choices, with $K > 2$. Each of the options is no better than the others *a priori* (*i.e.*, they are unordered), but agents can have preferences over them. Options in $\mathcal{K}$ with higher values, in contexts of unordered choice, do not necessarily represent better alternatives. Unlike settings with only 2 options (*e.g.*, treatment or no treatment) or in which choices are ordered (*e.g.*, years of schooling), where treatment effects most commonly refer to comparisons of counterfactual outcomes across two options or states of the world, in the case of unordered choice we can define treatment effects resulting from agents choosing from different choice sets. One typical candidate is the effect of choosing option k *versus* the next-best (which refers to the option that the agent would have chosen if faced with the choice set that does not contain k). Because options are unordered, the next-best option likely differs across agents.

2.1 Set Up

Let Y denote the outcome variable of interest. The researcher is interested in the effect of choosing k on Y *versus* another option $l \in \mathcal{K}$, or *versus* a combination of other options, but

k is endogenously chosen. For instance, the treatment effect of choosing option k *versus* option l can be defined as $Y_k - Y_l$ for $k \neq l$, where Y_k (Y_l) denotes the counterfactual outcome that the agent would derive from choosing option k (l). I refer to this type of counterfactual contrast, $Y_k - Y_l$, as the treatment effect of pairwise combinations of options in $\mathcal{K}$. We can also define the treatment effect of option k *versus* the next-best as $Y_k - Y_{\mathcal{K} \setminus k}$, where $Y_{\mathcal{K} \setminus k}$ denotes the potential outcome that the agent would have obtained if option k was not available. That is, $\mathcal{K} \setminus k$ refers to the choice set containing all options in $\mathcal{K}$ except k . Other treatment effects can be defined, as in Heckman et al. (2006) and Heckman and Vytlačil (2007).

As in the standard potential outcomes framework, Y_k is only observed for those who choose k . To model these choices, assume that agents select the option in choice set $\mathcal{K}$ with the highest latent utility. That is, assume that agents' preferences can be represented by a discrete choice problem where the latent utility from choosing option $k \in \mathcal{K}$ is

$$U_k(Z) = \mu_k - V_k(Z),$$

where μ_k represents unobserved preference heterogeneity for choice k , and $V_k(Z)$ is the disutility or cost of choosing k . Z is a random vector and contains L instrumental variables, such that $Z = \{Z_1, \dots, Z_L\}$ where each Z_l takes values in $\text{supp}(Z_l)$. There are no restrictions on the support of each Z_l , which can be continuous or discrete, nor on the joint support of the instruments in Z , denoted as $\mathcal{Z}$. For simplicity, I refer to an element $z \in \mathcal{Z}$ as a combination of evaluation points: $(Z_1 = z_1, \dots, Z_L = z_L)$.

Observed choices are represented by $D_{\mathcal{K},k}$, a binary variable indicating whether option k is chosen from choice set $\mathcal{K}$,

$$D_{\mathcal{K},k} = \mathbf{1}[U_k(Z) > \max_{l \neq k} \{U_l(Z)\} \forall l \in \mathcal{K}],$$

and, hence, the observed outcome is $Y = \sum_{k \in \mathcal{K}} D_{\mathcal{K},k} Y_k$. Define $D_{\mathcal{K}}(z)$ as a categorical variable containing potential choices at value z of the instruments, *i.e.*, the choice agents would select from choice set $\mathcal{K}$ when faced with value z . That is,

$$D_{\mathcal{K}}(z) = \underset{k \in \{1, \dots, K\}}{\text{argmax}} (\mu_k - V_k(z)), \quad (1)$$

which depends on Z only through V_k , or costs. Hence, changes in the instruments shift costs but do not alter the underlying preferences of agents for choice k .¹² Costs in V_k can depend on one or many instruments. In settings of multiple unordered choices and IVs, we are interested in agents who would change their choices as the instruments vary, and the margins at which they would change their behavior. To this end, define $D_{\mathcal{K},k}(z)$

¹²The assumption of separability between μ_k and V_k in equation (1) is standard in latent index selection models (Vytlačil, 2002; Heckman et al., 2006; Lee and Salanié, 2024).

as a binary variable indicating whether option k would have been chosen from choice set $\mathcal{K}$ when the instruments are fixed at $Z = z$; *i.e.*, $D_{\mathcal{K},k}(z) = \mathbf{1}[D_{\mathcal{K}}(z) = k]$. Similarly, $D_{\mathcal{K}\setminus k,l}(z)$ denotes a binary variable indicating whether option l would have been chosen from choice set $\mathcal{K}\setminus k$ when $Z = z$.

As is standard in the IV literature, I assume that, conditional on baseline variables, the instruments are independent of potential outcomes and choices:

A1 *Conditional Independence* - As in [Imbens and Angrist \(1994\)](#), the instruments in Z satisfy $(\{Y_k\}_{k \in 1, \dots, K}, \{D_{\mathcal{K}}(z)\}_{z \in \mathcal{Z}}) \perp\!\!\!\perp Z | X$, where X denotes a vector of covariates.

Observed outcomes in the multiple unordered options framework can be expressed, as in the standard binary choice setup, by using the approach in [Heckman et al. \(2006\)](#) and [Heckman et al. \(2008\)](#). That is, Y can be partitioned into two, rather than K components:

$$Y = D_{\mathcal{K},k}Y_k + (1 - D_{\mathcal{K},k})Y_{\mathcal{K}\setminus k},$$

where $Y_{\mathcal{K}\setminus k}$ is the potential outcome that would have been observed if option k had not been available, $Y_{\mathcal{K}\setminus k} = \sum_{l \neq k, D_{\mathcal{K},k} \neq 1} \frac{D_{\mathcal{K},l}}{1 - D_{\mathcal{K},k}} Y_l$. If the researcher is interested in the effect of option k versus the next-best alternative, a useful simplification is to focus on a scalar instrument $Z_k \in Z$ that satisfies [A1](#) and the following assumptions:

A2 *Relevance* $E[D_{\mathcal{K},k}|z'_k] \neq E[D_{\mathcal{K},k}|z_k]$.

A3 *Monotonicity* ([Imbens and Angrist, 1994](#)) - Let z_k, z'_k be two values in $\text{supp}(Z_k)$, with $Z_k \in Z$. For all $k \in \{1, \dots, K\}$ either $D_{\mathcal{K},k}(z_k) \geq D_{\mathcal{K},k}(z'_k)$ for all agents, or $D_{\mathcal{K},k}(z_k) \leq D_{\mathcal{K},k}(z'_k)$ for all agents.

In what follows, suppose [A1-A3](#) hold, without loss of generality, with $D_{\mathcal{K},k}(z_k) \leq D_{\mathcal{K},k}(z'_k)$. Then, the Wald estimator (left-hand side in the equation below) that results from employing Z_k as an IV,

$$\frac{E[Y|Z_k = z'_k] - E[Y|Z_k = z_k]}{E[D_{\mathcal{K},k}|Z_k = z'_k] - E[D_{\mathcal{K},k}|Z_k = z_k]} = E[Y_k - Y_{\mathcal{K}\setminus k} | D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K},k}(z_k) = 0],$$

identifies a local average treatment effect (LATE) for individuals induced to switch towards option k due to changes in Z_k . See [Appendix A.1](#) for the derivation of this result, which follows [Imbens and Angrist \(1994\)](#), adapted to the case of multiple alternatives as in [Heckman et al. \(2006\)](#). In the context of multiple unordered alternatives, it identifies the LATE of option k versus the next-best. In general, this LATE averages over multiple underlying pairwise treatment effects.

To decompose this LATE, I follow the approach in [Heckman and Pinto \(2018\)](#), where rules of behavior from revealed preference analysis define *vectors* and *matrices* to identify

the prevalence of response-types (*e.g.*, always-takers and compliers) and their counterfactuals. Let w denote the number of values of instrument Z_k at which agents' potential behavior is evaluated. For a binary instrument, w is equal to two. Similarly, for continuous instruments, the researcher could analyze marginal changes in the instrument such that $w = 2$. For discrete instruments with more than two possible realizations, such as assignment to multiple treatment arms, $w > 2$. Below I present a general definition of a response matrix, but focus on the case of $w = 2$ hereafter for simplicity.

Definition 1 (Unconditional response matrix). *Let R_k denote a matrix of dimensions $w \times T_k$, where T_k denotes the number of response-types. Each column of R_k corresponds to a distinct response-type defined by combinations of potential choices.*

If $w = 2$, each column of R_k corresponds to a response-type defined by a pair of potential choices $(D_{\mathcal{K}}(z_k), D_{\mathcal{K}}(z'_k))'$ for distinct values $z_k, z'_k \in \text{supp}(Z_k)$.

Lemma 1. *If A3 holds then $T_k = 2K - 1$.*

The proof of this lemma follows Heckman and Pinto (2018) and is presented in Appendix A.2.

From Lemma 1, for $w = 2$, the response matrix resulting from the variation in Z_k is given by:

$$R_k = \begin{bmatrix} 1 & \cdots & k & \cdots & K & 1 & \cdots & K \\ 1 & \cdots & k & \cdots & K & k & \cdots & k \end{bmatrix},$$

which contains all possible combinations of potential choices induced by Z_k , where the first (second) row denotes potential choices at $Z_k = z_k$ ($Z_k = z'_k$). The first K columns represent always-takers (*i.e.*, agents choosing the same option regardless of the value of the instrument) while the remaining $K - 1$ columns are compliers (*i.e.*, agents switching into option k from each of the $K - 1$ other alternatives in $\mathcal{K} \setminus k$).

Following the choice structure in matrix R_k , the LATE of option k versus the next-best can be decomposed into pairwise treatment effects,

$$\begin{aligned} \frac{E[Y|z'_k] - E[Y|z_k]}{E[D_{\mathcal{K},k}|z'_k] - E[D_{\mathcal{K},k}|z_k]} &= \sum_{l \in \mathcal{K} \setminus k} \frac{\Pr(D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K} \setminus k, l}(z_k) = 1)}{\Pr(D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K},k}(z_k) = 0)} \\ &\quad \times E[Y_k - Y_l | D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K} \setminus k, l}(z_k) = 1]. \end{aligned} \quad (2)$$

Equation (2) follows the logic of the classic LATE decomposition in Imbens and Angrist (1994), which is extended here to multivalued treatments. See Appendix A.3 for the derivation adapted to this setting, which follows Heckman et al. (2006). The right-hand side of equation (2) is a weighted average of pairwise treatment effects, with weights given by the prevalence of compliers across each margin. The first term inside the summation denotes the share of compliers switching into k out of state l , among all compliers switching into option k , as Z_k goes from z_k to z'_k . The second term corresponds to the (pairwise)

average treatment effect of option k *versus* l for compliers switching along that margin, as Z_k goes from z_k to z'_k . That is, the proportion of agents whose behavior matches a given column in R_k , who exhibit that response pattern, determines the weight assigned to the corresponding pairwise treatment effect in the LATE decomposition of equation (2).

Plus, imposing additional regularity conditions as in Heckman et al. (2006) for continuous instruments and Brinch et al. (2017) for binary instruments, the marginal treatment effect (MTE) of option k *versus* the next-best can be identified:

$$\begin{aligned} \text{MTE}_{k, \mathcal{K} \setminus k} &= E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z)] \\ &= \sum_{l \in \mathcal{K} \setminus k} \{ \Pr(D_{\mathcal{K} \setminus k, l} = 1 | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)) \\ &\quad \times E[Y_k - Y_l | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k), D_{\mathcal{K} \setminus k, l} = 1] \}, \end{aligned}$$

where the second line further decomposes the next-best-MTE into pairwise marginal treatment effects. See Appendix A.4 for an overview of the identification arguments in Brinch et al. (2017), applied to the MTE of k *versus* the next-best. The appendix also derives the corresponding decomposition following the arguments in Heckman et al. (2006).

In the case of the LATE as well as the MTE, if sufficiently many next-best states are unlikely to be chosen (for instance, all except option l), the pairwise effect of k *versus* l would be identified.

2.2 Identification of Pairwise Treatment Effects Using Conditional Responses

In what follows, I present a generalized set of restrictions stemming from economic theory under which some next-best states would disappear. This elimination cannot be achieved through variation in a single instrument alone. Therefore, I define a set of restrictions under which the researcher can select conditioning instruments—that is, instruments held fixed to isolate variation along a specific margin of interest.

In particular, eliminating next-best states requires understanding how agents switch into, and away, from choices in $\mathcal{K}$ as multiple instruments vary. To do so, and following the logic of matrix R_k , the researcher can define a set of rules of behavior. These rules stem directly from the properties of costs in $V_k(Z)$ and how they affect the relative mean utility of the different options. For instance, if increasing the distance to option 1 raises its cost relative to other options, one can assume that the likelihood of option 1 appearing in the set of next-best alternatives decreases with distance.

I focus on settings where, for example, the researcher has access to several costs for an option, such as tuition, fees, travel time or distance. In addition, experimental variation can also be available along with information on the costs of different options.

Heckman et al. (2006); Mogstad et al. (2021, 2024) and Mountjoy (2022) show that if the researcher simultaneously increases one instrument (e.g. tuition) while decreasing another (e.g. travel time), some agents can move towards, and some away from, the same option. This further motivates exploiting the variation in one instrument, while keeping the other instrument(s) fixed.

2.2.1 Cost Restrictions

To determine how the composition of the agent’s choice set varies across different values of the instruments, I propose imposing restrictions on the cost or disutility of choosing an option. Rooted in this idea is the logic of instruments as cost-shifters, such that as the cost of option l increases (as a result of, for instance, an increase in the value of an instrument), the likelihood of option l belonging to set $\mathcal{K} \setminus k$ decreases. How rapidly the increase in the cost of l rules out that option as belonging to the next-best set largely depends on the shape of the cost function, $V_l(Z)$. By imposing restrictions on the shape of the cost function, the researcher can rule out some states as next-best alternatives and thus reduce the combinations of potential choices to determine which margins of choice would prevail across the distribution of Z . I impose the following restrictions:

- R1 Let $V_l(Z)$ represent the costs of alternative $l \in \mathcal{K} \setminus k$, which depend on instruments in Z . Let $Z_l \in Z$ and define Z_{l-} as instruments in Z other than Z_l . Z_l is a conditioning instrument if there is at least one option $l \in \mathcal{K} \setminus k$ such that:
- R1.1 (*Convex costs*) If Z_l is continuous and V_l is twice differentiable, then $\frac{\partial^2 V_l(Z_l, Z_{l-})}{\partial Z_l^2} \geq 0$. If Z_l is discrete and ordered, then $V_l(Z_l, Z_{l-})$ is convex if the difference in costs between adjacent values of Z_l weakly increases (or decreases) along its support. If Z_l is binary, then $V_l(Z_l, Z_{l-})$ is non-decreasing (or non-increasing).
- R1.2 (*Dominance*) There exist $z_l, z'_l \in \text{supp}(Z_l)$ such that

$$V_l(z_l) = \min_{m \in \mathcal{K} \setminus k} \{V_m(z_l)\}, \quad \text{and} \quad V_l(z'_l) = \max_{m \in \mathcal{K} \setminus k} \{V_m(z'_l)\}.$$

The restriction **R1** on the cost function is defined for the set $\mathcal{K} \setminus k$ —that is, the choice set the agent would have to face if option k was not available—given that the goal is to define conditions under which some options in set $\mathcal{K} \setminus k$ would not be chosen (would not be part of the next-best states). **R1.1** introduces the definition of a convex cost function used hereafter, which need be only weakly element-wise convex on Z_l rather than on all elements in Z . Assuming (weak) convexity of costs is frequent in economic theory and can be extended, as is the case of this paper, to applications that employ exogenous variation in cost-shifters as instrumental variables. Moreover, the condition of weak convexity is necessary for a concave objective function, and therefore for a unique solution to the utility maximization problem. **R1.2** requires that for some evaluation regions of Z_l option

l is chosen (given that it had the minimum cost) whereas it is the least preferred option for some other evaluation regions of Z_l (and, therefore not in $\mathcal{K} \setminus k$ for that region of Z_l). Overall, **R1** implies that the choice set $\mathcal{K} \setminus k$ varies across Z_l . Namely, the composition of the next-best states in $\mathcal{K} \setminus k$ differs across evaluation points of Z_l . Although much of the discussion refers to variation in continuous instruments, these restrictions also apply when instruments are binary or discrete. In such cases, the cost function $V_l(Z)$ changes discretely across support points of Z_l . If these cost shifts are sufficiently large, they can alter the relative ranking of alternatives and, hence, the composition of $\mathcal{K} \setminus k$.

Convexity of the cost function serves two main purposes. To see its role, suppose the cost function is not convex but fluctuates over different values of Z_l , increasing and then decreasing for some values and vice versa. The argument that increasing costs would rule out some options as next-best states would then depend on the evaluation points in Z_l , and the composition of $\mathcal{K} \setminus k$ could vary unpredictably. That is, some next-best states would enter and exit set $\mathcal{K} \setminus k$ unpredictably for the researcher. In turn, convexity imposes restrictions such that, once an option becomes too costly to be in set $\mathcal{K} \setminus k$, its likelihood of being the next-best state goes to zero thereafter. These ideas can be formalized as follows:

Proposition 1. *Suppose restriction **R1** holds. Without loss of generality, in the case of increasing costs, as $Z_l \rightarrow z_l$, the next-best set satisfies $\mathcal{K} \setminus k = \{l\}$ with probability approaching one. As $Z_l \rightarrow z'_l$, $l \notin \mathcal{K} \setminus k$ with probability approaching one.*

See Appendix **A.6** for the proof.

Second, convexity and the conditions on **R1** impose restrictions such that agents who would be induced to change their choices because of a change in Z_k would do so toward or away from the same options, conditional on Z_l . The latter states that the assumption of convex costs translates into combinations of counterfactual choices satisfying a weaker version of **A3**:

A3a Conditional (Partial) Monotonicity - Let z_k, z'_k be two values in $\text{supp}(Z_k)$, $z_l \in \text{supp}(Z_l)$, with $Z_k, Z_l \in Z$. For all $k \in \{1, \dots, K\}$ either $D_{\mathcal{K},k}(z_k, z_l) \geq D_{\mathcal{K},k}(z'_k, z_l)$ for all agents, or $D_{\mathcal{K},k}(z_k, z_l) \leq D_{\mathcal{K},k}(z'_k, z_l)$ for all agents.¹³

A3a relaxes the monotonicity assumption in **A3** (Imbens and Angrist, 1994) such that changes in an instrument are required to induce agents towards, or against, the same choices conditional on a second instrument. For binary treatment, Mogstad et al. (2021) employs this assumption for identification. Similarly for binary treatment, Heckman et al. (2006) show that conditioning on other instruments can produce one-way flows in

¹³Monotonicity in Imbens and Angrist (1994) is a stronger assumption, since it has to hold regardless of changes in instruments other than Z_k .

the variation of a selected IV. Mountjoy (2022) extends partial monotonicity to the case of multiple unordered options, and secures identification with an additional assumption of comparable compliers. The latter requires that compliers induced along the same margin of choice by two different instruments would be comparable in terms of average potential outcomes.¹⁴ Instead, in my framework, I secure identification by imposing restrictions on the cost function such that changes in the IVs would satisfy *A3a*.

Proposition 2. *If the restrictions in *R1* hold, then Z_k satisfies *A3a*.*

See Appendix *A.7* for the proof. This proof underscores the central role of *R1* in settings where multiple instruments affect the same choices. For example, consider options k, l, m , and suppose that as Z_k changes, option k becomes more attractive while option l becomes less attractive. Such variation could induce two-way flows into and out of option m , violating monotonicity (*A3*). However, if Proposition 2 holds, partial monotonicity (*A3a*) can still be satisfied in some regions of $\text{supp}(Z_l)$. More broadly, restriction *R1* admits a richer set of cost functions where Z_k can affect multiple choices.

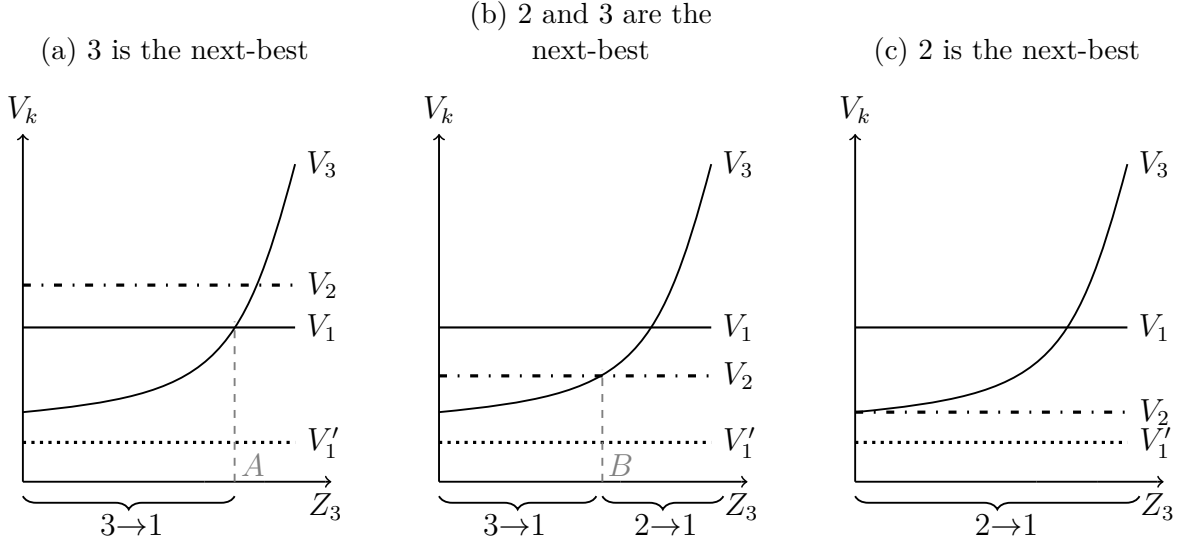
Example: To illustrate how changes in costs affect different margins of choice and, hence, the composition of the next-best set, consider the case of three choices, $k = \{1, 2, 3\}$, and two IVs, $Z = \{Z_1, Z_3\}$. Suppose that Z_1 is a discrete instrument which denotes random assignment with noncompliance to a treatment 1, and Z_3 refers to a measure of cost (e.g., price, travel time, travel distance) to a treatment/option 3. Let $V_k(Z)$ represent the cost of choosing option k which is a function of the instruments. Figure 1 depicts sets of costs for option 1, 2, and 3 under different orderings. V_3 is the cost of choosing option 3 and its convex in Z_3 , while options 1 and 2 are unaffected by this instrument. V_1 (V'_1) refers to the cost of option 1 evaluated at $Z_1 = 0$ ($Z_1 = 1$) with $V_1 > V'_1$ such that changing Z_1 reduces the cost of choosing 1, while not affecting options 2 and 3.

Each panel in Figure 1 displays a decrease in the cost of option 1 ($V_1 > V'_1$, as $Z_1 = 0 \rightarrow 1$), such that agents with the cost structures depicted would switch out of choosing options 2 or 3 and into choosing option 1. Moreover, as Z_3 increases, so does the cost of choosing 3, such that other options (2 in this example) are more likely to be the next-best alternative. Panel (a) of Figure 1 illustrates the case where an agent who would change her choice towards option 1 would do so by substituting away from option 3 (up to point A, thereafter the agent would be an option 1—always-taker). If all agents changing their choices because of changes in Z_1 have this type of cost structure, the next-best state would always be option 3. In contrast, Panel (b) shows the case where at relatively low realizations of Z_3 (below point B) an agent would substitute option 3

¹⁴Das (2005) employs a conditional monotonicity assumption to the case of nonlinear IV. The author shows that, conditioning on covariates, nonparametric estimation can recover a conditional LATE parameter.

for 1 but as Z_3 increases (above point B) the agent would substitute option 2 for 1. Last, Panel (c) presents a case where the cost of option 3 is not only convex but always above the cost of option 2 and hence an agent with these costs would switch out of 2 and into 1 at all values of Z_3 .

Figure 1: Example of relative costs that induce agents to switch along the 2 – 1 and 3 – 1 margins



Notes: The lines depict sets of costs for options $k \in \{1, 2, 3\}$ under different orderings. V_1 (V'_1) denotes the cost of option 1 evaluated at $Z_1 = 0$ ($Z_1 = 1$); V_2 denotes the cost of option 2; V_3 denotes the cost of option 3 and it is convex in Z_3 . In Panel (a), when $Z_1 = 0$ option 3 (1) is the cheapest option below (above) point A; when $Z_1 = 1$, option 1 becomes the cheapest option (V'_1 is below V_2 and V_3). Hence, as Z_1 changes, agents would switch from choosing 3 and into choosing 1 (depicted as $3 \rightarrow 1$ below the horizontal axis). A similar pattern emerges in Panel (b), such that below point B agents would switch from 3 towards 1, but above point B (since option 2 is the cheapest when $Z_1 = 0$) agents would switch from 2 and towards 1 (depicted as $2 \rightarrow 1$ below the horizontal axis). In Panel (c), when Z_1 changes (shifting costs from V_1 to V'_1) agents would switch from 2 to 1.

The costs depicted in Panel (c) violate restriction **R1** (**R1.2**, in specific), and in this case convexity plays no role on reducing the number of next-best states. Hence, assuming that V_3 is convex but always above the costs of other options has the same behavior implications as assuming that V_3 is constant over Z_3 . Whether the researcher should expect the patterns displayed in either panel depends on the sensitivity of costs to the instrument; if V_3 has a small slope, points A and B would be farther away to the right on the horizontal axis, making it less likely to eliminate option 3 from the next-best set. In contrast, in combination with **R1.2**, the higher the slope of the cost function of option 3, the closer points A and B would be to the origin, and the more likely is option 2 to be the next-best state.¹⁵ In the next section, I formalize the implications of the restrictions on the cost function for the identification of treatment effects.

¹⁵ **R1** requires at least one instrument affecting options other than k . For instance, in the example Z_3 affects option 3. Suppose that instead of Z_3 the researcher has available Z_2 which affects the cost of

2.2.2 Conditional Response Types

The restrictions on costs translate into admissible potential choices that agents could make as a response to changes in an instrument Z_k , or rules of behavior, conditional on an instrument Z_l . Here, I extend the response matrix in Heckman and Pinto (2018), R_k , defined previously, to the case of a *conditional response matrix* defined by an instrument Z_k , conditional on Z_l . The combinations of potential choices in this *conditional response matrix* change along the support of a second instrument Z_l .

Definition 2 (Conditional response matrix). $R_k(z_l)$ is a matrix of dimensions $w \times T_k(z_l)$. Where $T_k(z_l)$ denotes the number of conditional response types induced by variation in Z_k when Z_l is fixed at z_l .

If $w = 2$, each column of $R_k(z_l)$ corresponds to a conditional response-type defined by a pair of potential choices $(D(z_k, z_l), D(z'_k, z_l))'$ for distinct values $z_k, z'_k \in \text{supp}(Z_k)$ at a given $Z_l = z_l$.

Lemma 2. If [R1](#) holds, then $T_k(z_l) \leq 2K - 1$.

See Appendix [A.8](#) for the proof.

To illustrate, take the structure of choices and instruments in the example depicted in Figure [1](#). The unconditional response matrix,

$$R_1 = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 1 & 2 & 3 & 1 & 1 \end{bmatrix},$$

shows admissible (i.e., satisfying [A3](#)) combinations of potential choices for $Z_1 = 0$ (in the first row) and $Z_1 = 1$ (in the second row). Each column denotes a response-type. Columns (1), (2), and (3) are option 1, 2 and 3—always-takers, respectively. Column (4) denotes 3 – 1 compliers and column (5) denotes 2 – 1 compliers. As shown in section [2.1](#), the variation alone in Z_1 identifies the LATE of option 1 *versus* the next-best.

Based on the cost structure depicted in Panel (b) of Figure [1](#), where as Z_3 increases (decreases) it is less (more) likely that option 3 is the next best, the researcher can define two matrices containing conditional rules of behavior. Let $R_1(Z_3 \rightarrow z_3^{\min})$ and $R_1(Z_3 \rightarrow z_3^{\max})$ denote the rules of behavior as Z_3 decreases and increases, respectively. Hence,

$$R_1(Z_3 \rightarrow z_3^{\min}) = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 1 \end{bmatrix},$$

choosing option 1 jointly with Z_1 . Assume that V_1 is convex in Z_2 , and Z_1 has the same role as in the example above such that both V_1 and V'_1 are convex curves and $V_1 > V'_1$ always. As Z_2 increases option 1 would become more costly but the change in Z_1 also makes agents more likely to choose option 1 at all values of Z_2 from any next-best state. Hence, combining instruments that affect the same option would not necessarily rule out next-best states.

and,

$$R_1(Z_3 \rightarrow z_3^{\max}) = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 1 & 2 & 3 & 1 \end{bmatrix},$$

which capture the behavioral pattern such that $3 - 1$ compliers are more prevalent as Z_3 is approaching its minimum value, while $2 - 1$ compliers are more prevalent at higher values of Z_3 . In addition, if agents' costs follow scenarios as those depicted in Panel (c) of Figure 1 or where point B in Panel (b) is very close to $z_3^{\min}$, $R_1(Z_3 \rightarrow z_3^{\min})$ could contain responses along $2 - 1$. Whether the next-best state evolves sharply, as in Panel (b), is an empirical question. The following statements describe how this can be tested using observed patterns of choice.¹⁶

First, the reduction in the next-best set and in the rank of the unconditional response matrices $R_k(z_l)$ imply that, at certain values of Z_l , the probability of observing specific response types converges to zero or one.

Lemma 3. *Under the conditions of Proposition 1 and $w = 2$, as $Z_l \rightarrow z_l$,*

$$Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k,m}(z_k, z_l) = 1) \rightarrow 0, \forall m \in \mathcal{K} \setminus \{k, l\},$$

and $T_k(Z_l \rightarrow z_l) = 1 + K$ and the number of compliers decreases to 1.

As $Z_l \rightarrow z'_l$,

$$Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus k,l}(z_k, z'_l) = 1) \rightarrow 0,$$

and $T_k(Z_l \rightarrow z'_l) = (K - 1) \times 2$ and the number of compliers decreases to $K - 2$.

See Appendix A.9 for the proof. The following remark formalizes an empirical implication that can be tested nonparametrically.

Remark 2.1. *By Lemma 3, it follows that:*

$$E[D_{\mathcal{K},m}|z_k, z_l] - E[D_{\mathcal{K},m}|z'_k, z_l] = 0, \quad \forall m \in \mathcal{K} \setminus \{k, l\},$$

and,

$$E[D_{\mathcal{K},l}|z_k, z'_l] - E[D_{\mathcal{K},l}|z'_k, z'_l] = 0.$$

See Appendix A.10 for the proof.

Remark 2.2. *If $K = 3$ and $w = 2$, then under the conditions of Proposition 1, the two limits $Z_l \rightarrow z_l$ and $Z_l \rightarrow z'_l$ identify two distinct and non-overlapping complier groups.*

¹⁶One could define a threshold where the next-best composition changes (e.g., point B in the example) such that the conditional matrix changes. However, different agents can have distinct cost structures which can complicate the definition of a unique threshold. Instead, I propose following the theoretical implications of the assumptions on costs and evaluating effects as the conditioning instrument increases or decreases.

The latter follows directly from evaluating the results of Lemma 3 for $K = 3$. See Appendix C for the extension to the case of $w > 2$.

Next, I formalize the role of $R_k(z_l)$ in the identification of treatment effects.

2.2.3 Identification of Treatment Effects

I impose an additional assumption:

A2a *Extended Relevance* For $k \in \mathcal{K}$ and at least one option $l \in \mathcal{K} \setminus k$,

$$E[D_{\mathcal{K},k}|z'_k, z_l] \neq E[D_{\mathcal{K},k}|z_k, z_l] \quad \text{and} \quad E[D_{\mathcal{K},l}|z'_k, z_l] \neq E[D_{\mathcal{K},l}|z_k, z_l].$$

This condition extends the standard relevance requirement to multivalued treatments by imposing that Z_k induces variation in the relative probabilities of at least two alternatives, conditional on $Z_l = z_l$. The following proposition formalizes the identification of local average treatment effects:

Proposition 3. *Suppose assumptions A1-A2a and restriction R1 hold. Then, as $Z_l \rightarrow z_l$, the Wald estimator in equation (2) identifies the pairwise treatment effect of k versus l ,*

$$\frac{E[Y|z'_k, z_l] - E[Y|z_k, z_l]}{E[D_{\mathcal{K},k}|z'_k, z_l] - E[D_{\mathcal{K},k}|z_k, z_l]} = E[Y_k - Y_l | D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k, l}(z_k, z_l) = 1].$$

As $Z_l \rightarrow z'_l$,

$$\begin{aligned} \frac{E[Y|z'_k, z'_l] - E[Y|z_k, z'_l]}{E[D_{\mathcal{K},k}|z'_k, z'_l] - E[D_{\mathcal{K},k}|z_k, z'_l]} = \\ \sum_{m \in \mathcal{K} \setminus \{k, l\}} \frac{\Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus \{k, l\}, m}(z_k, z'_l) = 1)}{\Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K},k}(z_k, z'_l) = 0)} \\ \times E[Y_k - Y_m | D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus \{k, l\}, m}(z_k, z'_l) = 1]. \end{aligned}$$

See Appendix A.11 for the proof. The following proposition formalizes the identification of marginal treatment effects:

Proposition 4. *Suppose assumptions A1-A2a and restriction R1 hold. As $Z_l \rightarrow z_l$,*

$$\begin{aligned} \lim_{Z_l \rightarrow z_l} E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z, U_k(z) = U_{\mathcal{K} \setminus k}(z)] = \\ E[Y_k - Y_l | Z = z, U_k(z) = U_{\mathcal{K} \setminus k}(z), D_{\mathcal{K} \setminus k, l} = 1]. \end{aligned}$$

As $Z_l \rightarrow z'_l$,

$$\begin{aligned} \lim_{Z_l \rightarrow z'_l} E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z, U_k(z) = U_{\mathcal{K} \setminus k}(z)] \\ = \sum_{m \in \mathcal{K} \setminus \{k, l\}} \{Pr(D_{\mathcal{K} \setminus \{k, l\}, m} = 1 | Z = z, U_k(z) = U_{\mathcal{K} \setminus \{k, l\}}(z)) \\ \times E[Y_k - Y_m | Z = z, U_k(z) = U_{\mathcal{K} \setminus \{k, l\}}(z), D_{\mathcal{K} \setminus \{k, l\}, m} = 1]\}. \end{aligned}$$

See Appendix A.12 for the proof.

In general, with $K > 3$ the reduction in next-best states described in Propositions 3 and 4 is not sufficient to identify treatment effects of pairwise combinations of the options; *i.e.*, there would be more than one complier group remaining even after conditioning on $Z_l \rightarrow z'_l$.¹⁷ In contrast, when $K = 3$, sharp identification is achievable as follows:

Remark 4.1. *If $K = 3$, under the conditions of Proposition 3, the Wald estimator as $Z_l \rightarrow z_l$ and $Z_l \rightarrow z'_l$ identifies*

$$E[Y_k - Y_l | D_{\mathcal{K}, k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k, l}(z_k, z_l) = 1],$$

and,

$$E[Y_k - Y_m | D_{\mathcal{K}, k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus \{k, l\}, m}(z_k, z'_l) = 1],$$

respectively.

A similar argument follows for $MTE_{k, l}$ and $MTE_{k, m}$ following proposition 4.

Appendix B applies the sharp identification results to the setting in the Example in Section 2.2.

Alternatively, consider the case where Z_M denotes a vector of instruments for M options in $\mathcal{K} \setminus k, l$ (*i.e.*, the choice set containing all options in $\mathcal{K}$ other than k and l) and assume that all elements in Z_M satisfy R1. Then, only one complier remains for response matrix $R_k(z_m^*)$, where z_m^* denotes an evaluation point such that each instrument in Z_M is evaluated at its maximum value, meaning that all other M options are too costly to be the next-best state, and the pairwise treatment effect of option k versus option l can be identified. A similar argument could be made if M and L correspond to sets of options each with mutually exclusive characteristics (*e.g.*, M (L) denotes publicly (privately) provided options) and in that case the researcher could identify effects of option k versus

¹⁷In scenarios where conditional response types only rule out one complier group, the researcher could impose an assumption of homogeneity on counterfactuals as in Kline and Walters (2016); Hull (2018); Lee and Salanié (2024). Formally,

$$E[Y_k | D_{\mathcal{K}, k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus \{k, l\}, m}(z_k, z'_l) = 1] = E[Y_k | D_{\mathcal{K}, k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus \{k, l\}, j}(z_k, z'_l) = 1],$$

for all $j \in \mathcal{K} \setminus \{k, l\}$. In general, this assumption states that the potential outcome from choosing k would have been the same, on average, regardless of the next-best option.

the next-best public (or private) option, depending on the incentives of the instruments. Last, in the application in Section 3.3 I show how conditioning on more than one IV can also reduce the number of next-best states.¹⁸

In the next section, I present three empirical applications that illustrate the role of conditional response types across distinct settings and detail the empirical strategy.

3 EMPIRICAL APPLICATIONS

In this section, I present three empirical applications. First, I focus on subsidies for malaria treatment in Kenya, where households could choose to seek care at drug shops, at healthcare centers, or not at all. I show that, in this first case, conditional response types allow identifying two treatment parameters (*i.e.*, the effect of seeking care at drug shops *versus* health centers, and of seeking care at drug shops *versus* no care) in addition to the treatment effect of care at drug shops *versus* the next-best. In the second application, I study childcare choices in Colombia and their impact on the cognitive, socio-emotional, and nutritional development of children. Parents could choose to care for their children at home, or send them to either small or large centers. Conditional response types allow identifying the effect of care at small *versus* large centers; however, additional assumptions (*e.g.*, homogeneity) are necessary to disentangle effects for a second margin of choice: small centers *versus* home care.

Finally, in the third application, I focus on merit- and need-based scholarships and higher education outcomes in Colombia. In particular, I study the first cohort of the *Ser Pilo Paga* program which provided scholarships to high school graduates of 2014 based on need (*i.e.*, being below a poverty score threshold) and merit (*i.e.*, being above a cutoff on the score of the high school exit exam). Students could choose either not to enroll, or to enroll in high-quality or low-quality higher education institutions. Conditioning on two instrumental variables, rather than one as in the previous cases, allows identifying the effect of not enrolling *versus* enrolling in a high-quality university. I also show results for the margin of low- *versus* high-quality universities enrollment which is only identified after imposing the homogeneity assumption.

3.1 Subsidies for malaria treatment in Kenya

For my first application, I employ data in Cohen et al. (2015b) from Cohen et al. (2015a) who run a field experiment to study the targeting *versus* access trade-off of subsidizing antimalarial drugs in Kenya. Households were randomly assigned to receive subsidies to purchase an antimalarial (artemisinin combination therapies - ACTs) at drug shops. For these households, I study their health care choices for their first illness episode after

¹⁸Appendix C embeds the logic of conditional response vectors within the econometric model of Heckman and Pinto (2018), which extends to the case of $w > 2$, for discrete IVs taking on more than two values.

subsidy provision, and the effect of this choice on total illness episodes at endline.¹⁹ Households can choose between three alternatives: buying a subsidized drug at the drug shop (s), going to a formal health facility (h), or doing nothing or seeking other type of treatment (n).

There are three reasons why this setting is a good candidate for employing conditional response matrices to identify treatment effects of health care choices. First, we cannot assume that all households have the same ranking of preferences over these options. While h has medical professionals, it also has long waiting lines, stockouts of ACTs, as well as fees for diagnosis and treatment. In turn, s lacks medical staff and diagnosis tools but is, on average, more easily accessible to households (Cohen et al., 2015a). Second, two measures of arguably exogenous variation are available: the random subsidy and the distance to the nearest healthcare center. Third, subsidy provision was solely for purchases of ACTs at s , which can divert households from either seeking treatment at h or n and into choosing s . Arguably, households switching on the “intensive” margin might not benefit from choosing s in terms of their overall health, while those who are on the “extensive” margin (entering the health care system via s) might benefit most from seeking appropriate care for their illness. Therefore, disentangling treatment effects on each margin is relevant for policy.

3.1.1 Data

I use data from the replication package of Cohen et al. (2015a), which contains information on baseline characteristics and health related behaviors of households as well as subsidy assignment and endline variables (*e.g.*, records on illness episodes). It also contains information on health care choices (s , h , or n) for all illness episodes. To define the choice variable, I assign households to s , h , or n based on their health care choice for the first illness episode after subsidy provision. The estimating sample consists of 2,471 households; 5% belong to the control (no subsidy) group and 95% received an ACT subsidy. In terms of health care choice, 61.4% of households choose s , followed by 20% in h and 18.6% in n (see Table D1).

Households’ health care choices are correlated with their characteristics and health behaviors at baseline. Table D1 shows that, on average, households who chose h have younger and more educated household heads than those who chose s and n . In terms of health behaviors at baseline, households who chose n are, on average, less likely to have had malaria episodes, own fewer bednets, and visit s or h for prior illnesses less often than households choosing s and h . At endline, households who choose s report on average a higher number of illness episodes than those who choose h or n . Last, Table D2 shows

¹⁹To apply the logic of conditional response types and estimate treatment effects, I depart from the original question in Cohen et al. (2015a). While other outcomes, such as household wealth or labor market participation, might be potentially more interesting, I focus on the set of endline variables collected in the original study which pertain mainly to illness episodes and treatment seeking behavior.

that there is a higher share of subsidy recipients among households who choose s than among those who choose h or n . It also shows that households who choose h are closer to these options (row “Distance to h , in km”), on average, than households who choose n .

3.1.2 Conditional Response Types for Health Care Choices

Let k denote household choice of health care, where $k \in \{s, h, n\}$. This choice, k , affects households’ illness episodes at endline, denoted by Y . Furthermore, let $Z \in \{Z_1, Z_2\}$ contain instrumental variables where Z_1 is a binary variable and Z_2 is a continuous variable. Specifically, Z_1 is the outcome of the random subsidy assignment and Z_2 is the distance to the nearest h . Denote z_i as a realization of Z ; for instance, z_1 could be either zero or one.

Potential outcomes and choices are denoted by Y_k and $D(z_1, z_2)$, respectively. The former represents illness episodes that household i would experience from choosing option k , the latter denotes what household i would choose if facing instrument values (z_1, z_2) . I assume that [A1](#) and [A3a](#) hold.²⁰ To define potential choices at different values of the IVs, and illustrate the role of [R1](#), let $V_k(z_1, z_2)$ represent the disutility of choosing k at values $Z_1 = z_1$ and $Z_2 = z_2$ of the instruments. Assume that $V_s(0, z_2) \geq V_s(1, z_2)$, such that obtaining a subsidy to purchase ACTs at s lowers the cost of choosing s while the costs of h and n remain unaffected (*i.e.*, $V_h(0, z_2) = V_h(1, z_2)$ and $V_n(0, z_2) = V_n(1, z_2)$). Last, increases in Z_2 increase $V_{ih}(Z_1, z_2)$ but do not affect directly the costs of s and n . The unconditional response-matrix, containing combinations of potential choices at $Z_1 = 0$ (first row) and $Z_1 = 1$ (second row), is given by:

$$R_1 = \begin{bmatrix} s & h & n & h & n \\ s & h & n & s & s \end{bmatrix}.$$

Following the scenarios depicted in [Figure 1](#), replace option 3 with h , option 2 with n , and option 1 with s . I use these cost structures to define potential combinations of health care choices as the subsidy assignment changes which, following the assumptions on costs, only directly affects V_s (shown in the Figure as the dotted line). Below A and B, households would choose h in the absence of the subsidy but would choose s if they obtain the subsidy (that is, they would switch towards the cheapest option). In contrast, above B, households would switch from n and towards s as the subsidy assignment changes. There are other cost structures that satisfy the assumptions; however, the main implications on households’ behavior would remain the same: first, once the cost of choosing h is high,

²⁰Distance to the health center might be correlated with other factors that determine households’ overall health. While the variation to estimate treatment effects stems from the random subsidy, conditional on distance, violations of the exclusion restriction for Z_2 threaten identification if one group disappearing as the IV changes does so due to changes in gains from treatment that might correlate with the shifts in the distance. I recognize this limitation of the application, but assume in what follows that Z_2 is conditionally exogenous for illustration purposes.

such that the utility from choosing h becomes smaller, households are less likely to switch along the $h - s$ margin; second, if the cost of choosing h is low (near zero), households experiencing large decreases in the cost of choosing s are more likely to switch along the $h - s$ margin. These implications motivate the conditional response matrices in Table 1 which contain combinations of potential choices when Z_2 approaches zero (right matrix) and when Z_2 approaches its maximum value (left matrix). In both cases, the first three columns are always-takers, while the fourth column denotes switchers along the $h - s$ ($n - s$) margin. At any other point in the distribution of Z_2 we would observe both $h - s$ and $n - s$ switchers with some probability.

Table 1: Z_1 -responses at different evaluation points of Z_2

$$R_1(Z_2 \rightarrow 0) = \begin{bmatrix} s & h & n & h \\ s & h & n & s \end{bmatrix} \quad R_1(Z_2 \rightarrow z_2^{\max}) = \begin{bmatrix} s & h & n & n \\ s & h & n & s \end{bmatrix}$$

In what follows, I test empirically whether the behavior implied by Figure 1 and formalized in Table 1 holds in this setting. I also describe the empirical strategy for estimating local average treatment effects and MTEs.

3.1.3 Exploratory Analysis and Implementation

First stage: I estimate propensity scores locally around evaluation points of Z_2 with the following semiparametric regression:

$$\{\hat{\beta}_1^k(z_2), \hat{\beta}_2^k(z_2), \hat{\beta}_X^k(z_2)\} = \operatorname{argmin} \sum_{i=1}^N K\left(\frac{Z_{2i} - z_2}{h}\right) (D_{ki} - \beta_1^k Z_{1i} - \beta_2^k Z_{2i} - \beta_X^k \mathbf{X}), \quad (3)$$

where $\mathbf{X}$ denotes baseline variables such as number of young children in the household, other household members, and distance to the nearest drug shop. I estimate the equation for each choice separately with an Epanechnikov kernel with bandwidth 4.5km, and compute mean contrasts at each evaluation point, z_2 , of $Z_1 = 1$ *versus* $Z_1 = 0$ to obtain estimates of $\hat{\beta}_1^k(z_2)$. I employ the results in Appendix A.5 to estimate the probabilities of always-takers and compliers in Table 1. Moreover, I follow remark 2.1 to test the following hypothesis, $H_0 : \beta_1^n(0) = 0$ and $H_0 : \beta_1^h(z_2^{\max}) = 0$, where the former (latter) tests whether $n - s$ ($h - s$) switchers are not prevalent as $Z_2 \rightarrow 0$ ($Z_2 \rightarrow z_2^{\max}$).

First stage estimates: Panel A of Figure 2 presents results for $\hat{\beta}_1^k(z_2)$ at different evaluation points, z_2 for $k \in \{s, h, n\}$. In line with the implications for health care choice from the cost structure in the previous section, the effect of the subsidy on the probability of choosing h is negative (as households are less likely to choose it if they receive the subsidy) but decreases in magnitude as Z_2 increases. In particular, for $Z_2 > 7\text{km}$ the effect approaches zero and it is not statistically significant. On the other hand, the effect of the subsidy on the probability of choosing n is small and negative at smaller Z_2 but

increases in magnitude as Z_2 increases. Hence, these results are in line with the definition of response types in Table 1.

Treatment Effects: For each vector in $Y_o \in \{Y, YD_H, YD_S, YD_L\}$, I estimate the following semiparametric regression:

$$\{\hat{\beta}_1^{Y_o}(z_2), \hat{\beta}_2^{Y_o}(z_2), \hat{\beta}_X^{Y_o}(z_2)\} = \operatorname{argmin} \sum_{i=1}^N K\left(\frac{Z_{2i} - z_2}{h}\right) (Y_{oi} - \beta_1^{Y_o} Z_{1i} - \beta_2^{Y_o} Z_{2i} - \beta_X^{Y_o} X) \quad (4)$$

and I compute the counterfactuals for always-takers and compliers with equation (C3). The following parameters can be identified: (i) s versus the next-best, $E(Y_s - Y_{s_-} | \text{Compliers}, Z_2 = z_2)$, (ii) s versus h , $E(Y_s - Y_h | \text{Compliers: } h - s, Z_2 \rightarrow 0)$, and (iii) s versus n , $E(Y_s - Y_n | \text{Compliers: } n - s, Z_2 \rightarrow z_2^{max})$. Moreover, the following marginal treatment effects (MTEs) can be identified:

$$\lim_{Z_2 \rightarrow 0} E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)) = E(Y_s - Y_h | Z_2, U_s(Z_1) = U_h(Z_1)), \text{ and}$$

$$\lim_{Z_2 \rightarrow z_2^{max}} E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)) = E(Y_s - Y_n | Z_2, U_s(Z_1) = U_n(Z_1)),$$

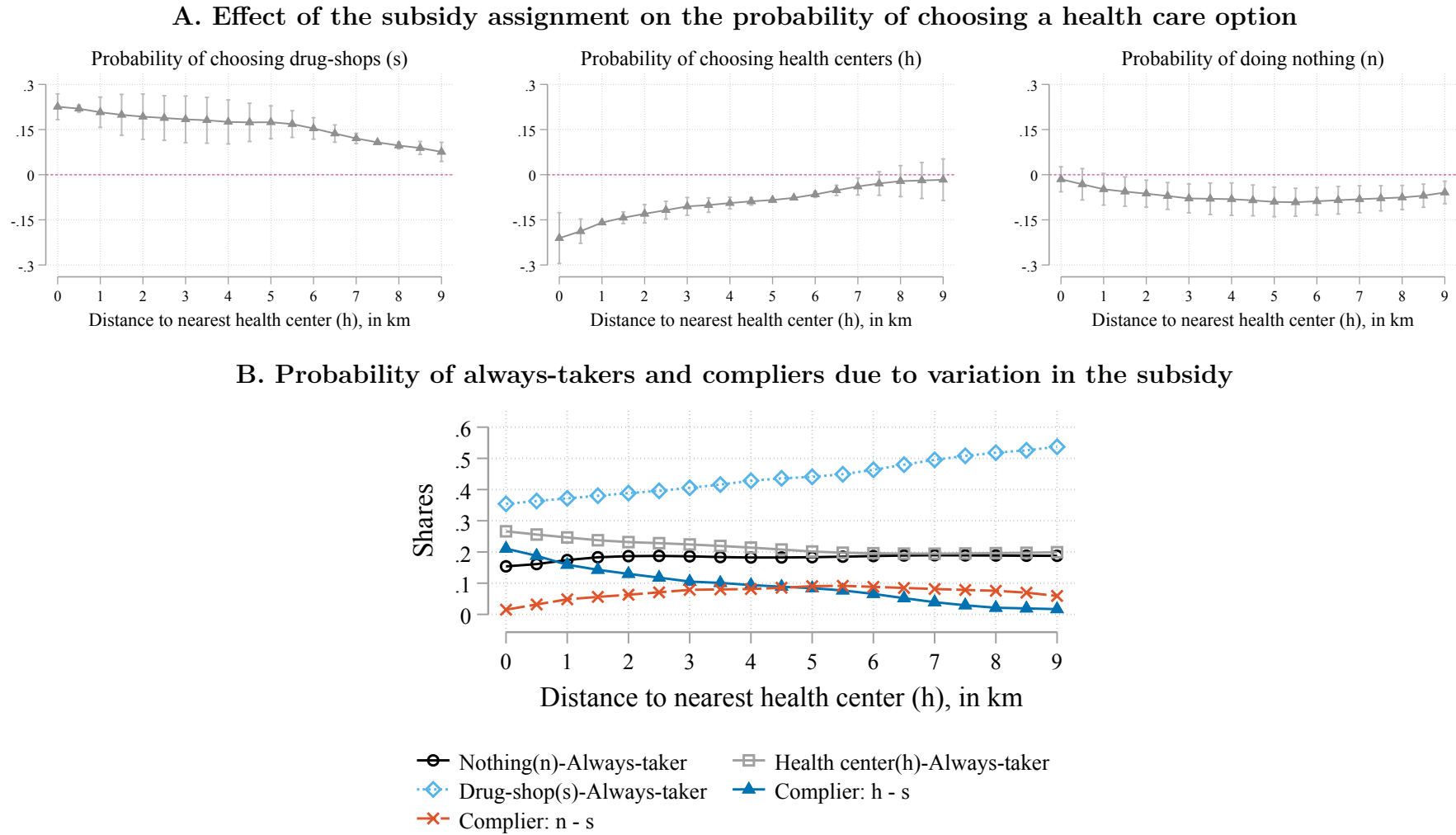
where s_- denotes the next-best alternative, and $U_{k \setminus s}$ denotes the value of choosing options different than s .

To estimate the MTEs, I exploit additional price variation in Z_1 embedded into Cohen et al. (2015a)'s experiment. That is, rather than using a binary indicator for subsidy assignment, I use a categorical variable that takes on four values (control - no subsidy, and 80%, 88%, and 92% subsidy). I then follow Brinch et al. (2017) and Mogstad et al. (2021) to estimate the MTEs using a categorical IV. First, I estimate the propensity score for s , denoted by p , with the first stage specification for s . Panel (a) of Figure D2 in the Appendix D.2 shows the common support, which is set as the interval $p \in [0.28, 0.8]$. Then I estimate the following semiparametric regression:

$$\{\hat{B}^Y(z_2)\} = \operatorname{argmin} \sum_{i=1}^N K_2\left(\frac{Z_{2i} - z_2}{h_2}\right) K_p\left(\frac{P_i - p}{h_p}\right) (Y_i - \beta_1^Y p_i - \beta_2^Y p_i^2 - \beta_3^Y Z_{2i} - \beta_X^Y \mathbf{X} * p), \quad (5)$$

where $K_2(\cdot)$ and $K_p(\cdot)$ denote the Epanechnikov kernel function and triangular kernel function, respectively. I evaluate the MTE at $z_2 \in \{0, 4.5, 8.5\}$, and at the sample average of the covariates in $\mathbf{X}$.

Figure 2: Effect of the subsidy assignment on the probability of choosing a health care option, by distance to the nearest health center



Source: subsample from [Cohen et al. \(2015a\)](#).

Notes: Panel A shows the effect of the subsidy on the probability of choosing the drug shop, health centers, and no care. Effects are computed with a semiparametric regression using an Epanechnikov kernel around different evaluations points of the distance to the nearest health center. All figures show confidence intervals at the 90% level. Panel B shows the estimated probability of always-takers and compliers induced by the variation of the subsidy assignment, by distance to the nearest health care center.

3.1.4 Results

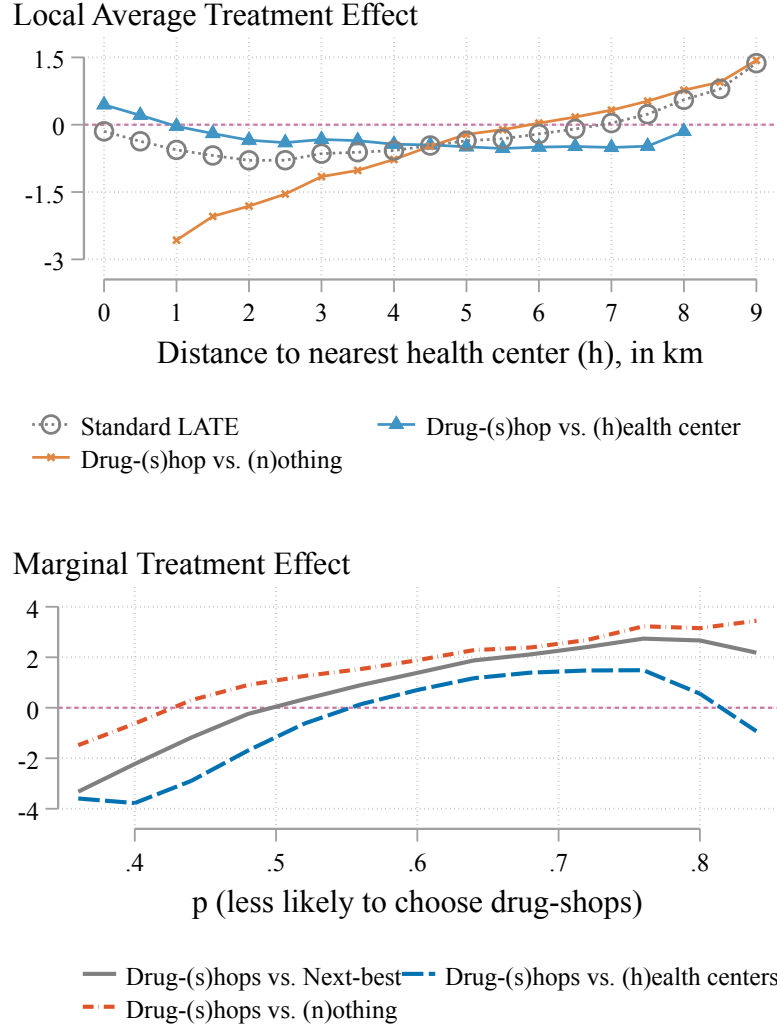
Panel B of Figure 2 shows the average prevalence of always-takers and compliers induced by the variation in the subsidy, at different evaluation points of the distance to the health center. The group with the largest share corresponds to s —always-takers (represented by the light blue diamonds), which goes from around 35% to above 50% as the distance to h increases. In contrast, the share of h —always-takers (that is, households who would have chosen the health center regardless of the subsidy assignment) decrease from 30% to 20% as the distance to h increases. Such behavior is consistent with the role of distance as a cost-shifter and households substituting h for relatively cheaper options. More important, the share of compliers along the $h - s$ ($n - s$) margin decreases (increases) as the distance to h increases which supports the definition of conditional response types in the previous section, and satisfies the conditions in remark 4.1 for sharp identification of pairwise treatment effects.

The top panel of Figure 3 shows estimates of local average treatment effects on the number of illness episodes experienced by the households 4 months after the subsidy provision. The figure presents estimates for standard LATEs (in gray) and subLATEs for each margin recovered by imposing an assumption of homogeneity.²¹ Following the identification results from Section 3.1.2, as the distance to h approaches zero (nine, which is the maximum distance in this setting), the LATE estimates in gray reflect the effect for the $h - s$ ($n - s$) margin. Hence, households who would switch from h and towards s as the subsidy assignment changes experience virtually no difference in illness episodes, on average. On the other hand, choosing s due to receiving the subsidy when the fallback option is n increases illness episodes by 1-1.5. Note that the standard LATE (in gray) differs from the homogeneity-assumption results for the s vs. h margin (in blue): the former is smaller than the latter. This is in line with households in the $h - s$ margin having a lower expected outcome from choosing s than those in the $n - s$ margin (if switching into s means lower quality care than h), which would result in the homogeneity assumption overestimating the effect of s versus h .

The bottom panel of Figure 3 shows results of estimating the marginal treatment effect for s versus the next-best, as well as for the separate margins of $h - s$ and $n - s$. Higher values along the horizontal axis represent a lower likelihood of choosing s ; the vertical axis measures the effect on the number of illness episodes at endline. The black line shows

²¹I include homogeneity estimates to compare the results for this application to those of the other two applications. In this case, the homogeneity assumption states that the mean potential outcome of choosing s would have been the same regardless of the original state (h or n). Note that separate identification of pairwise treatment effects does not rely on this assumption here. Nonetheless, I also present estimates of average of baseline variables for complier groups to assess the validity of homogeneity in Figure D4 of Appendix D.4 which shows that households who would have chosen h in the absence of the subsidy are on average more concerned about their health despite having a similar likelihood of experiencing malaria episodes, compared to households who would have chosen n in the absence of the subsidy.

Figure 3: Treatment Effect of choosing drug shops on the number of illness episodes at endline, by distance to the nearest health center



Source: subsample from [Cohen et al. \(2015a\)](#). *Notes:* The figure at the top shows the estimated local average treatment effects for compliers induced by the variation of the subsidy assignment, by distance to the nearest health care center. The figure at the bottom shows the estimated marginal treatment effects for s vs. the next-best (in black), s vs. h (in blue), and s vs. n (in red), by distance to the nearest health care center. Higher (lower) values on the horizontal axis represent households who are less (more) likely to choose s . The vertical axis measures effects on the number of illness episodes of the household at baseline.

estimates of the MTE for s vs. the next-best, indicating that households who are more likely to choose s (lower p values) rather than the next-best alternative experience fewer illness episodes. A similar pattern emerges for s vs. h (blue dashed line). However, when the fallback option is h (the orange dotted line), only those with the highest likelihood of choosing s would benefit from seeking care at the drug shop; households with $p > 0.4$ would lose from choosing s .²² These results highlight the importance of disentangling

²²Panel (a) of Figure D3 in Appendix D.3 presents estimates of the weights that the IV estimator

effects at the different margins as those in the margin of indifference between s and n might be worse off in terms of illness episodes than households who are choosing s instead of seeking care at health centers.

3.2 Parental choice of childcare in Colombia

This second application studies parental childcare choices for children aged six months to five years, in the context of a public-care expansion in Colombia during 2011. From the 1980s to 2011, small nurseries (s) were the main providers of public childcare for low income families. Small nurseries are run by one caretaker that serves 12-15 children in the same space, typically the home of the provider. In 2011 the public provision of childcare was expanded to offer services through both s and large centers (l). The latter serve around 300 children who receive care in age-specific classrooms, with one teacher per 25 children, a nutritionist and a psychologist in each center, and administrative staff.²³

Although l centers offer nutritional and psychological services and have better infrastructure than s centers, other factors could improve or deter children’s development. Age-specific classrooms may improve cognitive development, but higher teacher to student ratios may have a negative impact. Having a nutritionist could be beneficial for children at risk, but they can be supervised more closely in smaller classrooms. The heterogeneity of the potential benefits of s and l makes it difficult to rank them *a priori*. For instance, parents with preferences for centers that provide nutritional services could prefer l over s , while those concerned with socio-emotional development could rank s (where children of all ages are interacting) above l . Parents with strong preferences for low caretaker-to-children ratios could also prefer home care to either s or l . Rather than imposing a rank that would restrict parent’s preference heterogeneity, I assume that these alternatives are unordered (*e.g.*, l centers are no better than s ones across all dimensions of children’s development).

3.2.1 Data

I exploit rich data containing a random sample of children who were enrolled in s , before l was available (see Bernal et al., 2019). That is, all children in my sample were enrolled in s at baseline, and I observe their enrollment at follow-up: home care (h), s , or l . The dataset contains socioeconomic variables collected at baseline such as children’s age and sex, mother’s years of education, number of children in the household, and household income. It also has age-standardized measures of child development on cognitive, socio-emotional, and nutritional dimensions, which were collected before and after l was available. At

would place on the MTEs at each value of p . In this case, IV places more weight on households less likely to choose drug shops when s is compared *versus* the next-best. When the fallback option is h , there is more weight on the left of the distribution meaning that households less likely to choose s receive relatively more weight than when the fallback option is n .

²³For further details on the supply of childcare in Colombia and the characteristics of each center, see Bernal et al. (2019) and Bernal and Ramírez (2019).

follow-up, more than half of the children in the sample attend l , compared to 32% in s and 13% who receive care at home (h) (see Table D3).

To estimate treatment effects, I exploit two sources of exogenous variation: an experiment providing information to families about the supply of centers, and the geographical distance between the child's home and the different options. The former is a lottery assigned at the s level that offered the chance to transfer to l , for caretakers and children at winner s . The lottery did not provide financial incentives to transfer, but it increased the likelihood of being informed about l and transferring to l . This transfer was not enforced. Distance to l is the straight-line distance from the child's home to the nearest l center, measured in kilometers.

About 74% of children in the sample were randomly assigned to transfer with their caretaker directly from s to l ("Wins lottery", Column 5 in Table D4); the remaining 26% are control children who did not win the lottery. Information on this transfer opportunity was given to caretakers in winner s centers, who could decide: (i) whether or not to contact and inform parents, and (ii) whether to switch to work at l centers since both s and l are publicly funded. Thus, there is noncompliance in the parental response to the lottery, and some parents might have lost their original s center if the caretaker decided to work for l .

3.2.2 Conditional Response Types for Childcare Choices

Let k denote parental choice of childcare, with $k \in \{h, s, l\}$. The choice of childcare, k , affects children's development, denoted by Y . I focus on three dimensions: cognitive, socio-emotional, and nutritional. Furthermore, let $Z \in \{Z_1, Z_2\}$ contain instrumental variables where Z_1 is a binary variable and Z_2 is a continuous variable. Specifically, Z_1 is the outcome of the random lottery and Z_2 is the distance to l . I assume that the instruments in Z satisfy A1 and A3a.

As in the previous application, I assume that parents choose the option with the highest latent utility and that IVs affect choices through their effects on costs. In particular, as the distance to the large center increases (and because some careworkers in winning s centers transition to work at l centers), parents would choose s without the lottery but would switch to h or l when winning the lottery. In contrast, when distance to l decreases, parents would have strong incentives to switch out of h and s and into l centers due to the lottery outcome.

Table 2: Z_1 -responses at different evaluation points of Z_2

$$R_1(Z_2 \rightarrow 0) = \begin{bmatrix} h & s & l & h & s \\ h & s & l & l & l \end{bmatrix} \quad R_1(Z_2 \rightarrow z_2^{\max}) = \begin{bmatrix} h & s & l & s & s \\ h & s & l & h & l \end{bmatrix}$$

Unlike the case of malaria subsidies in Kenya, the combination of the experimental

variation and the institutional context (where the lottery also changes the supply of childcare) generates sets of pairs of complier groups across the distance to l , rather than unique groups at different evaluation points. Moreover, pooling all the variation from the lottery would violate [A3](#) (monotonicity), since some parents would be induced towards– and away– from h as the lottery outcome changes. Below, I empirically establish that as $Z_2 \rightarrow 0$, the share of households along the $h - l$ margin is negligible; thus, for low values of Z_2 , the pairwise LATE of $s - l$ can be identified. In contrast, as $Z_2 \rightarrow z_2^{\max}$, identifying pairwise treatment parameters requires implementing conditional response types in combination with homogeneity assumptions.

3.2.3 Exploratory Analysis and Implementation

First stage: I estimate propensity scores locally around evaluation points of the distance to the large center (Z_2) with the semiparametric regression in equation [\(3\)](#), with an Epanechnikov kernel with bandwidth 0.8km. Baseline variables included in $\mathbf{X}$ are child’s age in months, sex, a binary variable for lowest income level, mother’s years of education, number of children aged 0-5 living at home, binary variables for birth order, and scales of cognitive, socio-emotional, and nutritional development. Estimates of $\hat{\beta}_1^k(z_2)$ for $k \in \{h, s, l\}$ are obtained in a similar manner as in the previous example. Using the results in [Appendix A.5](#), I estimate the probabilities of always-takers and compliers in [Table 2](#).

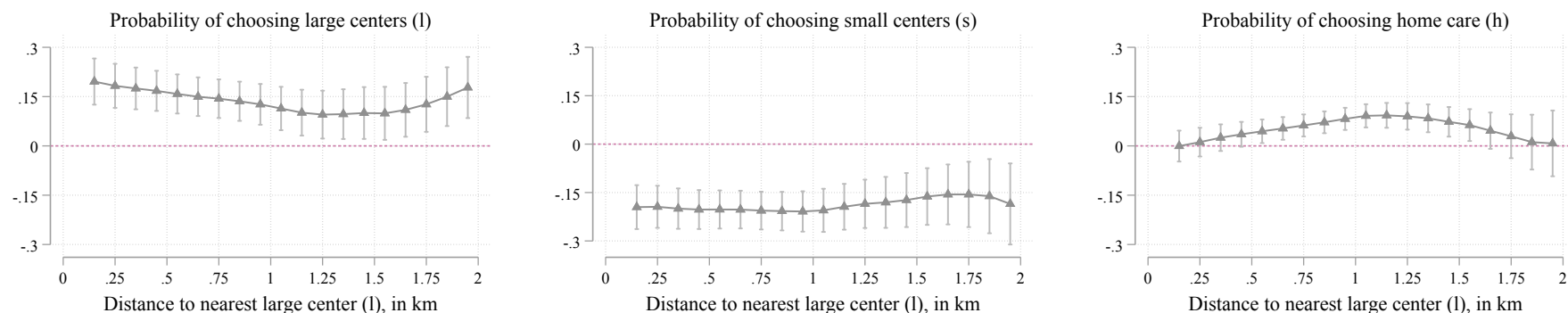
First stage estimates: Panel A of [Figure 4](#) shows results for $\hat{\beta}_1^k(z_2)$ at different evaluation points of distance to l , with each column referring to a childcare choice (l , s , and h). I observe that the lottery outcome’s effect on the probability of choosing l (s) is positive (negative) and relatively constant at all distances. In contrast, winning the lottery increases the probability of choosing h for distances below 1.25km but loses magnitude and significance after 1.7km. These patterns are consistent with the conditional response types in [Table 2](#) and provide empirical support to assume that $\Pr(\text{Complier: } h - l \mid Z_2 \rightarrow 0) = 0$, since the effect of the lottery outcome is not statistically significant for low values of Z_2 .

Treatment Effects: I estimate semiparametric regressions in equation [\(4\)](#) for each vector in $Y_o \in \{Y, YD_L, YD_S, YD_H\}$, and compute the counterfactuals for always-takers and compliers with equation [\(C3\)](#). Note that variation in the lottery outcome induces parents to switch along the $s - l$ and $s - h$ margins, implying that univariate IV (*i.e.*, using only Z_1 as instrument) would identify the effect of l vs. s plus a bias component from the s vs. h margin. The same result holds for estimating effects using Z_1 and conditioning on different points of Z_2 , in particular:

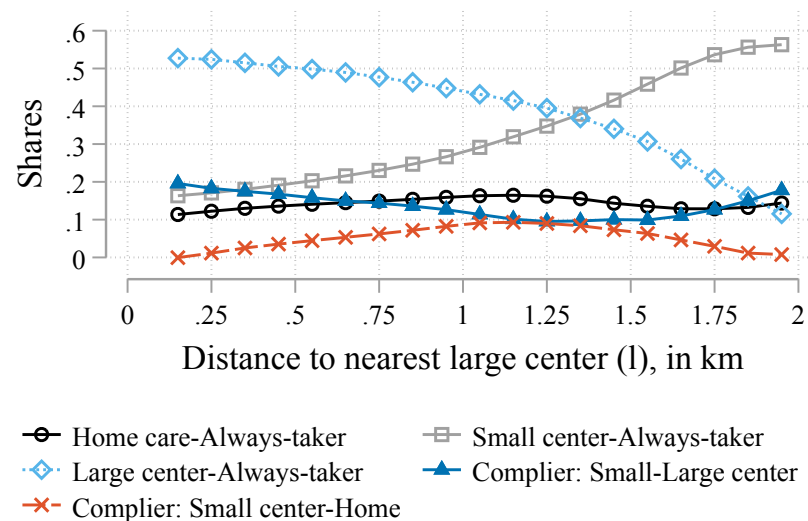
$$\begin{aligned} \frac{E[Y|Z_1 = 1, Z_2] - E[Y|Z_1 = 0, Z_2]}{E[D_l|Z_1 = 1, Z_2] - E[D_l|Z_1 = 0, Z_2]} &= E[Y_l - Y_s|s - l, Z_2] \\ &+ \frac{E[Y_h - Y_s|s - h, Z_2]\Pr(s - h|Z_2)}{\Pr(s - l|Z_2)}, \end{aligned}$$

Figure 4: Effect of the lottery outcome on the probability of choosing a type of childcare, by distance to the nearest large center

A. Effect of the lottery outcome on the probability of choosing a type of childcare



B. Probability of always-takers and compliers due to variation in the lottery outcome



Source: subsample from [Bernal et al. \(2019\)](#).

Notes: Panel A shows the effect of the lottery outcome on the probability of choosing large centers, small centers, and home care. Effects are computed with a semiparametric regression using an Epanechnikov kernel around different evaluations points of the distance to the nearest large center. All figures show confidence intervals at the 90% level. Panel B shows the estimated probability of always-takers and compliers induced by the variation of the lottery outcome, by distance to the nearest large center.

where the right-hand side of the equation above corresponds to the Wald estimator, evaluated at different values of distance to l . To separately identify LATEs for different margins, I assume homogeneity—*i.e.*, that potential children’s development at s centers would have been the same, on average, regardless of their next-best option: $E[Y_s | \text{Complier}: s - l, Z_2] = E[Y_s | \text{Complier}: s - h, Z_2]$, which can be identified from the ratio of $\beta_1^{YD_s}(z_2)$, in equation (4), over $\beta_1^s(z_2)$, in equation (3). Last, from the first-stage estimates, at $Z_2 < 0.25$ and $Z_2 > 1.7$, $E[Y_l - Y_s | \text{Complier}: s - l, Z_2]$ can be separately identified.

Since the lottery outcome induced compliers along the $s - l$ and $s - h$ margins, in what follows and for the purpose of interpretation of effects, I focus on the choice of s vs. the next-best (l or n) or vs. l . The following marginal treatment effects (MTEs) can be identified:

$$E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)), \text{ and}$$

$$\lim_{Z_2 \rightarrow z_2} E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)) = E(Y_s - Y_l | Z_2, U_s(Z_1) = U_l(Z_1)),$$

for $z_2 = 0$ and $z_2 = z_2^{\max}$. Where s_- denotes the next-best alternative, and $U_{k \setminus s}$ denotes the value of choosing options other than s . To estimate the MTEs using a binary IV, I follow Brinch et al. (2017) and Mogstad et al. (2021) as in the previous application. I first estimate the propensity score for s , and denote it as p , with the first stage specification for s . Panel (b) of Figure D2 in the Appendix D.2 shows the common support, which is set as the interval $p \in [0.16, 0.76]$. Then, I follow the semiparametric regression in equation (5) and evaluate the MTE at $z_2 \in \{0, 1.25, 1.9\}$, and at the sample average of the covariates in $\mathbf{X}$.

3.2.4 Results

Panel B of Figure 4 shows that, as distance to l increases, the fraction of l -always-takers decreases (from 50% to 10%) while the fraction of s -always-takers increases (from about 20% to almost 60%). This is consistent with parents substituting their childcare choices toward the relatively cheaper option; in this case, as distance to l increases, bringing children to l becomes more costly in relative terms. The fraction of compliers along the $s - l$ margin is around 10 – 20%, while the share of $s - h$ compliers is below 10% and tends toward zero for distances below 0.7km and above 1.5km.

The top row of Figure 5 presents local average treatment effects on cognitive, socio-emotional, and nutritional development. Estimates using the variation in the lottery outcome, evaluated at different distances to l , recover the effect of s vs. l plus a bias component from the s vs. h margin (denoted as “Standard LATE” in each subfigure). I employ the homogeneity assumption to decompose subLATEs for each of those margins (s vs. l in blue, and s vs. h in orange).²⁴ Overall, the results indicate no benefit for children

²⁴In Appendix D.4, I present estimates of average of baseline variables for compliers along the $s - l$

switching away from s centers and into home care. For the s vs. l margin, the effect of large centers is negative on cognitive development but positive on socio-emotional and, for some children, on nutritional development. While the results lack predictive power, they are indicative of sizable differences in the effects of choosing alternative childcare choices.

The bottom row of Figure 5 presents marginal treatment effects on cognitive, socio-emotional, and nutritional development. Higher values on the horizontal axis denote a lower propensity to choose small centers. The gray line denotes the MTE of s vs. the next-best (evaluated nonparametrically at $Z_2 = 1.25\text{km}$) where half of compliers switch along $s - h$ and the other half along $s - l$. The blue dashed and dotted lines evaluate nonparametrically the MTE at Z_2 equal to 0.1km and 1.9km, where the share of $s - h$ compliers goes to zero. Those more likely to choose s (with low values of p) would lose in cognitive and socio-emotional development and slightly benefit in nutrition, regardless of the fallback option. However, comparisons of the estimates for the MTE of s vs. l and s vs. the next-best suggest that the fallback option matters for those unlikely to choose s (with high values of p): if the fallback option is l , children less likely to choose s would gain in cognitive development but lose in socio-emotional and nutritional development; if the fallback option is either l or n , the cognitive benefits disappear, the behavioral problems remain, with minor gains in nutrition. Last, those with values of p in the interval (0.35,0.55) seem to do better across all development dimensions, in particular if the fallback option is l .²⁵

3.3 Higher education choices in Colombia

The last application focuses on the *Ser Pilo Paga* program (SPP, henceforth), which provided scholarships for attending any 4 or 5-year higher education program in any high-quality university in Colombia. Eligibility was need (*i.e.*, being below a poverty score) and merit (*i.e.*, being above a cutoff score in the high school exit exam) based. To analyze this case, I use an anonymized subsample of eligible students from the universe of high school graduates in 2014 from Londoño-Vélez et al. (2025). The results I present here are not intended to evaluate SPP in any dimension, which has been documented rigorously by Londoño-Vélez et al. (2020) and Londoño-Vélez et al. (2025), rather they illustrate the role of conditional response types in the context of higher education choices. To do so, I deviate from identifying SPP’s impact on access to higher education and labor market outcomes with a Regression Discontinuity Design, and instead estimate the effect of choosing a higher education option (*i.e.*, not enrolling in higher education, enrolling

and $s - h$ margin. Figure D5 shows that $s - l$ compliers have, on average, more educated mothers but lower baseline levels of cognitive and socio-emotional development compared to $s - h$ compliers. This finding casts doubt on the validity of the homogeneity assumption in this context.

²⁵Panel (b) of Figure D3 in Appendix D.3 presents estimates of the weights that the IV estimator would place on the MTEs at each value of p . In this case, IV places more weight on households less likely to choose small centers.

in a low-quality university, or enrolling in a high-quality university) on the probability of working in 2021, using SPP eligibility as an instrumental variable.

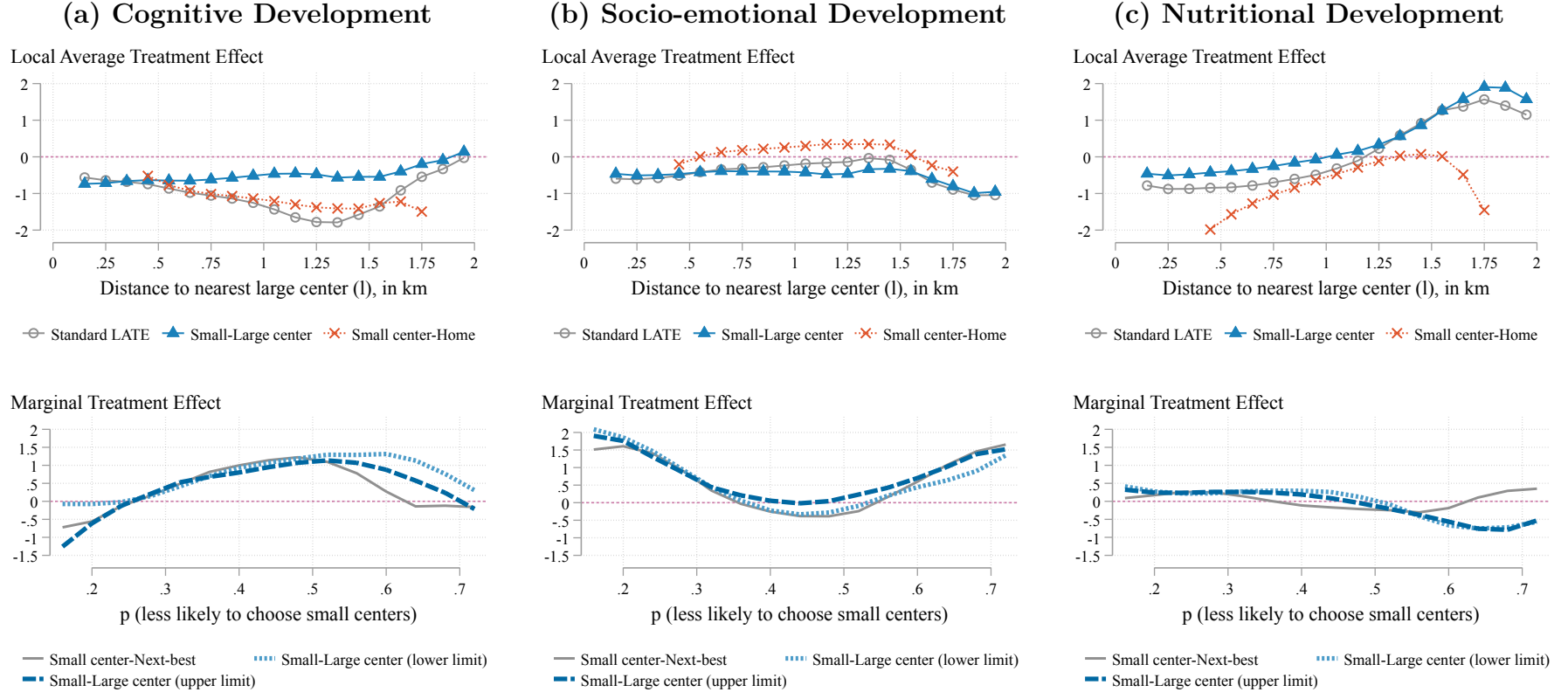
3.3.1 Data

The subsample from [Londoño-Vélez et al. \(2025\)](#) contains student level information for the first cohort of SPP on test scores in the high school exit exam, the poverty score, socioeconomic characteristics such as household size and mother’s educational level. All these variables are measured in 2014, prior to the higher education enrollment choice and to the announcement of SPP by the government. It also contains the higher education enrollment choice students made which I categorize into three alternatives: not enrolling in higher education (n), enrolling in a low quality university (l), and enrolling in a high quality university (h). Following the logic of the previous applications, I assume that these options are unordered, such that one is not better than the other *for all* students.

I split the data into two estimating subsamples based on whether students are need-eligible (*i.e.*, below the poverty score cutoff for SPP) or merit-eligible (*i.e.*, above the high school exit exam score cutoff for SPP), as has been done in other studies analyzing SPP. To estimate treatment effects of higher education choices, I use the remaining dimension of SPP eligibility in each subsample as a source of exogenous variation. That is, for the need-eligible (merit-eligible) students, I construct a binary indicator, Z_1 , taking the value of one if the student is above the high school exit exam cutoff (below the poverty score cutoff), and zero otherwise. In both groups, being eligible for SPP would induce students towards enrolling in h instead of choosing l or n . Moreover, I georeference all higher education institutions in the country and compute the geographic distance from the student’s high school to the nearest low- and high-quality university. I define Z_2 , the second instrument, as the distance to the nearest l .

Table [D5](#) shows average characteristics at baseline, average outcomes, and the distribution of enrollment choices for the need- and merit-eligible subsamples. About 70% (36%) of need-eligible (merit-eligible) students do not enroll in a higher education program, while 17% (18%) enroll in a low-quality program, and 14% (47%) enroll in a high-quality program. Students choosing h are on average younger, less likely to be female, and more likely to have more educated mothers, than those choosing n . As expected, those in the need-eligible subsample are about twice as likely to belong to the lowest socio-economic level than those in the merit-based subsample. In terms of outcomes, students who choose h are more likely to be formally employed in 2021 (7 years after completing high school) than those choosing l or n . Among the need-eligible, students choosing h are 6 percentage points more likely to work formally than those choosing l , while this difference is only of 4 p.p. between l and n . In contrast, h merit-eligible students are only 2 p.p. more likely to work while these difference increases to 14 p.p. between l and n students.

Figure 5: Treatment Effects for compliers along the large *versus* small center margin and the home-care *versus* small center margin, by distance to the nearest large center (l)



Source: subsample from Bernal et al. (2019).

Notes: The figure presents estimated treatment effects for cognitive development in column (a), socio-emotional development in column (b), and nutritional development in column (c). Figures in the top row show the estimated local average treatment effects for compliers induced by the variation in the lottery outcome, by distance to the nearest large center. The bottom row shows estimates of MTEs for small-center versus the next-best, evaluated at the minimum and maximum distances to the large center. Higher (lower) values on the horizontal axis represent households less (more) likely to choose small centers. The vertical axis measures effects on standard deviations of developmental measures.

Table D6 shows the average of Z_1 and the distance to the nearest h and l , for the subsamples of merit- and need-based eligible students. More than two thirds of students choosing h satisfy both eligibility criteria for SPP (72% among the need-eligible, and 75% among the merit-eligible), as expected given that the program covers full tuition for a program at a high-quality university. In terms of distance to h and l , regardless of the subsample, students choosing h (l) are on average at least as near to those options than students choosing l or n (h or n). Moreover, l universities are on average closer than h universities regardless of the subsample, and need-eligible students are on average farther away from any type of HEIs than merit-eligible students.

3.3.2 Conditional Response Types for Higher Education Choices

Let k denote the higher education enrollment choice, with $k \in \{n, l, h\}$. I assume that these options are unordered, meaning that no option is better than another *for all* students. Higher education enrollment choice, k , affects the probability of working in the formal sector in 2021 given by Y . Let $Z \in \{Z_1, Z_2\}$ contain instruments where Z_1 is a binary variable of eligibility (either being above the Saber 11 or Sisben cutoff) and Z_2 is the distance to the nearest low quality HEI (measured at baseline). I assume that the instruments in Z satisfy A1 and A3a.

Since SPP required students to secure a slot at any h , I assume that satisfying the eligibility criteria would shift students across two margins of choice: n *versus* h , and l *versus* h . That is, SPP eligibility would shift students away from not enrolling or enrolling in a low-quality university, and toward enrolling in a high-quality university. Moreover, following the logic of cost-shifters and applying it to the second instrument, Z_2 , as distance to l increases it is more costly to choose l relative to h or n . Using the variation from Z_1 conditional on Z_2 , one would expect that as distance to l increases the probability of observing students switching along the $l - h$ margin, because of changes in SPP eligibility, decreases (see Table 3 below).

Table 3: Z_1 -responses at different evaluation points of Z_2 (distance to nearest low-quality university)

$$R_1(Z_2 \rightarrow 0) = \begin{bmatrix} n & l & h & l \\ n & l & h & h \end{bmatrix} \quad R_1(Z_2 \rightarrow z_2^{\max}) = \begin{bmatrix} n & l & h & n \\ n & l & h & h \end{bmatrix}$$

However, students might be less sensitive to distance than in previous applications; a considerable fraction usually migrates across the country to pursue a higher education degree even without SPP. If students are insensitive to the distance to the HEI, then restriction R1 does not hold, the cost function of enrolling in higher education does not depend on Z_2 , and the set of conditional response types would be the same as the

set of unconditional response types. Moreover, note that the sensitivity of students to Z_1 and Z_2 might differ for need- versus merit-eligible students. Need-eligible students have higher constraints and, on average, less access to HEIs, implying that, regardless of changes in SPP eligibility or a decrease in the distance to l , they would be less likely to attend l or h and more likely to choose n without SPP than merit-eligible students. Hence, conditioning on meeting the need-based criteria leads to a low likelihood of option l belonging to the set of next-best states, implicitly resulting from the logic of conditional response types.

3.3.3 Exploratory Analysis and Implementation

First stage: I estimate propensity scores locally around evaluation points of the distance to the nearest low-quality university (Z_2) with the semiparametric regression in equation (3), with an Epanechnikov kernel with bandwidth 20km. Baseline variables included in $\mathbf{X}$ are student’s age, sex, family size, mother’s educational level, distance to the nearest h , and a cubic polynomial in the high school exit exam (poverty) score for the need-eligible (merit-eligible) subsample. Estimates of $\hat{\beta}_1^k(z_2)$ for $k \in \{n, l, h\}$ are obtained similarly to the previous examples. With the results in Appendix A.5, I estimate the probabilities of always-takers and compliers in Table 3.

First stage estimates: Figure D1 shows results for $\hat{\beta}_1^k(z_2)$ at different evaluation points of distance to the nearest l , with each column referring to an enrollment choice (h , l , and n) for the need-eligible students (in Panel A) and the merit-eligible students (in Panel B). First, Panel A shows that, for the need-eligible subsample (column “Probability of choosing high-quality HEI (h)”), SPP eligibility increases enrollment in h by about 60 percentage points, with most of this effect (around 90%) stemming from a decrease in the probability of choosing n rather than l . In contrast, for the merit-eligible subsample (in Panel B), SPP increases the probability of choosing h by about 40 percentage points; 60% of this effect comes from reductions on the probability of choosing n , while the remaining effect stems from reductions on the probability of choosing l . In both subsamples, the effect of SPP eligibility is constant across different values of distance to l , $Z_2 \in \{0\text{km}, 77\text{km}\}$, implying that students are insensitive to changes in this instrument. Unlike the previous applications, changes in the prevalence of complier groups do not stem from a single additional instrument, Z_2 , but from analyzing distinct subsamples of students with more or less binding constraints resulting from combinations of SPP eligibility.

Treatment Effects: For each vector in $Y_o \in \{Y, YD_n, YD_l, YD_h\}$, I estimate the semiparametric regression in equation (4) and I compute the counterfactuals for always-takers and compliers with equation (C3). Regardless of the subsample, the following parameters can be identified: (i) h vs. the next-best, $E(Y_h - Y_{h_-} | \text{Compliers}, Z_2)$ where h_- denotes the next-best alternative. To separately identify effects at distinct

margins, I employ the homogeneity assumption which allows identifying: (i) n vs. h , $E(Y_h - Y_n | \text{Compliers: } n - h, Z_2)$, and (ii) l vs. h , $E(Y_h - Y_l | \text{Compliers: } l - h, Z_2)$. In this case, homogeneity assumes that labor market participation from choosing h would have been the same, on average, regardless of the next-best option: $E[Y_h | \text{Complier: } h - l, Z_2] = E[Y_h | \text{Complier: } h - n, Z_2]$, which can be identified as the ratio of $\beta_1^{Y_{D_h}}(z_2)$, in equation (4), over $\beta_1^h(z_2)$, in equation (3). It is worth noting that, among the need-eligible students, $\Pr(\text{Compliers: } l - h | Z_2)$ is very small, such that $E(Y_h - Y_{h_-} | \text{Compliers}, Z_2)$ mostly captures the effect of n vs. l .

The marginal treatment effect (MTE) of h vs. the next-best can be identified, $E(Y_h - Y_{h_-} | Z_2, U_h(Z_1) = U_{k \setminus h}(Z_1))$ where h_- denotes the next-best alternative and $U_{k \setminus h}$ denotes the value of choosing options different than h . Rather than employing a homogeneity assumption, I evaluate MTEs at the minimum and maximum values of Z_2 in each subsample to illustrate how a modest increase in the $\Pr(\text{Compliers: } l - h | Z_2)$ might affect the shape and range of the MTE. That is, I estimate two additional MTE parameters:

$$\text{MTE (lower limit)} = \lim_{Z_2 \rightarrow z_2^{\min}} E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)), \text{ and,}$$

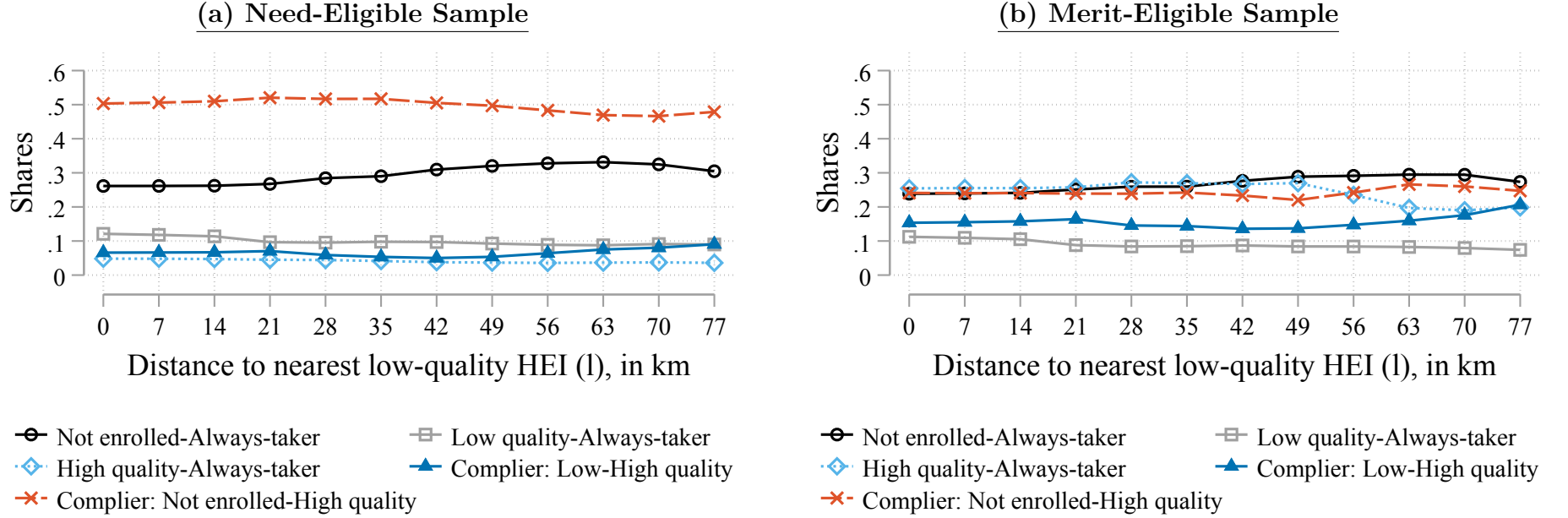
$$\text{MTE (upper limit)} = \lim_{Z_2 \rightarrow z_2^{\max}} E(Y_s - Y_{s_-} | Z_2, U_s(Z_1) = U_{k \setminus s}(Z_1)).$$

I estimate the MTEs using a binary IV in a similar manner as in the previous applications, by first estimating the propensity score for h , denoted by p , with the first stage specification for h . Panels (c) and (d) of Figure D2 in the Appendix D.2 shows the common support for each subsample, which is set as the interval $p \in [0.12, 0.8]$. I then estimate the semiparametric regression in equation (5). I evaluate the MTE at $z_2 \in \{7, 15, 65\}$ for the need-eligible subsample (and at $z_2 \in \{7, 9, 65\}$ for the merit-eligible subsample) where each value corresponds to the minimum, average, and maximum of Z_2 respectively, and at the sample average of the covariates in $\mathbf{X}$.

3.3.4 Results

Figure 6 shows the estimates of always-takers and compliers for the subsample of need-eligible students (in Panel A) and merit-eligible students (in Panel B) at different values of Z_2 . The share of l - and n -always-takers (in black and gray, respectively) is about 30% and 10% across values of the distance to l , regardless of the subsample. In contrast, the share of h -always-takers (*i.e.*, students who would have chosen h regardless of SPP eligibility) is of less than 5% among the need-eligible compared to 20 – 30% among the merit-eligible. Moreover, the majority of students among the need-eligible are $n - h$ compliers (about 50% at different values of Z_2) and less than 5 – 10% are compliers along the $l - h$ margin. Among the merit-eligible, the share of $n - h$ compliers is 20 – 30% and the share of $l - h$ compliers is 15 – 20%.

Figure 6: Probability of always-takers and compliers due to variation in eligibility, by distance to the nearest low-quality HEI (l)



Source: subsample from [Londoño-Vélez et al. \(2025\)](#).

Notes: The figure shows the estimated probability of always-takers and compliers induced by the variation in eligibility, by sample and distance to the nearest low-quality HEI. Panel (a) presents estimates of group shares for the need-eligible students, using as IV a binary indicator taking the value of one if the student is above the high school exit exam cutoff, and zero otherwise. Panel (b) presents estimates of group shares for the merit-eligible students, using as IV a binary indicator taking the value of one if the student is below the poverty score cutoff, and zero otherwise.

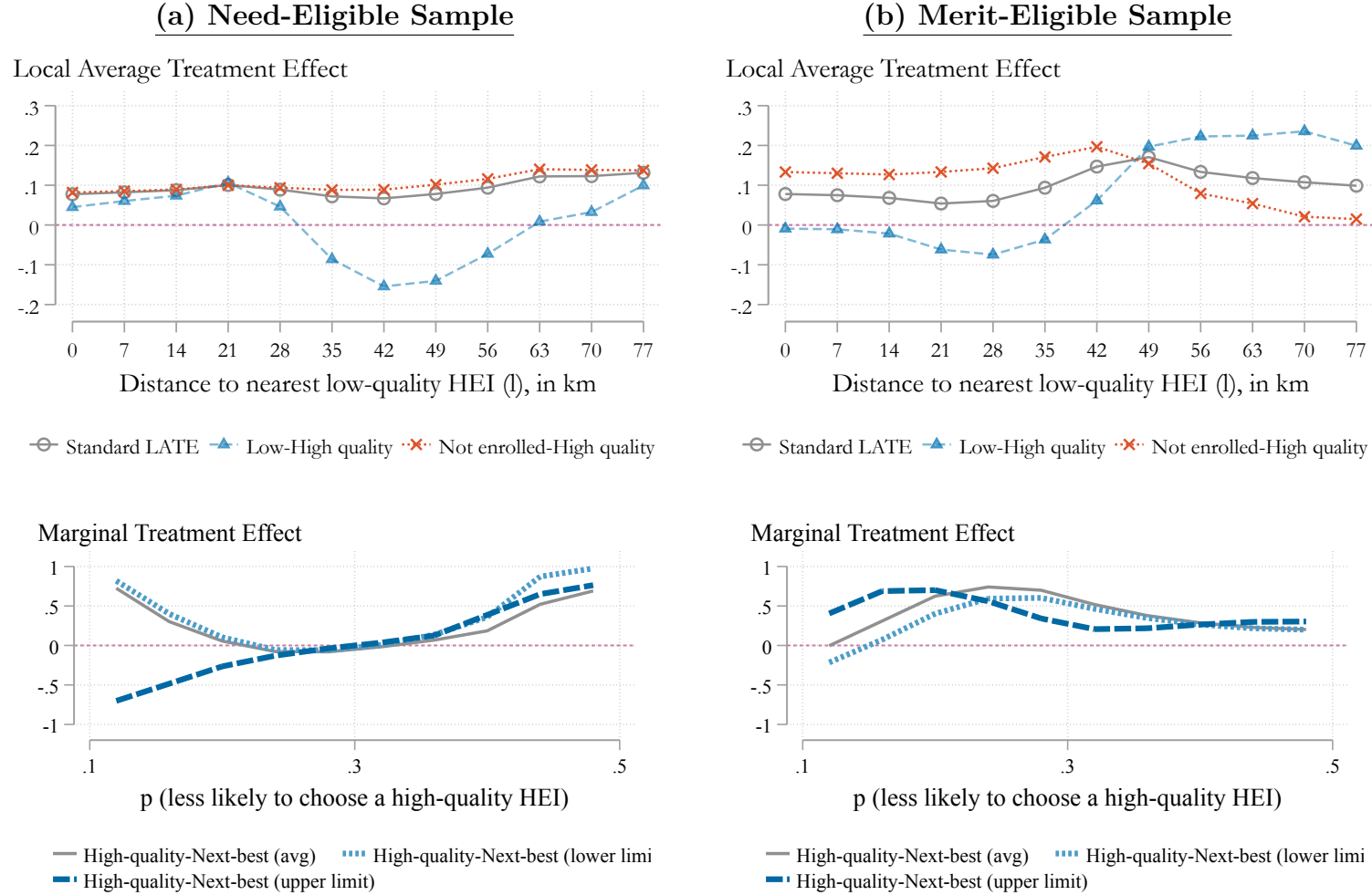
The first row of Figure 7 shows estimates of LATEs for the need-eligible (in column A) and merit-eligible (in column B) subsamples. The effect of h vs. the next-best (denoted as “Standard LATE” in the first row of subfigures) is positive, such that choosing a high-quality program instead of l or n increases the probability of working in the formal labor market in 2021 by about 10p.p., irrespective of the subsample. Among the need-eligible, the estimates of h vs. the next-best (in gray) closely follow those of the effect of n vs. h (in orange), identified by imposing the homogeneity assumption. The estimates of h vs. l (in blue) are negative for some values of Z_2 , but the small share of compliers along this margin might render these estimates less informative. Among the merit-eligible (first subfigure in Panel B), the estimates of h vs. the next-best are a combination of estimates for the $n - h$ and $l - h$ margins (a byproduct of employing the homogeneity assumption); however, the magnitude and sign of the subLATEs changes after 49km of distance to the nearest l . For $Z_2 < 49\text{km}$, choosing h instead of l has a negative effect on the probability of working formally in 2021, while the effect for $n - h$ compliers is positive and between 10-20 percentage points. After 49km, the latter effect decreases to zero, while the effect for $l - h$ compliers increases to around 20 percentage points.²⁶

The second row of Figure 7 shows estimates of MTEs for the need-eligible (in column A) and merit-eligible (in column B). Higher values along the horizontal axis represent a lower likelihood of choosing h , while the vertical axis measures the effect on the probability of working formally in 2021. Among the need-eligible (Panel A), those with a low or high likelihood of choosing h vs. the next-best would experience a higher likelihood of working formally in 2021 (the gray and dashed blue line), and this effect is largely driven by the n vs. h margin. Meanwhile, when the share of $l - h$ compliers increases marginally, the effect for those more likely to choose h instead of the next-best is negative (the upper-limit MTE in the blue dotted line). This might reflect effects for students who would have been better off by choosing l than h . Among the merit-eligible, the MTE is positive regardless of whether it is evaluated at the minimum (the dashed blue line), the average (in gray), or the maximum (the blue dotted line) of distance to l . In practice, students on the margin between choosing h or the next-best benefit from choosing a high-quality program, increasing their likelihood of working formally in 2021. This effect nonetheless decreases as individuals become less likely to choose h (which intuitively would denote individuals with higher family or financial constraints to choose h).²⁷

²⁶Differences on the subLATEs across evaluation points of Z_2 for the merit-eligible subsample might be explained by differences in their average characteristics at baseline. Figure D5 in Appendix D.4 shows that compliers along the $n - h$ have similar average shares of female students but are older, on average, than $l - h$ compliers. Moreover, for $Z_2 < 49\text{km}$ both groups seem comparable in average household size, maternal education, with some differences in terms of SES in favor of $n - h$ compliers. However, for $Z_2 > 49\text{km}$ $n - h$ compliers are on average less vulnerable with more educated mothers while the share of $l - h$ compliers belonging to low SES households increases.

²⁷Panels (c) and (d) of Appendix D.3 Figure D3 shows estimates of IV weights on the MTEs at each value of p for the need-eligible and the merit-eligible subsamples, respectively. In the need-eligible (merit-eligible), IV places more (less) weight on students more likely to choose high-quality HEI.

Figure 7: Treatment Effects on the probability of formal employment for compliers along the low *versus* high-quality HEI margin and the margin of not enrolled *versus* high-quality HEI, by distance to the nearest low-quality HEI (l)



Source: subsample from [Londoño-Vélez et al. \(2025\)](#)

Notes: The figure presents estimated treatment effects on the probability of formal work. The top row shows the estimated local average treatment effects for compliers induced by the variation in the SPP eligibility, by distance to the nearest low-quality HEI. The hollow gray circles show estimates of the LATE of enrolling in a high-quality HEI versus the next best. The blue triangles and red x show estimates of LATEs for the high- versus low-quality HEI margin and the high-quality versus not enrolling margin, respectively. The bottom row shows estimates of MTEs for high-quality HEI versus the next-best, evaluated at the minimum and maximum distances to the nearest low-quality HEI. Higher (lower) values on the horizontal axis represent students less (more) likely to choose high-quality HEI.

4 CONCLUDING REMARKS

In this paper, I study the identification and estimation of treatment effects in contexts where agents can choose between multiple options and the researcher has access to multiple instrumental variables that can affect the same options. I focus on settings with two key features. First, agents can have preferences over the alternatives and self-select into their preferred choice. The latter motivates employing IVs as an estimation strategy. Second, the options are unordered, meaning that we cannot say that one option is better than the other, *a priori*. As a result, agents can have different fallback options and derive different gains or losses from their choices.

The IV approach has inherent limitations. It estimates local effects for agents who change their behavior as the IV changes (i.e., compliers). In the presence of multiple unordered choices, compliers are heterogeneous, and agents can switch along many margins. For instance, if households receive a subsidy for purchases at drug shops, they could switch between healthcare centers and drug shops, or seeking no care and drug shops. The set of compliers becomes more complex when multiple instruments are available. In the first application I show that healthcare choices are highly sensitive to the availability of alternative options, and whether households respond to the subsidy by switching to drug shops depends on their proximity to the nearest health center. While this type of joint response to multiple instruments has been recognized in the literature, current methods implicitly assume that the behavior of compliers (or responses) to one instrument is the same across the distribution of other instruments. Furthermore, the case of multiple instruments that affect the same margins of choice is relevant for many empirical applications.

My main contribution is an identification strategy that addresses these challenges. First, I account for the joint effects of multiple instruments on the probability of choosing an option. That is, I allow for the response to the variation in one instrument (for example, a subsidy for drug shop purchases) to differ depending on other instruments (for example, proximity to the health center) that affect the same margins of choice. To do so, I employ a latent utility framework and model responses to the instruments through their effect on the costs of each option. I impose restrictions on the shape of the cost function, assuming in particular that it is convex. This allows me to define *conditional response types* that satisfy partial unordered monotonicity, locally.

I apply these tools to healthcare and childcare choices and to decisions about pursuing different higher education paths. I show the scope of *conditional response types* in identifying treatment effects across these settings, which differ in their sources of variation and institutional features. Namely, I combine experimental or quasi-experimental variation with the distance to one of the options available. A key takeaway from the applications' results is that identifying margin-specific treatment effects depends on the sensitivity of the cost function to the conditioning instrument—namely, distance to one of the options

in each application. Notably, while healthcare and childcare choices exhibit strong sensitivity to distance, higher education decisions are less so. A considerable proportion of students usually migrate to pursue a degree even without financial aid or incentives. In such contexts, researchers might need to supplement *conditional response types* with the standard homogeneity assumption or derive bounds on the estimates to account for the weaker role of the available conditioning instrument.

The application results underscore key caveats of the proposed identification strategy. First, it requires combining variation from multiple instrumental variables, which might be prohibitive as researchers often have access to a single source of variation. Second, the estimated effects are inherently local, as they are evaluated at specific values or ranges of the conditioning instrument. However, in the context of multiple treatments, (local) evaluation points of the instruments define the set of next-best states. As a result, the results I present can be informative of treatment effects under different combinations of next-best states. Third, the evaluation points are often defined at extreme values of the conditioning instrument, but in practice changes in the composition of *conditional response types* depend on the slope of the cost function. That is, not every context requires extreme values, as is evident from the first two applications.

Moreover, the strategy rests on the standard no-defiers assumption extended to the case of partial monotonicity, but, as shown by Kirkebøen et al. (2016) and Heinesen et al. (2024) in the case of observed next-best alternatives, the estimates can be biased if unordered defiers prevail. Similarly, the standard limitations apply to employing the distance to options as an IV, as it might be correlated with other factors that can affect potential outcomes. In this case, the conditioning IV serves an illustrative purpose, and the results I present are not intended to evaluate or provide evidence on the effectiveness of each policy scenario. That is, I focus on each application context to show the role of *conditional response types* and their importance in estimating effects in complex settings. Finally, while my results pertain to identification without inference (an important direction for future research), they indicate the importance of uncovering variation along different margins of choice in estimating multivalued treatment effects.

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A Derivations and Proofs

A.1 Identification of effects of k *versus* the next-best

For Z_k with $z_k, z'_k \in \text{supp}(Z_k)$ satisfying [A1-A3](#), inducing agents toward k and away from all other options, $l \in \mathcal{K} \setminus k$, it follows that:

$$\begin{aligned} E[Y|Z_k = z'_k] &= E[D_{\mathcal{K},k}Y_k + (1 - D_{\mathcal{K},k})Y_{\mathcal{K} \setminus k} | Z_k = z'_k] \\ &= E[D_{\mathcal{K},k}Y_k | Z_k = z'_k] + E[(1 - D_{\mathcal{K},k})Y_{\mathcal{K} \setminus k} | Z_k = z'_k] \\ &= E[Y_k | Z_k = z'_k, D_{\mathcal{K},k} = 1] \Pr(D_{\mathcal{K},k} = 1 | Z_k = z'_k) \\ &\quad + E[Y_{\mathcal{K} \setminus k} | Z_k = z'_k, D_{\mathcal{K},k} = 0] \Pr(D_{\mathcal{K},k} = 0 | Z_k = z'_k), \end{aligned}$$

and,

$$\begin{aligned} E[Y|Z_k = z_k] &= E[D_{\mathcal{K},k}Y_k + (1 - D_{\mathcal{K},k})Y_{\mathcal{K} \setminus k} | Z_k = z_k] \\ &= E[D_{\mathcal{K},k}Y_k | Z_k = z_k] + E[(1 - D_{\mathcal{K},k})Y_{\mathcal{K} \setminus k} | Z_k = z_k] \\ &= E[Y_k | Z_k = z_k, D_{\mathcal{K},k} = 1] \Pr(D_{\mathcal{K},k} = 1 | Z_k = z_k) \\ &\quad + E[Y_{\mathcal{K} \setminus k} | Z_k = z_k, D_{\mathcal{K},k} = 0] \Pr(D_{\mathcal{K},k} = 0 | Z_k = z_k). \end{aligned}$$

By [A1](#) and [A3](#),

$$\begin{aligned} E[Y|Z_k = z'_k] &= E[Y_k | D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1) \\ &\quad + E[Y_k | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1) \\ &\quad + E[Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 0] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 0), \end{aligned}$$

and,

$$\begin{aligned} E[Y|Z_k = z_k] &= E[Y_k | D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1) \\ &\quad + E[Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 0] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 0) \\ &\quad + E[Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1). \end{aligned}$$

Hence,

$$\begin{aligned} E[Y|Z_k = z'_k] - E[Y|Z_k = z_k] &= \\ &= E[Y_k | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1) \\ &\quad - E[Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1) \\ &= E[Y_k - Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1), \end{aligned}$$

where the first term on the right-hand side is the local average treatment effect of choosing option k *versus* the next-best (*i.e.*, what the agent would have chosen if option k was not

available).

Similarly, by [A1](#) and [A3](#),

$$\begin{aligned} E[D_{\mathcal{K},k}|Z_k = z'_k] &= \Pr(D_{\mathcal{K},k} = 1, Z_k = z'_k) \\ &= \Pr(D_{\mathcal{K},k}(z'_k) = 1) \\ &= \Pr(D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1) \\ &\quad + \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1), \end{aligned}$$

and,

$$\begin{aligned} E[D_{\mathcal{K},k}|Z_k = z_k] &= \Pr(D_{\mathcal{K},k} = 1, Z_k = z_k) \\ &= \Pr(D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1) \end{aligned}$$

Thus,

$$E[D_{\mathcal{K},k}|Z_k = z'_k] - E[D_{\mathcal{K},k}|Z_k = z_k] = \Pr(D_{\mathcal{K},k}(z_k) = 1, D_{\mathcal{K},k}(z'_k) = 1).$$

Then, the Wald estimator identifies the local average treatment effect of choosing option k versus the next-best alternative,

$$\frac{E[Y|Z_k = z'_k] - E[Y|Z_k = z_k]}{E[D_{\mathcal{K},k}|Z_k = z'_k] - E[D_{\mathcal{K},k}|Z_k = z_k]} = E[Y_k - Y_{\mathcal{K} \setminus k} | D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1].$$

A.2 Proof of Lemma 1

First, by definition [1](#) with $w = 2$, T_k denotes the number of response-types, that is, combinations of potential choices at $z_k, z'_k \in \text{supp}(Z_k)$. Following the argument in [Heckman and Pinto \(2018\)](#), without any restrictions on admissible combinations of potential choices, the total number of response-types would be K^2 . That is, all possible combinations of K choices over two possible values z_k, z'_k in $\text{supp}(Z_k)$.

Imposing [A3](#), either $D_{\mathcal{K},k}(z_k) \geq D_{\mathcal{K},k}(z'_k)$ for all agents, or $D_{\mathcal{K},k}(z_k) \leq D_{\mathcal{K},k}(z'_k)$ for all agents. Without loss of generality, take the latter case, then the only admissible combinations of potential choices are: (i) those with $D_{\mathcal{K},k}(z_k) = D_{\mathcal{K},k}(z'_k)$, namely always-takers; and (ii) those with $D_{\mathcal{K},k}(z'_k) = 1$ and $D_{\mathcal{K},k}(z_k) = 0$, namely switchers or compliers. There are a total of K always-takers, while the number of switchers is $K - 1$, bringing the total number of response-types to $T_k = 2K - 1$.

A.3 Decomposition into pairwise local average treatment effects

Note that $Y_{\mathcal{K} \setminus k} = \sum_{l \neq k, D_{\mathcal{K},k} \neq 1} \frac{D_{\mathcal{K},l}}{1 - D_{\mathcal{K},k}} Y_l$. Hence,

$$\begin{aligned} E[(1 - D_{\mathcal{K},k})Y_{\mathcal{K} \setminus k} | Z_k = z'_k] &= \sum_{l \in \mathcal{K} \setminus k} E[D_{\mathcal{K},l}Y_l | Z_k = z'_k] \\ &= \sum_{l \in \mathcal{K} \setminus k} E[Y_l | Z_k = z'_k, D_{\mathcal{K},l} = 1] \Pr(D_{\mathcal{K},l} = 1 | Z_k = z'_k). \end{aligned}$$

From Lemma 1 for $w = 2$,

$$R_k = \begin{bmatrix} 1 & \cdots & k & \cdots & K & 1 & \cdots & K \\ 1 & \cdots & k & \cdots & K & k & \cdots & k \end{bmatrix}.$$

Hence,

$$\begin{aligned} E[(1 - D_{\mathcal{K},k})Y_{\mathcal{K}\setminus k}|Z_k = z'_k] \\ = \sum_{l \in \mathcal{K}\setminus k} E[Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 1] \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 1) \end{aligned}$$

Similarly,

$$\begin{aligned} E[(1 - D_{\mathcal{K},k})Y_{\mathcal{K}\setminus k}|Z_k = z_k] &= \sum_{l \in \mathcal{K}\setminus k} E[Y_l|Z_k = z_k, D_{\mathcal{K},l} = 1] \Pr(D_{\mathcal{K},l} = 1|Z_k = z_k) \\ &= \sum_{l \in \mathcal{K}\setminus k} E[Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 1] \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 1) \\ &\quad + E[Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0] \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0) \end{aligned}$$

Plus,

$$\Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1) = \sum_{l \in \mathcal{K}\setminus k} \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0)$$

Hence,

$$\begin{aligned} E[Y|Z_k = z'_k] - E[Y|Z_k = z_k] &= \\ &E[Y_k|D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1] \Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1) \\ &- \sum_{l \in \mathcal{K}\setminus k} E[Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0] \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0) \\ &= \sum_{l \in \mathcal{K}\setminus k} E[Y_k - Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0] \Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0), \end{aligned}$$

and the Wald estimator,

$$\begin{aligned} \frac{E[Y|Z_k = z'_k] - E[Y|Z_k = z_k]}{E[D_{\mathcal{K},k}|Z_k = z'_k] - E[D_{\mathcal{K},k}|Z_k = z_k]} &= \sum_{l \in \mathcal{K}\setminus k} \frac{\Pr(D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0)}{\Pr(D_{\mathcal{K},k}(z_k) = 0, D_{\mathcal{K},k}(z'_k) = 1)} \\ &\quad \times E[Y_k - Y_l|D_{\mathcal{K},l}(z_k) = 1, D_{\mathcal{K},l}(z'_k) = 0]. \end{aligned} \tag{A1}$$

A.4 Decomposition into pairwise marginal treatment effects

In terms of the Marginal Treatment Effect, note that identification relies on the variation of the binary instrument, Z_k . Invoking the identification arguments in Brinch et al. (2017) and explicitly conditioning on X , the MTE of option k versus the next-best can

be identified. In particular, imposing an auxiliary assumption (namely, Assumption 2 in Brinch et al. (2017)) under which the MTE is additively separable in X and μ_k (that is, the unobserved preference heterogeneity for choice k). By Proposition 2 in Brinch et al. (2017),

$$\begin{aligned}\frac{\partial E(Y|P(Z_k) = p, X = x)}{\partial p} &= E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z, U_1(z_k) = U_{\mathcal{K} \setminus k}(z_k)] \\ &= E[Y_k - Y_{\mathcal{K} \setminus k} | X = x, Z = z_k, \mu_k = p],\end{aligned}$$

where p a realization of the propensity score, $P(Z_k)$, of choosing option k vs. the next-best, and μ_k denotes the unobserved component in the latent utility of choosing option k . That is, using the variation in Z_k and imposing parametric restrictions on the shape of the MTE, the average treatment effect of option k for those on the margin of indifference between choosing k and not choosing it, *i.e.*, $\text{MTE}_{k, \mathcal{K} \setminus k}$, can be identified.

Similarly, the MTE of option k *versus* the next-best can be decomposed into pairwise MTEs. Recall that the MTE identifies the average treatment effect of option k for those on the margin of indifference between option k and the next-best option, $\text{MTE}_{k, \mathcal{K} \setminus k} = E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)]$. First, replacing $Y_{\mathcal{K} \setminus k}$ and $1 - D_{\mathcal{K}, k} = \sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K}, l}$ in $Y_k - Y_{\mathcal{K} \setminus k}$,

$$\begin{aligned}Y_k - Y_{\mathcal{K} \setminus k} &= Y_k - \sum_{l \in \mathcal{K} \setminus k} \frac{D_{\mathcal{K}, l}}{1 - D_{\mathcal{K}, k}} Y_l \cdot \mathbb{1}[D_{\mathcal{K}, k} \neq 1] \\ &= \frac{\mathbb{1}[D_{\mathcal{K}, k} \neq 1]}{1 - D_{\mathcal{K}, k}} \left[Y_k(1 - D_{\mathcal{K}, k}) - \sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K}, l} Y_l \right] \\ &= \frac{\mathbb{1}[D_{\mathcal{K}, k} \neq 1]}{1 - D_{\mathcal{K}, k}} \left[\sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K}, l} Y_k - \sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K}, l} Y_l \right] \\ &= \frac{\mathbb{1}[D_{\mathcal{K}, k} \neq 1]}{1 - D_{\mathcal{K}, k}} \left[\sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K}, l} (Y_k - Y_l) \right] = \sum_{l \in \mathcal{K} \setminus k} \frac{\mathbb{1}[D_{\mathcal{K}, k} \neq 1] \cdot D_{\mathcal{K}, l}}{1 - D_{\mathcal{K}, k}} (Y_k - Y_l) \\ &= \sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K} \setminus k, l} (Y_k - Y_l)\end{aligned}$$

Replacing in the MTE formula,

$$\begin{aligned}E[Y_k - Y_{\mathcal{K} \setminus k} | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)] &= E \left[\sum_{l \in \mathcal{K} \setminus k} D_{\mathcal{K} \setminus k, l} (Y_k - Y_l) \middle| Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k) \right] \\ &= \sum_{l \in \mathcal{K} \setminus k} E[D_{\mathcal{K} \setminus k, l} (Y_k - Y_l) | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)] \\ &= \sum_{l \in \mathcal{K} \setminus k} \Pr(D_{\mathcal{K} \setminus k, l} = 1 | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)) E[Y_k - Y_l | Z = z_k, U_k(z_k) = U_{\mathcal{K} \setminus k}(z_k)].\end{aligned}\tag{A2}$$

A.5 Probabilities of always-takers and compliers

Take the unconditional response-matrix,

$$R_k = \begin{bmatrix} 1 & \cdots & k & \cdots & K & 1 & \cdots & K \\ 1 & \cdots & k & \cdots & K & k & \cdots & k \end{bmatrix}.$$

The following identifies the share of m -always-takers:

$$\begin{aligned} E[D_{\mathcal{K},l}|z'_k] &= \Pr(D_{\mathcal{K},l} = 1|z'_k) \\ &= \Pr(D_{\mathcal{K},l}(z'_k) = 1, D_{\mathcal{K},l}(z_k) = 1), \end{aligned}$$

for all $l \in \mathcal{K} \setminus k$. The share of k -always-takers is identified as follows:

$$\begin{aligned} E[D_{\mathcal{K},k}|z_k] &= \Pr(D_{\mathcal{K},k} = 1|z_k) \\ &= \Pr(D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K},k}(z_k) = 1). \end{aligned}$$

Last,

$$E[D_{\mathcal{K},l}|z_k] - E[D_{\mathcal{K},l}|z'_k] = \Pr(D_{\mathcal{K},l}(z'_k) = 0, D_{\mathcal{K},l}(z_k) = 1),$$

identifies the share of compliers (switchers) along the $l - k$ margin as Z_k goes from z_k to z'_k , for all $l \in \mathcal{K} \setminus k$. While,

$$E[D_{\mathcal{K},k}|z'_k] - E[D_{\mathcal{K},k}|z_k] = \Pr(D_{\mathcal{K},k}(z'_k) = 1, D_{\mathcal{K},k}(z_k) = 0),$$

identifies the total share of k compliers.

A.6 Proof of Proposition 1

Under restriction **R1**, $V_l(Z)$ is (weakly) convex in Z_l . Without loss of generality, take $z_l, z'_l \in \text{supp}(Z_l)$ and $z_l < z'_l$. Therefore, as $Z_l \rightarrow z_l$, convexity implies that $V_l(Z)$ becomes arbitrarily small relative to $V_m(Z)$ for all $m \in \mathcal{K} \setminus k$, since $V_m(Z)$ is held fixed in Z_l while $V_l(Z)$ decreases. That is, as $Z_l \rightarrow z_l$, $V_l(Z) < V_m(Z)$ for all $m \in \mathcal{K} \setminus k$. By equation (1), then $D_{\mathcal{K} \setminus k}(Z_k, z_l) \rightarrow l$ and $D_{\mathcal{K} \setminus k, l}(Z_k, z_l) \rightarrow 1$ with $D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \rightarrow 0$ for all $m \neq l \in \mathcal{K} \setminus k$. It follows that $\mathcal{K} \setminus k = \{l\}$, as $Z_l \rightarrow z_l$.

Conversely, as $Z_l \rightarrow z'_l$, convexity implies that $V_l(Z)$ increases (again holding $V_m(Z)$ fixed for $m \neq l$), so that $V_l(Z) > V_m(Z)$ for all $m \in \mathcal{K} \setminus k$. By equation (1), then $D_{\mathcal{K} \setminus k}(Z_k, z'_l) \rightarrow m$ and $D_{\mathcal{K} \setminus k, l}(Z_k, z'_l) \rightarrow 0$ and $\sum_{m \neq l \in \mathcal{K} \setminus k} D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \rightarrow 1$. It follows that $l \notin \mathcal{K} \setminus k$, as $Z_l \rightarrow z'_l$.

A.7 Proof of Proposition 2

Let $k \in \{1, 2, \dots, K\}$ and $\mathcal{K} \setminus k$ represent choices other than k . Recall that $U_k(z) = \mu_k - V_k(z)$ which denotes the utility of choice k evaluated at realization z of the instruments in Z , where μ_k represents unobserved preferences for choice k and $V_k(z)$ denote the costs of choosing option k . Let (z_k, z_l) and (z'_k, z_l) be two realizations of $Z_k, Z_l \in Z$, and, w.o.l.g, suppose $z_k > z'_k$.

Suppose that $Z_k \cap Z_l = \emptyset$, such that there is no intersection in the instruments that affect each choice. That is, Z_k and Z_l only affect the cost of option k and l , respectively. If $z_k > z'_k$, then $V_k(z_k, z_l) \geq V_k(z'_k, z_l)$ while $V_l(z_k, z_l) = V_l(z'_k, z_l)$ for all $l \in \mathcal{K} \setminus k$. That is, $U_k(z_k, z_l) \leq U_k(z'_k, z_l)$, and $U_l(z_k, z_l) = U_l(z'_k, z_l)$ for all $l \in \mathcal{K} \setminus k$. Thus, $(z_k, z_l) \rightarrow (z'_k, z_l)$ weakly induces agents towards k such that $D_{\mathcal{K},k}(z_k, z_l) \leq D_{\mathcal{K},k}(z'_k, z_l)$. In this case, partial monotonicity holds trivially as the choice problem is represented by a latent index model as in Vytlačil (2002).

Now consider the case of $Z_k \cap Z_l \neq \emptyset$ and let $Z_l \in Z_k \cap Z_{\mathcal{K} \setminus k}$, such that there is at least one instrument that affects k and options in set $\mathcal{K} \setminus k$. To simplify the notation, assume that $V_l(z_k, z_l) \leq V_l(z'_k, z_l)$ such that as $(z_k, z_l) \rightarrow (z'_k, z_l)$ option l becomes less attractive, while $V_m(z_k, z_l) = V_m(z'_k, z_l)$ for all $m \in \mathcal{K} \setminus \{k, l\}$. Note that, under these conditions, assumption A3 does not hold as agents could switch along the margins $l \rightarrow k$, $l \rightarrow m$, and $m \rightarrow k$ for all $m \in \mathcal{K} \setminus \{k, l\}$. Impose restrictions in R1, such that $V_l(Z_k, Z_l)$ is convex in Z_l , $\frac{\partial^2 V_k(Z_k, Z_l)}{\partial Z_l^2} \geq 0$, and restriction R1.2 also holds. By proposition 1, as $Z_l \rightarrow z_l$, $D_{\mathcal{K} \setminus k, l}(Z_k, z_l) \rightarrow 1$ with $D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \rightarrow 0$ for all $m \neq l \in \mathcal{K} \setminus k$ which rules out $m \rightarrow k$ switches for all $m \in \mathcal{K} \setminus \{k, l\}$. Then, $D_{\mathcal{K},k}(z_k, z_l) \leq D_{\mathcal{K},k}(z'_k, z_l)$. The argument for $Z_l \rightarrow z'_l$ is symmetric. By proposition 1, $D_{\mathcal{K} \setminus k, l}(Z_k, z'_l) \rightarrow 0$ which rules out margins $l \rightarrow k$ and $l \rightarrow m$ for all $m \in \mathcal{K} \setminus \{k, l\}$. Then, $D_{\mathcal{K},k}(z_k, z'_l) \leq D_{\mathcal{K},k}(z'_k, z'_l)$.

A.8 Proof of Lemma 2

Recall that $T_k(z_l)$ denotes the number of response-types (i.e., distinct columns) in the response matrix $R_k(z_l)$ induced by variation in Z_k and evaluated at different values of Z_l . By Proposition 1, as $Z_l \rightarrow z_l$, the next-best set reduces to $\{l\}$ with probability approaching one, that is $\mathcal{K} \setminus k = \{l\}$. This implies that among switchers into k , the only relevant counterfactual option is l . Similarly, as $Z_l \rightarrow z'_l$, the option l is excluded from the next-best set with high probability, and switchers into k must come from the remaining alternatives $m \in \mathcal{K} \setminus \{k, l\}$.

Therefore, $R_k(z_l)$ contains at most one switcher column (from l), in addition to the always-takers. On the other hand, $R_k(z'_l)$ contains at most $(K-2)$ switcher columns, one for each $m \in \mathcal{K} \setminus \{k, l\}$. For an internal evaluation point $z_l^* \in \text{supp}(Z_l)$, $R_k(z_l^*)$ contains at most $(K-1)$ switcher columns. Thus, by Proposition 2 and the argument in Lemma 1, the number of response-types is bounded above by $T_k(Z_l) \leq 2K - 1$.

A.9 Proof of Lemma 3

By Proposition 1, as $Z_l \rightarrow z_l$, $D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \rightarrow 0$ for all $m \neq l \in \mathcal{K} \setminus k$. It follows that:

$$\Pr(D_{\mathcal{K}, k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k, m}(z_k, z_l) = 1) \rightarrow 0 \quad \forall m \in \mathcal{K} \setminus \{k, l\},$$

and,

$$R_k(Z_l \rightarrow z_l) = \begin{bmatrix} 1 & \cdots & k & \cdots & K & l \\ 1 & \cdots & k & \cdots & K & k \end{bmatrix}.$$

Hence, $T_k(Z_l \rightarrow z_l) = 1 + (K - 1) = 1 + K$, with K always-takers and one complier.

By Proposition 1, as $Z_l \rightarrow z'_l$, $D_{\mathcal{K} \setminus k, l}(Z_k, z'_l) \rightarrow 0$ and:

$$\Pr(D_{\mathcal{K}, k}(z'_k, z'_l) = 1, D_{\mathcal{K} \setminus k, l}(z_k, z'_l) = 1) \rightarrow 0,$$

while $D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \geq 0$ for all $m \neq l \in \mathcal{K} \setminus k$, with $\sum_{m \neq l \in \mathcal{K} \setminus k} D_{\mathcal{K} \setminus k, m}(Z_k, z_l) \rightarrow 1$. The number of possible complier groups now excludes those coming from l into k , and includes all others $m \in \mathcal{K} \setminus \{k, l\}$.

$$R_k(Z_l \rightarrow z'_l) = \begin{bmatrix} 1 & \cdots & k & \cdots & K & 1 & \cdots & K \\ 1 & \cdots & k & \cdots & K & k & \cdots & k \end{bmatrix},$$

Thus, the number of compliers decreases to $K - 2$, and the total number of response-types is $T_k(Z_l \rightarrow z'_l) = 1 + [2(K - 1)] - 1 = 2(K - 1)$.

A.10 Proof of Remark 2.1

By A1 and matrix R_k ,

$$\begin{aligned} E[D_{\mathcal{K}, m}|z'_k, z_l] &= \Pr(D_{\mathcal{K}, m} = 1|z'_k, z_l) \\ &= \Pr(D_{\mathcal{K}, m}(z'_k, z_l) = 1, D_{\mathcal{K}, m}(z_k, z_l) = 1), \end{aligned}$$

and,

$$\begin{aligned} E[D_{\mathcal{K}, m}|z_k, z_l] &= \Pr(D_{\mathcal{K}, m} = 1|z_k, z_l) \\ &= \Pr(D_{\mathcal{K}, m}(z'_k, z_l) = 1, D_{\mathcal{K}, m}(z_k, z_l) = 1) \\ &\quad + \Pr(D_{\mathcal{K}, m}(z'_k, z_l) = 0, D_{\mathcal{K}, m}(z_k, z_l) = 1). \end{aligned}$$

Hence,

$$E[D_{\mathcal{K}, m}|z_k, z_l] - E[D_{\mathcal{K}, m}|z'_k, z_l] = \Pr(D_{\mathcal{K}, m}(z'_k, z_l) = 0, D_{\mathcal{K}, m}(z_k, z_l) = 1),$$

identifies the share of compliers (switchers) along the $m - k$ margin as Z_k goes from z_k to z'_k . By Lemma 3, and matrix $R_k(Z_l \rightarrow z_l)$,

$$E[D_{\mathcal{K},m}|z_k, z_l] - E[D_{\mathcal{K},m}|z'_k, z_l] = 0.$$

Likewise,

$$\begin{aligned} E[D_{\mathcal{K},l}|z'_k, z'_l] &= \Pr(D_{\mathcal{K},l} = 1|z'_k, z'_l) \\ &= \Pr(D_{\mathcal{K},l}(z'_k, z'_l) = 1, D_{\mathcal{K},l}(z_k, z'_l) = 1), \end{aligned}$$

and,

$$\begin{aligned} E[D_{\mathcal{K},l}|z_k, z'_l] &= \Pr(D_{\mathcal{K},l} = 1|z_k, z'_l) \\ &= \Pr(D_{\mathcal{K},l}(z'_k, z'_l) = 1, D_{\mathcal{K},l}(z_k, z'_l) = 1) \\ &\quad + \Pr(D_{\mathcal{K},l}(z'_k, z'_l) = 0, D_{\mathcal{K},l}(z_k, z'_l) = 1) \end{aligned}$$

Thus,

$$E[D_{\mathcal{K},l}|z_k, z'_l] - E[D_{\mathcal{K},l}|z'_k, z'_l] = \Pr(D_{\mathcal{K},l}(z'_k, z'_l) = 0, D_{\mathcal{K},l}(z_k, z'_l) = 1),$$

identifies the share of compliers (switchers) along the $l - k$ margin as Z_k goes from z_k to z'_k . By Lemma 3 and matrix $R_k(Z_l \rightarrow z'_l)$,

$$E[D_{\mathcal{K},l}|z_k, z'_l] - E[D_{\mathcal{K},l}|z'_k, z'_l] = 0.$$

A.11 Proof of Proposition 3

Suppose A1, A2a and R1 hold. Abusing notation,

$$\begin{aligned} \Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k, m}(z_k, z_l) = 1) \\ &= \Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K},m}(z_k, z_l) = 1) \\ &= \Pr(D_{\mathcal{K},m}(z'_k, z_l) = 0, D_{\mathcal{K},m}(z_k, z_l) = 1) \quad \forall m \in \mathcal{K} \setminus k. \end{aligned}$$

Recall that

$$\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K},k}(z_k, z_l) = 0) = \sum_{l \in \mathcal{K} \setminus k} \Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K},l}(z_k, z_l) = 1).$$

By Lemma 3, as $Z_l \rightarrow z_l$,

$$\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K} \setminus k, m}(z_k, z_l) = 1) \rightarrow 0 \quad \forall m \in \mathcal{K} \setminus \{k, l\},$$

and,

$$\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z_l) = 1) \rightarrow 1.$$

Thus,

$$\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K},k}(z_k, z_l) = 0) = \Pr(D_{\mathcal{K},l}(z'_k, z_l) = 0, D_{\mathcal{K},l}(z_k, z_l) = 1).$$

And equation (2) evaluated as $Z_l \rightarrow z_l$,

$$\begin{aligned} \frac{E[Y|z'_k, z_l] - E[Y|z_k, z_l]}{E[D_{\mathcal{K},k}|z'_k, z_l] - E[D_{\mathcal{K},k}|z_k, z_l]} &= \sum_{l \in \mathcal{K} \setminus k} \frac{\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z_l) = 1)}{\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K},k}(z_k, z_l) = 0)} \\ &\quad \times E[Y_k - Y_l | D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z_l) = 1], \end{aligned}$$

reduces to:

$$\frac{E[Y|z'_k, z_l] - E[Y|z_k, z_l]}{E[D_{\mathcal{K},k}|z'_k, z_l] - E[D_{\mathcal{K},k}|z_k, z_l]} = E[Y_k - Y_l | D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z_l) = 1].$$

By a similar argument, following Lemma 3, as $Z_l \rightarrow z'_l$,

$$\Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z'_l) = 1) \rightarrow 0,$$

and,

$$\begin{aligned} \frac{E[Y|z'_k, z'_l] - E[Y|z_k, z'_l]}{E[D_{\mathcal{K},k}|z'_k, z'_l] - E[D_{\mathcal{K},k}|z_k, z'_l]} &= \\ &\sum_{m \in \mathcal{K} \setminus \{k, l\}} \frac{\Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K}\setminus \{k, l\}, m}(z_k, z'_l) = 1)}{\Pr(D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K},k}(z_k, z'_l) = 0)} \\ &\quad \times E[Y_k - Y_m | D_{\mathcal{K},k}(z'_k, z'_l) = 1, D_{\mathcal{K}\setminus \{k, l\}, m}(z_k, z'_l) = 1]. \end{aligned}$$

A.12 Proof of Proposition 4

By the same logic in Appendix A.11, as $Z_l \rightarrow z_l$, $\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,m}(z_k, z_l) = 1) \rightarrow 0 \quad \forall m \in \mathcal{K} \setminus \{k, l\}$, and, $\Pr(D_{\mathcal{K},k}(z'_k, z_l) = 1, D_{\mathcal{K}\setminus k,l}(z_k, z_l) = 1) \rightarrow 1$. Hence, invoking a limit argument as in Heckman and Vytlačil (2007) and Heckman et al. (2008, 2006),

$$\begin{aligned} \lim_{Z_l \rightarrow z_l} E[Y_k - Y_{\mathcal{K}\setminus k} | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)] &= \Pr(D_{\mathcal{K}\setminus k,l} = 1 | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)) E[Y_k - Y_l | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)] \\ &+ \sum_{m \in \mathcal{K}\setminus k, l} \Pr(D_{\mathcal{K}\setminus k,m} = 1 | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)) E[Y_k - Y_l | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)] \\ &= E[Y_k - Y_l | Z = z, U_k(z) = U_{\mathcal{K}\setminus k}(z)]. \end{aligned}$$

A similar argument trivially follows for $Z_l \rightarrow z'_l$.

B Example: Panel (b) in Figure 1

This section follows the Example in Section 2.2, specifically for the case displayed in Panel (b) of Figure 1 and the combination of potential choices summarized in the unconditional response matrix:

$$R_1 = \begin{bmatrix} 1 & 2 & 3 & 3 & 2 \\ 1 & 2 & 3 & 1 & 1 \end{bmatrix},$$

In this case, the Wald estimator identifies:

$$\begin{aligned} \frac{E[Y|Z_1 = 1] - E[Y|Z_1 = 0]}{E[D_{\mathcal{K},1}|Z_1 = 1] - E[D_{\mathcal{K},1}|Z_1 = 0]} &= \sum_{l=2}^3 \frac{\Pr(D_{\mathcal{K},l}(0) = 1, D_{\mathcal{K},l}(1) = 0)}{\Pr(D_{\mathcal{K},1}(0) = 0, D_{\mathcal{K},1}(1) = 1)} \\ &\quad \times E[Y_1 - Y_l | D_{\mathcal{K},l}(0) = 1, D_{\mathcal{K},l}(1) = 0]. \end{aligned}$$

The MTE can be decomposed into:

$$\begin{aligned} E[Y_1 - Y_{\mathcal{K} \setminus 1} | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\ &= E[D_{\mathcal{K} \setminus 1,2}(Y_1 - Y_2) + D_{\mathcal{K} \setminus 1,3}(Y_1 - Y_3) | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\ &= \Pr(D_{\mathcal{K} \setminus 1,2} = 1 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)) E[Y_1 - Y_2 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\ &\quad + \Pr(D_{\mathcal{K} \setminus 1,3} = 1 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)) E[Y_1 - Y_3 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)]. \end{aligned}$$

The role of conditional response vectors: The following matrix summarizes conditional response vectors as $Z_3 \rightarrow z_3^{\min}$,

$$R_1(Z_3 \rightarrow z_3^{\min}) = \begin{bmatrix} 1 & 2 & 3 & 3 \\ 1 & 2 & 3 & 1 \end{bmatrix},$$

which, by lemma 3, implies that $\Pr(D_{\mathcal{K},2}(0) = 1, D_{\mathcal{K},2}(1) = 0) = 0$. That is, following the cost structure in Panel (b) of Figure 1, the likelihood of option 2 being the next-best alternative approaches zero as $Z_3 \rightarrow z_3^{\min}$. Hence, by proposition 3, equation (A1) evaluated as $Z_3 \rightarrow z_3^{\min}$ reduces to:

$$\begin{aligned} \frac{E[Y|Z_1 = 1, z_3^{\min}] - E[Y|Z_1 = 0, z_3^{\min}]}{E[D_{\mathcal{K},1}|Z_1 = 1, z_3^{\min}] - E[D_{\mathcal{K},1}|Z_1 = 0, z_3^{\min}]} &= \\ E[Y_1 - Y_3 | D_{\mathcal{K},3}(0, z_3^{\min}) = 1, D_{\mathcal{K},3}(1, z_3^{\min}) = 0]. \end{aligned}$$

By proposition 4, equation (A2) evaluated as $Z_3 \rightarrow z_3^{\min}$ reduces to:

$$\begin{aligned}
& \lim_{Z_3 \rightarrow z_3^{\min}} E[Y_1 - Y_{\mathcal{K} \setminus 1} | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\
&= \lim_{Z_3 \rightarrow z_3^{\min}} \Pr(D_{\mathcal{K} \setminus 1, 2} = 1 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)) E[Y_1 - Y_2 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\
&+ \lim_{Z_3 \rightarrow z_3^{\min}} \Pr(D_{\mathcal{K} \setminus 1, 3} = 1 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)) E[Y_1 - Y_3 | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] \\
&= E[Y_1 - Y_3 | Z = z, U_1(z) = U_3(z)],
\end{aligned}$$

which is the marginal treatment effect of option 1 vs. 3 for those at the margin of indifference between option 1 and 3.

By a similar argument, following propositions 3 and 4, as $Z_3 \rightarrow z_3^{\max}$,

$$R_1(Z_3 \rightarrow z_3^{\max}) = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

which implies that $\Pr(D_{\mathcal{K}, 3}(0) = 1, D_{\mathcal{K}, 3}(1) = 0) = 0$, and equation (A1) evaluated as $Z_3 \rightarrow z_3^{\max}$ reduces to:

$$\begin{aligned}
& \frac{E[Y | Z_1 = 1, z_3^{\max}] - E[Y | Z_1 = 0, z_3^{\max}]}{E[D_{\mathcal{K}, 1} | Z_1 = 1, z_3^{\max}] - E[D_{\mathcal{K}, 1} | Z_1 = 0, z_3^{\max}]} = \\
& E[Y_1 - Y_2 | D_{\mathcal{K}, 2}(0, z_3^{\max}) = 1, D_{\mathcal{K}, 2}(1, z_3^{\max}) = 0],
\end{aligned}$$

and,

$$\lim_{Z_3 \rightarrow z_3^{\max}} E[Y_1 - Y_{\mathcal{K} \setminus 1} | Z = z, U_1(z) = U_{\mathcal{K} \setminus 1}(z)] = E[Y_1 - Y_2 | Z = z, U_1(z) = U_2(z)],$$

the marginal treatment effect of option 1 vs. 2 for those at the margin of indifference between option 1 and 2.

C Generalized Response Matrices

This appendix follows the econometric model and the identification results in Heckman and Pinto (2018). The authors develop a general framework for identifying treatment effects in settings with unordered discrete choices and essential heterogeneity, using response matrices to characterize agent behavior induced by categorical instruments. Section C.1 shows that, when the instrument has support $w = 2$, their identification results correspond to the formulation used in the main text. Section C.2 extends the framework to the case $w > 2$, generalizing the definition of conditional response-types to accommodate categorical instruments.

C.1 Equivalence for $w = 2$

Here, I show how the econometric model and results in [Heckman and Pinto \(2018\)](#) can be employed to identify treatment effects for $w = 2$, as well as identifying counterfactuals for always-takers or imposing the homogeneity assumption. Recall that, following lemma 1, the unconditional response matrix is:

$$R_k = \begin{bmatrix} 1 & \cdots & k & \cdots & K & 1 & \cdots & l & \cdots & K \\ 1 & \cdots & k & \cdots & K & k & \cdots & k & \cdots & k \end{bmatrix}.$$

Define a binary matrix B^k where the superscript denotes an option $k \in \mathcal{K}$. B^k takes the value of one each time k appears in R_k . For instance,

$$B^1 = \begin{bmatrix} 1 & \cdots & 0 & \cdots & 0 & 1 & \cdots & 0 & \cdots & 0 \\ 1 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 0 \end{bmatrix}.$$

Stacking option-specific B matrices results in:

$$B = \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & 1 & 0 & \cdots & 0 \\ 1 & 0 & \cdots & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & 0 & 1 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 1 & \cdots & 0 & 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 1 & 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & 0 & \cdots & 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{matrix} \left. \begin{matrix} \\ \\ \end{matrix} \right\} B^1 \\ \left. \begin{matrix} \\ \\ \end{matrix} \right\} B^2 \\ \\ \left. \begin{matrix} \\ \\ \end{matrix} \right\} B^k \\ \\ \left. \begin{matrix} \\ \\ \end{matrix} \right\} B^K \end{matrix}$$

Now define $\sum_k(q)$ for $q \in \{1, 2\}$, which denotes sets of responses in R_k where k appears q times. For every $l \in \mathcal{K} \setminus k$, $\sum_l(1) = \{S_{l-k}\}$ and $\sum_l(2) = \{S_l\}$, where S_l denotes option l -always-takers and S_{l-k} denotes $l - k$ compliers. For option k , $\sum_k(1) = \{S_{1-k}, S_{2-k}, \dots, S_{K-k}\}$ and $\sum_k(2) = \{S_k\}$. That is, for option k the set of responses where k appears exactly one time (*i.e.*, compliers) contains $K - 1$ elements. Let $b_k(q)$ be a binary vector that takes the value of one every time option k appears q times in matrix R_k . It follows that $b_k(1) = [0, 0, \dots, 0, \dots, 0, 1, 1, \dots, 1]$, which indicates which response types belong to $\sum_k(1)$ (in this case, all complier response types), while $b_k(2) = [0, 0, \dots, 1, \dots, 0, 0, 0, \dots, 0]$ where the value of one is associated with the option k -always-taker.

Let YD_k denote a vector of interactions between outcome Y and choice $k \in \{1, 2, \dots, K\}$, with $D_k = 1$ if option k is chosen. Moreover, define D as a categorical variable equal to

k if option k is chosen. Define $L_{Y_k} = [E[YD_k|z_k], E[YD_k|z'_k]]'$ as the average of outcome Y when option k is chosen, evaluated at realizations $z_k, z'_k \in \text{supp}(Z_k)$. Hence, following Theorem T-6 in Heckman and Pinto (2018), the following counterfactuals are identified:

$$E\left(Y_k \mid S \in \sum_{k,k}(2)\right) = \frac{b_k(2)B^{k+}L_{Y_k}^k}{b_k(2)B^{k+}\Pr(D=k)},$$

for all groups of always-takers. Where, B^{k+} is the Moore-Penrose pseudoinverse, and $\Pr(D=k)$ is the propensity score of choice k evaluated at realizations $z_k, z'_k \in \text{supp}(Z_k)$. That is, $\Pr(D=k) = [\Pr(D=k|z_k), \Pr(D=k|z'_k)]$.²⁸ For instance, for $k=1$,

$$\begin{aligned} E(Y_1 \mid S_1) &= \frac{b_1(2)B^{1+}L_{Y_1}}{b_1(2)B^{1+}\Pr(D=1)} \\ &= \frac{E(YD_1|z'_k)}{E(D_1|z'_k)}, \end{aligned}$$

which follows from applying the definitions of B^{1+} , L_{Y_1} , and $\Pr(D=1)$ above. Moreover,

$$\begin{aligned} E\left(Y_k \mid S \in \sum_k(1)\right) &= \frac{b_k(1)B^{k+}L_{Y_k}^k}{b_k(1)B^{k+}\Pr(D=k)} \\ &= \frac{E(YD_k|z'_k) - E(YD_k|z_k)}{E(D_k|z'_k) - E(D_k|z_k)}, \end{aligned}$$

which identifies the counterfactual of option k for compliers switching into k , that is, $\{S_{1-k}, S_{2-k}, \dots, S_{K-k}\}$; and,

$$\begin{aligned} E(Y_l \mid S_{l-k}) &= \frac{b_l(1)B^{l+}L_{Y_l}}{b_l(1)B^{l+}\Pr(D=l)} \\ &= \frac{E(YD_l|z_k) - E(YD_l|z'_k)}{E(D_l|z_k) - E(D_l|z'_k)}, \end{aligned}$$

which denotes the counterfactual of choosing option l for compliers along the $l-k$ margin, for each $l \in \mathcal{K} \setminus k$. Given that the counterfactual for option k is only identified for the combination of compliers, rather than for one complier alone, only the effect of k vs. the next-best can be identified as a weighted average of margin specific treatment effects. Direct identification of this parameter also follows from Corollary C-5 in Heckman and Pinto (2018). These results are equivalent to those presented in equation (2).

From lemma 3, the prevalence of some complier groups goes to zero as $Z_l \rightarrow z_l$ or $Z_l \rightarrow z'_l$. This implication can be tested empirically by estimating the share of compliers conditional on Z_l . Let $P_g(Z_l)$ represent a vector containing the shares of each response type in matrix R_k , of dimensions $2K-1 \times 1$, conditional on Z_l . These shares can be

²⁸If B has full column rank then $B^+ = (B'B)^{-1} B'$.

identified from:

$$P_g(Z_l) = B^+ \Pr(D|Z_l). \quad (C1)$$

Vector $\Pr(D|Z_l)$ contains the propensity score for choices $k \in \{1, 2, \dots, K\}$ evaluated at instrument Z_k , conditional on Z_l . I denote the shares as functions of Z_l , to indicate that the prevalence of different types of responses can vary depending on the evaluation point in Z_l . Let $\Pr(D = k|Z_l) = [\Pr(D = k|z_k, Z_l), \Pr(D = k|z'_k, Z_l)]$ be the propensity score of choice k evaluated at realizations $z_k, z'_k \in \text{supp}(Z_k)$, conditional on Z_l . Thus,

$$\Pr(D|Z_l) = [\Pr(D = 1|Z_l), \Pr(D = 2|Z_l), \dots, \Pr(D = K|Z_l)]' \quad (C2)$$

and its dimension is $(2 \cdot K) \times 1$. Intuitively, the identification of shares of responses is a weighed combination of observed choices and potential behavior at different values of the instruments, given Z_l . From lemma 3, as $Z_l \rightarrow z_l$, all elements of the column vector $P_g(Z_l)$ from position $K + 1$ onward converge to zero, except for the entry at position $K + l$. As $Z_l \rightarrow z'_l$, the element at position $K + l$ of the column vector $P_g(Z_l)$ converges to zero. These results are equivalent to those presented in remark 2.1.

C.2 Extension to $w > 2$

The previous subsection showed that, when the instrument has binary support ($w = 2$), the identification results in the main text align with the general framework in Heckman and Pinto (2018). While their model accommodates categorical instruments, it does not explicitly distinguish response types conditional on a second instrument. As shown in the binary case, conditioning helps isolate substitution patterns along a specific margin. This distinction is also empirically relevant when $w > 2$, as variation in the conditioning instrument may affect the composition of switchers. In what follows, I formalize the conditional response matrix for $w > 2$ and outline how it can aid in the identification of treatment effects for subsets of response types.

To start, consider definitions 1 and 2 in the general case where the instrument takes on w values. In this case, the unconditional response matrix R_k has dimensions $w \times T_k$, while the conditional response matrix $R_k(z_l)$ has dimensions $w \times T_k(z_l)$. The following result extends the logic of convexity of costs to the case of multivalued instruments, where variation in Z_k can induce different response types depending on the value of Z_l .

Remark 4.2 (Elimination of response-types). *Suppose the conditions of Proposition 1 hold. For discrete Z_k with $w > 2$, as $Z_l \rightarrow z_l$, only response types switching between options k and l remain. As $Z_l \rightarrow z'_l$, response types switching between options k and l are not admissible.*

As in the main argument of the paper, increasing the cost of option l reduces its likelihood of being chosen as a next-best alternative, thereby reducing the number of

response-types that switch into option k from l . To illustrate this logic in a setting with a multivalued instrument, suppose the choice set is $\mathcal{K} = \{1, 2, 3\}$, and Z_1 is an IV denoting assignment status in a multi-arm RCT, with $\text{supp}(Z_1) = \{0, 1, 2\}$, where 0 is the control group, and 1 and 2 are treatment arms that provide incentives to choose option 1. Assume that both treatment arms encourage selecting of option 1, but the incentive is stronger under arm 2. Further assume that if option 1 is chosen under weaker incentives ($Z_1 = 1$), then it should be chosen under stronger incentives ($Z_1 = 2$); and that Z_1 does not affect the utility of choosing options 2 or 3. Thus, variation across Z_1 can induce different response-types into 1. The unconditional response-matrix is:

$$R_1 = \begin{bmatrix} 1 & 2 & 3 & 2 & 2 & 3 & 3 \\ 1 & 2 & 3 & 2 & 1 & 3 & 1 \\ 1 & 2 & 3 & 1 & 1 & 1 & 1 \end{bmatrix},$$

which contains combinations of potential choices at different values of the instrument. The first, second, and third rows correspond to potential choices when $Z_1 = 0, 1$, and 2 , respectively. Columns 1–3 represent always-takers for options 1, 2, and 3 (denote these response-types as g_1, g_2 , and g_3). Columns 4–7 denote *switchers*, or agents who would change their choices towards option 1 as a result of the variation in Z_1 (denote these response-types as g_4, g_5, g_6 , and g_7). Variation in Z_1 alone does not allow identification of the effect of choosing option 1 versus option 2 or 3; instead, it allows identification of treatment effects for switchers into option 1, averaging over different next-best alternatives.

Suppose Z_3 satisfies the restrictions in [R1](#). Then as $Z_3 \rightarrow z_3^{\min}$, option 3 has the lowest cost and would have been chosen instead of 2, while as $Z_3 \rightarrow z_3^{\max}$ option 3 is too costly to have been chosen over option 2. Hence, the conditional response-matrices are given by:

$$R_1(Z_3 \rightarrow z_3^{\min}) = \begin{bmatrix} 1 & 2 & 3 & 3 & 3 \\ 1 & 2 & 3 & 3 & 1 \\ 1 & 2 & 3 & 1 & 1 \end{bmatrix}, \quad R_1(Z_3 \rightarrow z_3^{\max}) = \begin{bmatrix} 1 & 2 & 3 & 2 & 2 \\ 1 & 2 & 3 & 2 & 1 \\ 1 & 2 & 3 & 1 & 1 \end{bmatrix}.$$

The remaining combinations of potential choices in $R_1(Z_3 \rightarrow z_3^{\min})$ and $R_1(Z_3 \rightarrow z_3^{\max})$ have the following implications: $\Pr(g_4) \rightarrow 0$ and $\Pr(g_5) \rightarrow 0$ as $Z_3 \rightarrow z_3^{\min}$, and $\Pr(g_6) \rightarrow 0$ and $\Pr(g_7) \rightarrow 0$ as $Z_3 \rightarrow z_3^{\max}$. In general, these implications on the prevalence of response-types can be tested empirically using the general version of equation [C1](#). In this case, $\Pr(D|Z_l)$ has dimensions $(w \cdot K) \times 1$.

To define the counterfactuals that can be identified, let $\Sigma_k(q)$ be a set that contains responses in matrix R_k where k appears q times, with $q \in \{1, \dots, w\}$. Let $b_k(q)$ be a binary vector that takes the value of one every time option k appears q times in

matrix R_k . For general $w > 2$, $L_{Y_k} = [E[YD_k|z_{1k}], E[YD_k|z_{2k}], \dots, E[YD_k|z_{wk}]]'$ as the average of outcome Y when option $k \in \{1, 2, \dots, K\}$ is chosen, evaluated at realizations $z_{1k}, z_{2k}, \dots, z_{wk}$. If unordered monotonicity holds, then the following counterfactuals can be identified:

$$E(Y_k | S \in \sum_k(q)) = \frac{b_k(q)B^{k+}L_{Y_k}}{b_k(q)B^{k+}\Pr(D = k)} \quad (C3)$$

For unconditional response matrix R , [Heckman and Pinto \(2018\)](#) show that R is unordered monotonic (that is, unordered monotonicity is satisfied) if and only if each $B^k = \mathbf{1}[R = k]$ is lonesum such that there are no two-way patterns in any sub-matrix of dimension 2×2 . That is, there is no 2×2 sub-matrix in R that takes the form of:

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \quad \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

If the set $\sum_k(q)$ contains exactly one element, then response specific counterfactuals can be identified. Otherwise, the counterfactuals for k , such that set $\sum_k(q)$ is not unique, are a combination weighted over next-best alternatives.

D Empirical Applications

D.1 Additional Tables and Figures

Table D1: Average characteristics, by type of health care

Variable	Drug-shop (s)	Health center (h)	Nothing (n)
Age of hh head	38.338	37.015	43.170
Years of educ of hh head	5.536	5.878	4.311
Household size	5.558	5.455	4.730
<i>Health at baseline</i>			
Number of bednets owned	1.867	1.879	1.483
Any malaria episode in hh	0.719	0.735	0.558
Drugs from drug-shop	0.309	0.345	0.214
Drugs from health center	0.361	0.305	0.235
<i>Outcomes</i>			
Illness episode reported at endline	3.314	3.139	2.174
%	61.4	20	18.6
N	1,517	495	459

Source: subsample from [Cohen et al. \(2015a\)](#).

Notes: The sample consists of households in control and ACT subsidies groups, excluding those who also received testing subsidies. All socioeconomic variables were collected at baseline.

Table D2: Average of instruments, by type of health care

Variable	Drug-shop (s)	Health center (h)	Nothing (n)
ACT Subsidy	0.950	0.915	0.913
Distance to the health center (h), in km	6.471	6.563	6.623
Distance to the drug-shop (s), in km	1.647	1.646	1.670
N	1,517	495	459

Source: subsample from [Cohen et al. \(2015a\)](#). *Notes:* The sample consists of households in control and ACT subsidies groups, excluding those who also received testing subsidies.

Table D3: Average characteristics, by type of childcare

Variable	Large centers (l)	Small centers (s)	Home care (h)
Age (months)	36.980	36.241	37.553
Male	0.515	0.526	0.465
Children 0-5 at home	1.477	1.536	1.730
Mother's years of education	8.986	8.697	7.791
Low income household	0.409	0.442	0.535
Distance to small center (s), in km	0.299	0.331	0.290
Children development at baseline			
Cognitive	0.027	-0.094	0.121
Socio-emotional	-0.022	0.091	-0.135
Nutritional	0.035	0.068	-0.323
Outcomes [Obs.]			
Cognitive [1231]	0.166 [674]	-0.021 [401]	-0.662 [156]
Socio-emotional [1238]	-0.064 [678]	-0.035 [403]	0.368 [157]
Nutritional [927]	0.127 [513]	0.007 [299]	-0.584 [115]
%	54.9	32.34	12.76
N	684	403	159

Source: subsample from [Bernal et al. \(2019\)](#). *Notes:* The sample consists of children who were initially at small centers. All socioeconomic variables were collected at baseline. Cognitive development is a composite of scores from the ASQ. Socio-emotional development is a composite of behavioral components of the ASQ. All scores from the ASQ are age standardized. Nutritional development corresponds to z-scores for weight-for-age, height-for-age, and weight for height.

Table D4: Average of instruments, by type of childcare

Variable	Large centers (l)	Small centers (s)	Home care (h)	All
Wins lottery	0.805	0.615	0.802	0.743
Distance to large center (l), in km	0.668	1.125	0.912	0.848

Source: subsample from [Bernal et al. \(2019\)](#). *Notes:* The sample consists of children who were initially at small centers. Each column represents a childcare choice; the last column presents averages for the full sample for comparison. The row "Wins lottery" refers to the fraction of households that were assigned to receive information and the option to transfer to large centers. Distance to large centers is the geographic distance from the child's home to the nearest large center, in km.

Table D5: Average characteristics, by enrollment choice and eligibility sample

Variable	Need-eligible			Merit-eligible		
	High-quality HEI (<i>h</i>)	Low-quality HEI (<i>l</i>)	Not enrolled (<i>n</i>)	High-quality HEI (<i>h</i>)	Low-quality HEI (<i>l</i>)	Not enrolled (<i>n</i>)
Age at Saber 11	16.875	16.937	17.476	16.807	16.767	17.499
Female	0.469	0.538	0.526	0.437	0.423	0.445
Household size	4.520	4.611	4.739	4.380	4.232	4.329
Mother's educ: primary	0.208	0.222	0.359	0.155	0.137	0.191
Mother's educ: secondary	0.472	0.485	0.485	0.439	0.444	0.464
Mother's educ: 2-year college	0.168	0.150	0.093	0.181	0.196	0.159
Mother's educ: 4-year college+	0.152	0.143	0.062	0.225	0.223	0.186
Household SES: lowest	0.304	0.385	0.386	0.178	0.147	0.141
<i>Outcomes</i>						
Formally employed in 2021	0.549	0.484	0.443	0.583	0.568	0.428
%	13.7	16.8	69.5	46.7	17.7	35.6
N	11,926	14,607	60,608	6,842	2,599	5,217

Source: subsample from [Londoño-Vélez et al. \(2025\)](#).

Notes: The sample consists of students belonging to the first cohort of SPP, and all the socioeconomic variables are measured in 2014 prior to the enrollment choice and the SPP announcement. Household SES consists of 6 socioeconomic stratum, with strata 1 denoting the lowest socioeconomic level. The outcome is a binary variable that indicates if the student was working in the formal labor market in 2021 which is measured by whether the student contributed to social security and was recorded in PILA during that year. Need-eligible (merit-eligible) denotes students who were below the poverty score cutoff for SPP (above the high school exit exam score cutoff for SPP).

Table D6: Average of instruments, by enrollment choice and eligibility sample

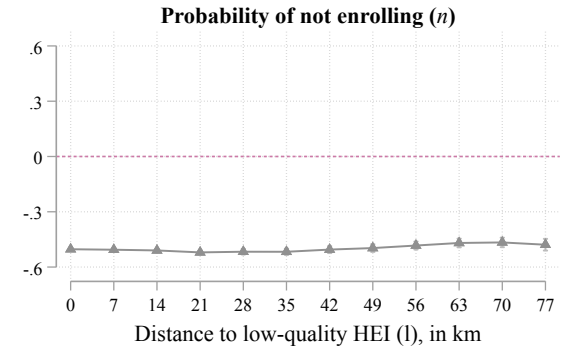
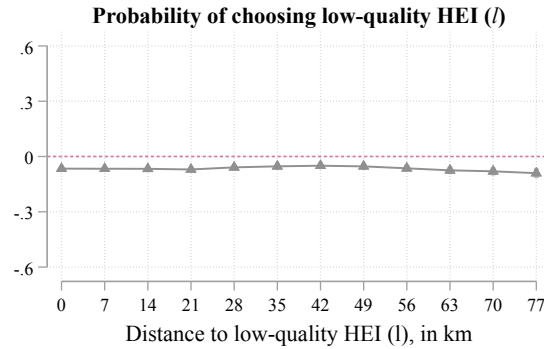
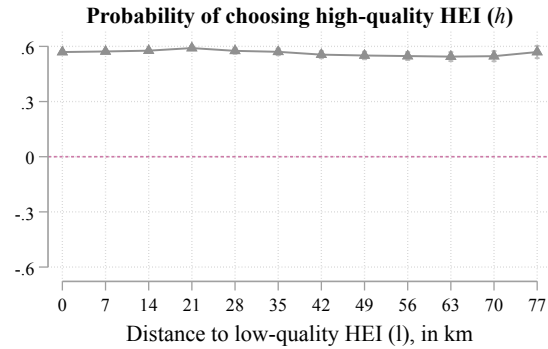
Variable	Need-eligible			Merit-eligible		
	High-quality HEI (<i>h</i>)	Low-quality HEI (<i>l</i>)	Not enrolled (<i>n</i>)	High-quality HEI (<i>h</i>)	Low-quality HEI (<i>l</i>)	Not enrolled (<i>n</i>)
Eligible	0.721	0.108	0.062	0.746	0.317	0.371
Distance to high-quality HEI	50.945	74.091	62.645	38.640	44.032	37.333
Distance to low-quality HEI	12.751	14.572	16.332	10.052	6.901	8.792
N	11,926	14,607	60,608	6,842	2,599	5,217

Source: subsample from [Londoño-Vélez et al. \(2025\)](#).

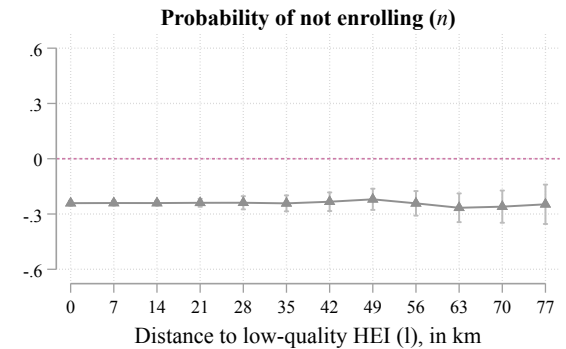
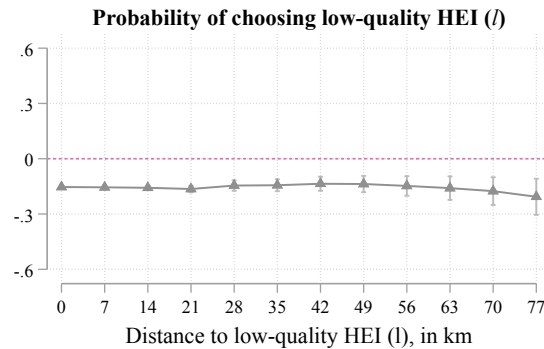
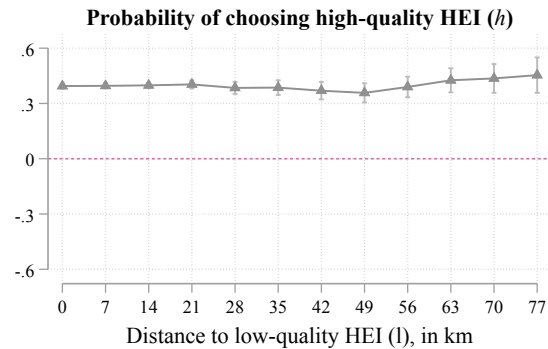
Notes: The sample consists of students belonging to the first cohort of SPP. Need-eligible (merit-eligible) denotes students who were below the poverty score cutoff for SPP (above the high school exit exam score cutoff for SPP). The row “Eligible” for the need-eligible (merit-eligible) refers to the fraction of students above the high school exit exam cutoff (below the poverty score cutoff). Distance to high- and low-quality HEI is the geographic distance from the student’s high school to the nearest high- and low-quality HEI respectively, in km.

Figure D1: Effect of SPP eligibility on the probability of enrolling in higher ed., by eligibility and distance to the nearest low-quality HEI

A. Need-Eligible Sample



B. Merit-Eligible Sample



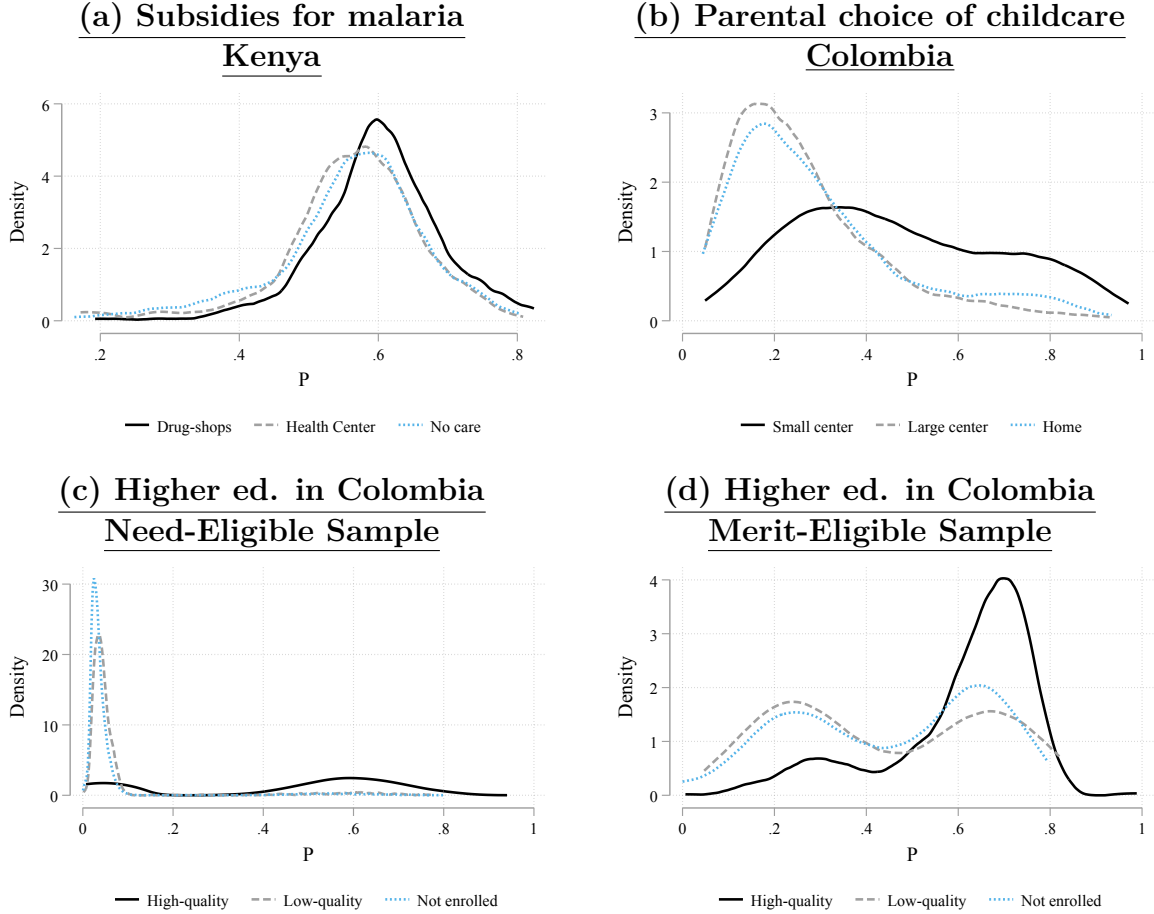
Source: subsample from [Londoño-Vélez et al. \(2025\)](#).

Notes: Panel A shows the effect of being above the high school exit exam cutoff for need-eligible students on the probability of choosing a high-quality HEI, a low-quality HEI, and not enrolling. Panel B shows the effect of being below the poverty score cutoff for merit-eligible students on the probability of choosing a high-quality HEI, a low-quality HEI, and not enrolling. Effects are computed with a semiparametric regression using an Epanechnikov kernel around different evaluations points of the distance to the nearest high-quality HEI. All figures show confidence intervals at the 90% level.

D.2 Common Support

Figure D2 plots propensity scores estimated with equation (3). Panel (a) displays the predicted probability of choosing the drug shop for those who chose the drug shop (in black), the health center (the dashed line in gray), and no care (the dotted line in blue). In this case, the common support is defined as the set of values $P \in [0.28, 0.8]$. Panel (b) shows the predicted probability of choosing small childcare centers and its distribution across childcare choices. The common support for this application is the interval $P \in [0.16, 0.76]$. Panels (c) and (d) show the predicted probability of enrolling in a high-quality university, as well as its distribution across choices. The common support for this application is the interval $P \in [0.12, 0.8]$.

Figure D2: Propensity Score



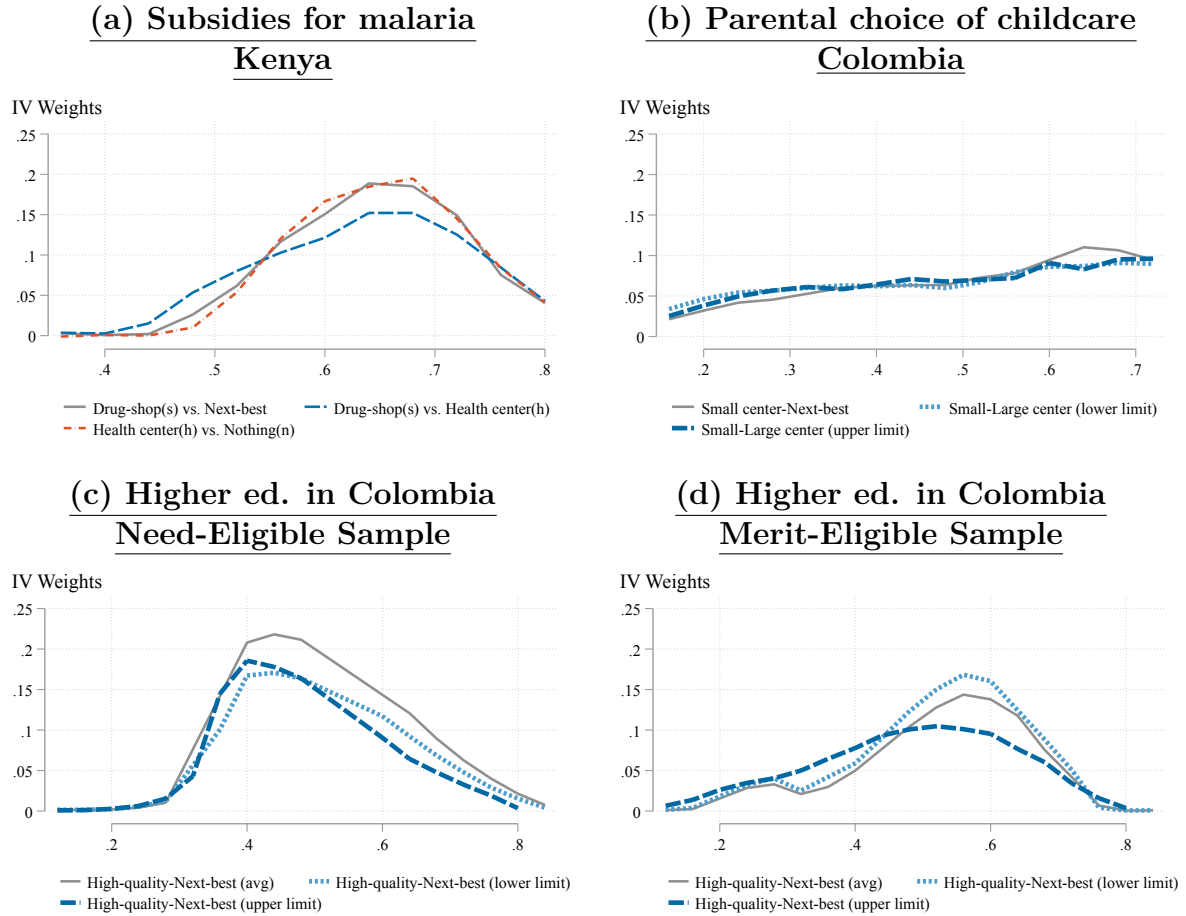
Source: Panel (a) subsample from [Cohen et al. \(2015a\)](#). Panel (b) subsample from [Bernal et al. \(2019\)](#). Panels (c) and (d) subsample from [Londoño-Vélez et al. \(2025\)](#).

Notes: The figure shows the distribution of agents across values of the propensity score for each application. Panel (a) presents a density plot for the propensity score of choosing drug shops. Panel (b) presents a density plot for the propensity score of choosing small childcare centers. Panels (c) and (d) present density plots for the propensity score of choosing high-quality universities.

D.3 IV Weights

The estimation of IV Weights follows [Heckman et al. \(2006\)](#) and its supplemental material. Figure D3 shows estimates of the weights that IV would place on the marginal treatment effects estimated across values of the propensity score.

Figure D3: IV Weights



Source: Panel (a) subsample from [Cohen et al. \(2015a\)](#). Panel (b) subsample from [Bernal et al. \(2019\)](#). Panels (c) and (d) subsample from [Londoño-Vélez et al. \(2025\)](#).

Notes: The figure shows estimated IV weights. Panel (a) presents IV weights for malaria treatment in Kenya. Panel (b) presents IV weights for parental choice of childcare in Colombia. Panels (c) and (d) present IV weights for higher education choices in Colombia.

D.4 Average of Baseline Variables

Subsidies for malaria treatment in Kenya:

The following averages of baseline variables is identified. For simplification, conditioning on Z_2 is left implicit. First, for h -always-takers:

$$\begin{aligned} E[XD_h|Z_1 = 1] &= E[X|D_h = 1, Z_1 = 1]\Pr(D_h = 1|Z_1 = 1) \\ &= E[X|D_h(0) = 1, D_h(1) = 1]\Pr(D_h(0) = 1, D_h(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of h -always-takers estimated using the results in Appendix A.5. For n -always-takers:

$$\begin{aligned} E[XD_n|Z_1 = 1] &= E[X|D_n = 1, Z_1 = 1]\Pr(D_n = 1|Z_1 = 1) \\ &= E[X|D_n(0) = 1, D_n(1) = 1]\Pr(D_n(0) = 1, D_n(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of n -always-takers estimated using the results in Appendix A.5. For s -always-takers:

$$\begin{aligned} E[XD_s|Z_1 = 0] &= E[X|D_s = 1, Z_1 = 0]\Pr(D_s = 1|Z_1 = 0) \\ &= E[X|D_s(0) = 1, D_s(1) = 1]\Pr(D_s(0) = 1, D_s(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of s -always-takers estimated using the results in Appendix A.5.

For $h - s$ compliers,

$$E[XD_h|Z_1 = 0] - E[XD_h|Z_1 = 1] = E[X|D_h(0) = 1, D_s(1) = 1]\Pr(D_h(0) = 1, D_s(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $h - s$ compliers estimated using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_h|Z_1 = 0] &= E[X|D_h = 1, Z_1 = 0]\Pr(D_h = 1|Z_1 = 0) \\ &= E[X|D_h(0) = 1, D_h(1) = 1]\Pr(D_h(0) = 1, D_h(1) = 1) \\ &\quad + E[X|D_h(0) = 1, D_s(1) = 1]\Pr(D_h(0) = 1, D_s(1) = 1). \end{aligned}$$

For $n - s$ compliers,

$$E[XD_n|Z_1 = 0] - E[XD_n|Z_1 = 1] = E[X|D_n(0) = 1, D_s(1) = 1]\Pr(D_n(0) = 1, D_s(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $h - s$ compliers estimated using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_n|Z_1 = 0] &= E[X|D_n = 1, Z_1 = 0]\Pr(D_n = 1|Z_1 = 0) \\ &= E[X|D_n(0) = 1, D_n(1) = 1]\Pr(D_n(0) = 1, D_n(1) = 1) \\ &\quad + E[X|D_n(0) = 1, D_s(1) = 1]\Pr(D_n(0) = 1, D_s(1) = 1). \end{aligned}$$

Figure D4 presents estimates of average baseline characteristics for compliers.

Parental choice of childcare in Colombia:

The following average of baseline variables is identified. For simplification, conditioning on Z_2 is left implicit. First, for h -always-takers (home care always takers):

$$\begin{aligned} E[XD_h|Z_1 = 0] &= E[X|D_h = 1, Z_1 = 0]\Pr(D_h = 1|Z_1 = 0) \\ &= E[X|D_h(0) = 1, D_h(1) = 1]\Pr(D_h(0) = 1, D_h(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of h -always-takers using the results in Appendix A.5. For s -always-takers:

$$\begin{aligned} E[XD_s|Z_1 = 1] &= E[X|D_s = 1, Z_1 = 1]\Pr(D_s = 1|Z_1 = 1) \\ &= E[X|D_s(0) = 1, D_s(1) = 1]\Pr(D_s(0) = 1, D_s(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of s -always-takers using the results in Appendix A.5. For l -always-takers:

$$\begin{aligned} E[XD_l|Z_1 = 0] &= E[X|D_l = 1, Z_1 = 0]\Pr(D_l = 1|Z_1 = 0) \\ &= E[X|D_l(0) = 1, D_l(1) = 1]\Pr(D_l(0) = 1, D_l(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of l -always-takers using the results in Appendix A.5.

For $s - h$ compliers,

$$E[XD_h|Z_1 = 0] - E[XD_h|Z_1 = 1] = E[X|D_s(0) = 1, D_h(1) = 1]\Pr(D_s(0) = 1, D_h(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $s - h$ compliers using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_h|Z_1 = 1] &= E[X|D_h = 1, Z_1 = 1]\Pr(D_h = 1|Z_1 = 1) \\ &= E[X|D_h(0) = 1, D_h(1) = 1]\Pr(D_h(0) = 1, D_h(1) = 1) \\ &\quad + E[X|D_s(0) = 1, D_h(1) = 1]\Pr(D_s(0) = 1, D_h(1) = 1). \end{aligned}$$

For $s - l$ compliers,

$$E[XD_l|Z_1 = 1] - E[XD_l|Z_1 = 0] = -E[X|D_s(0) = 1, D_l(1) = 1]\Pr(D_s(0) = 1, D_l(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $s - l$ compliers using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_l|Z_1 = 1] &= E[X|D_l = 1, Z_1 = 1]\Pr(D_l = 1|Z_1 = 1) \\ &= E[X|D_l(0) = 1, D_l(1) = 1]\Pr(D_l(0) = 1, D_l(1) = 1) \\ &\quad + E[X|D_s(0) = 1, D_l(1) = 1]\Pr(D_s(0) = 1, D_l(1) = 1). \end{aligned}$$

Figure D5 presents estimates of average baseline characteristics for compliers.

Higher education choices in Colombia:

The following average of baseline variables is identified. For simplification, conditioning on Z_2 is left implicit. First, for n -always-takers:

$$\begin{aligned} E[XD_n|Z_1 = 1] &= E[X|D_n = 1, Z_1 = 1]\Pr(D_n = 1|Z_1 = 1) \\ &= E[X|D_n(0) = 1, D_n(1) = 1]\Pr(D_n(0) = 1, D_n(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of n -always-takers using the results in Appendix A.5. For l -always-takers:

$$\begin{aligned} E[XD_l|Z_1 = 1] &= E[X|D_l = 1, Z_1 = 1]\Pr(D_l = 1|Z_1 = 1) \\ &= E[X|D_l(0) = 1, D_l(1) = 1]\Pr(D_l(0) = 1, D_l(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of l -always-takers using the results in Appendix A.5. For h -always-takers:

$$\begin{aligned} E[XD_h|Z_1 = 0] &= E[X|D_h = 1, Z_1 = 0]\Pr(D_h = 1|Z_1 = 0) \\ &= E[X|D_h(0) = 1, D_h(1) = 1]\Pr(D_h(0) = 1, D_h(1) = 1) \end{aligned}$$

identifies the average of baseline variables after the RHS is divided by the share of h -always-takers using the results in Appendix A.5.

For $n - h$ compliers,

$$E[XD_n|Z_1 = 0] - E[XD_n|Z_1 = 1] = E[X|D_n(0) = 1, D_h(1) = 1]\Pr(D_n(0) = 1, D_h(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $n - h$ compliers using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_n|Z_1 = 0] &= E[X|D_n = 1, Z_1 = 0]\Pr(D_n = 1|Z_1 = 0) \\ &= E[X|D_n(0) = 1, D_n(1) = 1]\Pr(D_n(0) = 1, D_n(1) = 1) \\ &\quad + E[X|D_n(0) = 1, D_h(1) = 1]\Pr(D_n(0) = 1, D_h(1) = 1). \end{aligned}$$

For $l - h$ compliers,

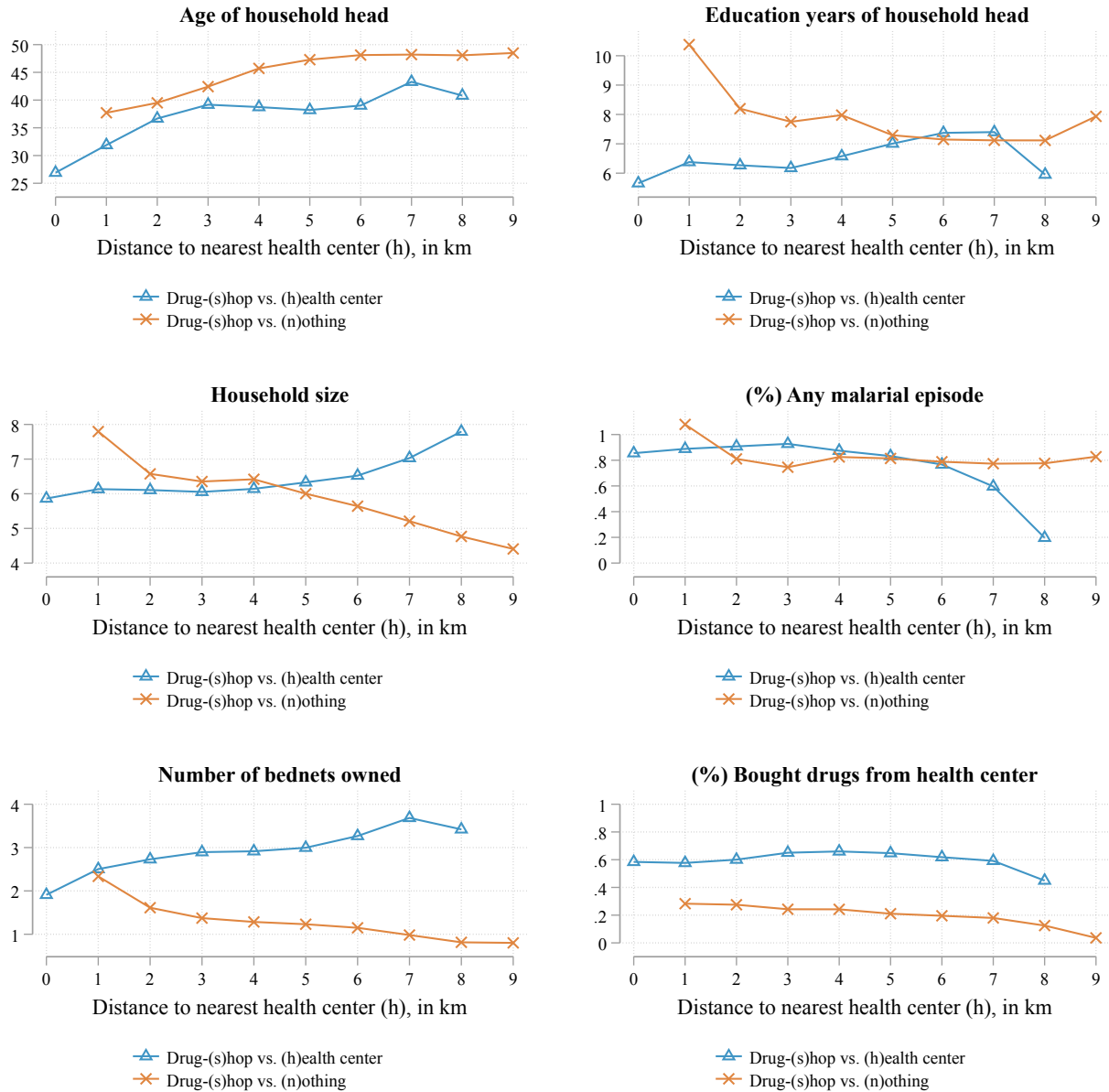
$$E[XD_l|Z_1 = 0] - E[XD_l|Z_1 = 1] = E[X|D_l(0) = 1, D_h(1) = 1]\Pr(D_l(0) = 1, D_h(1) = 1)$$

identifies the average of baseline variables after the RHS is divided by the share of $l - h$ compliers using the results in Appendix A.5. Note that

$$\begin{aligned} E[XD_l|Z_1 = 0] &= E[X|D_l = 1, Z_1 = 0]\Pr(D_l = 1|Z_1 = 0) \\ &= E[X|D_l(0) = 1, D_l(1) = 1]\Pr(D_l(0) = 1, D_l(1) = 1) \\ &\quad + E[X|D_l(0) = 1, D_h(1) = 1]\Pr(D_l(0) = 1, D_h(1) = 1). \end{aligned}$$

Figures D6 and D7 present estimates of average baseline characteristics for compliers for the need-eligible and merit-eligible students, respectively.

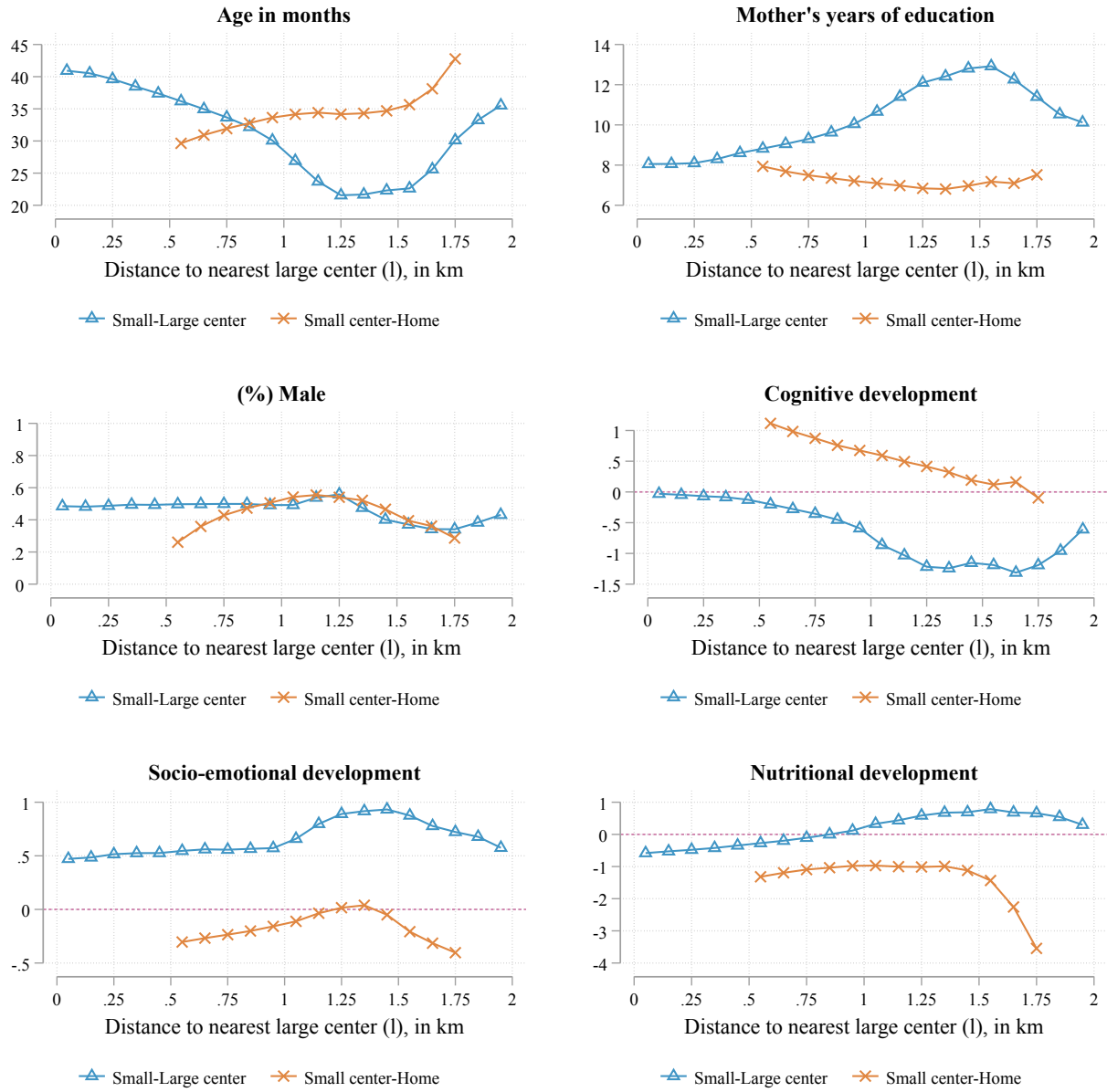
Figure D4: Average of baseline characteristics for compliers, by distance to the nearest health center (h)



Source: subsample from [Cohen et al. \(2015a\)](#)

Notes: The figure shows the average of baseline characteristics for compliers along the drug shop *versus* healthcare centers margin (hollow triangles in blue) and compliers along the drug shop *versus* no care margin (x in red), by distance to the nearest health center.

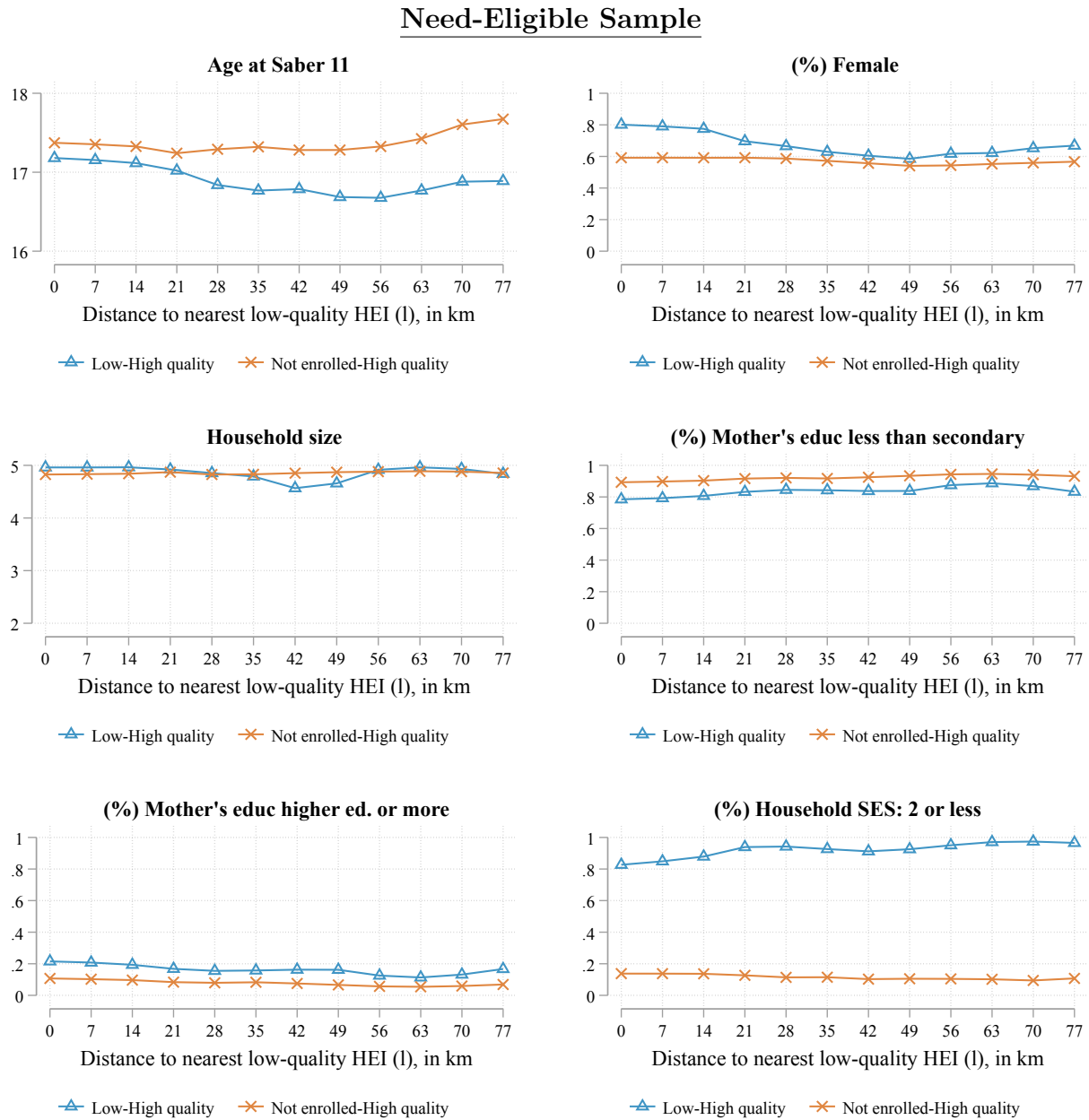
Figure D5: Average of baseline characteristics for compliers, by distance to the nearest large center (l)



Source: subsample from Bernal et al. (2019)

Notes: The figure shows the average of baseline characteristics for compliers along the small *versus* large centers margin (hollow triangles in blue) and compliers along the small center *versus* home margin (x in red), by distance to the nearest large center.

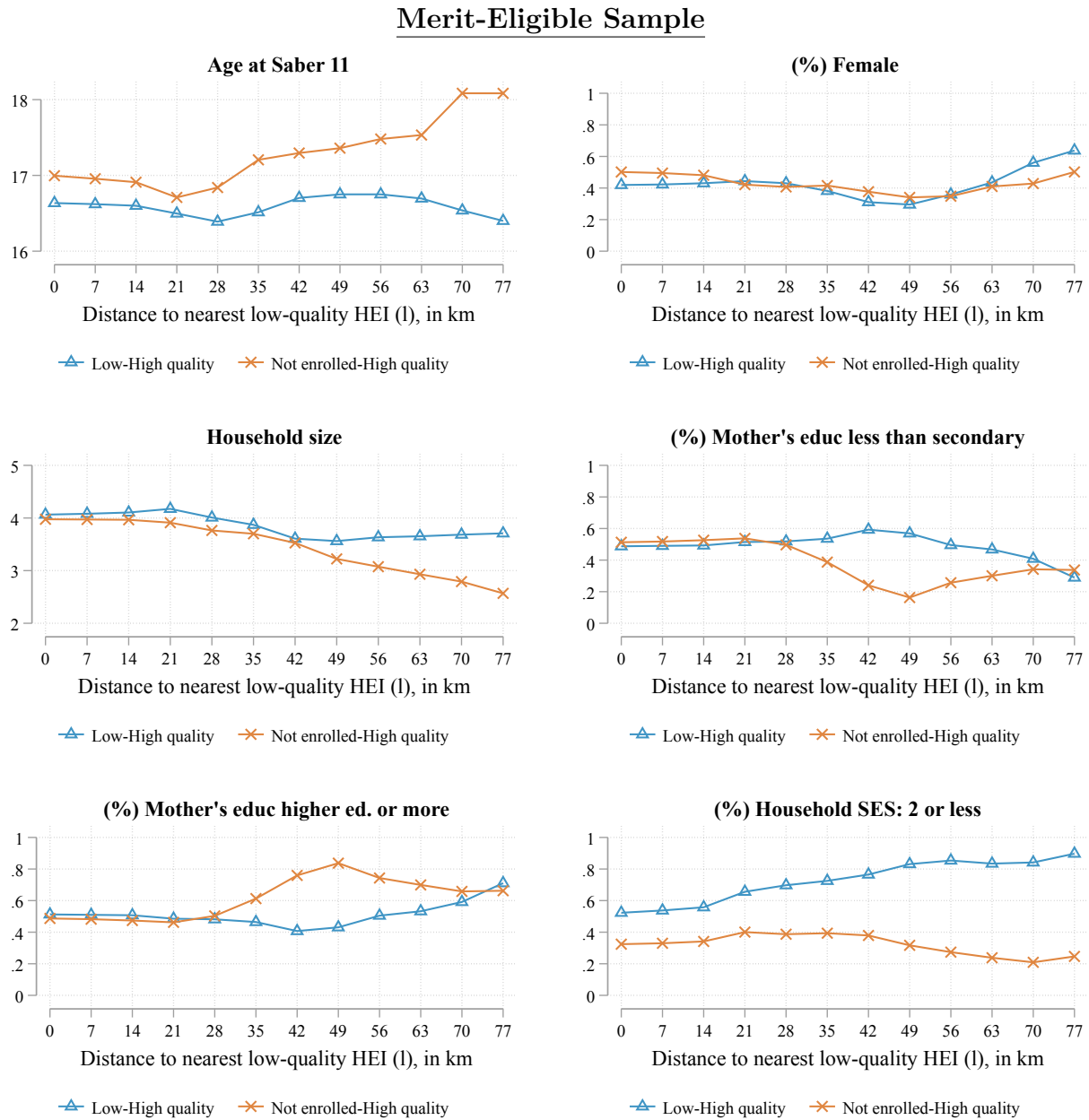
Figure D6: Average of baseline characteristics for compliers, by distance to the nearest low-quality HEI (l) in km



Source: subsample from [Londoño-Vélez et al. \(2025\)](#)

Notes: The figure shows the average of baseline characteristics for compliers along the low *versus* high-quality HEI margin (hollow triangles in blue) and compliers along the not enrolled *versus* high-quality HEI (x in red), by distance to the nearest low-quality HEI.

Figure D7: Average of baseline characteristics for compliers, by distance to the nearest low-quality HEI (l) in km



Source: subsample from [Londoño-Vélez et al. \(2025\)](#)

Notes: The figure shows the average of baseline characteristics for compliers along the low *versus* high-quality HEI margin (hollow triangles in blue) and compliers along the not enrolled *versus* high-quality HEI (x in red), by distance to the nearest low-quality HEI.